ETH zürich

ETH Library

CMOS integrated circuits for microelectrode arrays and design of experimental setups

Doctoral Thesis

Author(s):
Shadmani, Amir

Publication date:
2016

Permanent link:
https://doi.org/https://doi.org/10.3929/ethz-a-010874087

Rights / license:
In Copyright - Non-Commercial Use Permitted

This page was generated automatically upon download from the ETH Zurich Research Collection.
For more information, please consult the Terms of use.

DISS. ETH NO. 24004

CMOS INTEGRATED CIRCUITS FOR MICROELECTRODE ARRAYS AND DESIGN OF EXPERIMENTAL SETUPS

A thesis submitted to attain the degree of

DOCTOR OF SCIENCES of ETH ZURICH

(Dr. sc. ETH Zurich)

presented by

Amir Shadmani

M.Sc. in System-on-Chip, Lund University, Sweden

born on 30.08.1982 citizen of Iran

accepted on the recommendation of

Prof. Dr. Andreas Hierlemann
Prof. Dr. Hanspeter Schmid
Prof. Dr. Carlotta Guiducci

2017

Contents

Abstract ..... **vii**

Zusammenfassung ..... **ix**

Chapter 1 ..... **1**

Introduction **1**

1.1 Multifunctional MEA 1

1.2 Experimental setup design considerations 2

1.3 Scope and structure of the thesis 2

1.4 Author contributions 4

1.5 Major results 6

References 9

Chapter 2 ..... **11**

In-Vitro Multi-Functional Microelectrode Array Featuring 59760 Electrodes, 2048 Electrophysiology Channels, Stimulation, Impedance Measurement and Neurotransmitter Detection Channels **11**

2.1 Abstract 12

2.2 Introduction 13

2.3 System architecture 15

2.4 Functional units 18

2.4.1 Electrode array 18

2.4.2 Voltage recording channels 20

2.4.3 Neurotransmitter detection channels 24

2.4.4 Impedance measurement channels 26

2.4.5 Stimulation channels 28

2.5 System integration 29

2.5.1 Digital control 29

2.5.2 Chip layout and fabrication 30

2.6 Electrical characterization 31

2.6.1 Voltage recording channels 31

2.6.2 Neurotransmitter detection channels 32

2.6.3 Impedance measurement channels 33

2.6.4 Stimulation channels 34

2.7 Biological measurement results 35

2.8 Conclusion 37
Acknowledgments 39
References 40

Chapter 3 45

Integrated Current/Voltage Controlled Stimulation Buffer and Artifact Suppression Circuits for Multifunctional High-density Microelectrode Arrays 45

3.1 Abstract 46
3.2 Introduction 47
3.3 System overview 50
3.4 Circuit implementation 52
3.4.1 Near rail-to-rail OTA for the voltage-only controlled stimulation mode 52
3.4.2 Current conveyor with output voltage limiters for the current/voltage controlled stimulation mode 54
3.4.3 Pseudo resistors and stimulation artifact suppression 56
3.5 Electrical characterization 58
3.6 Result of biological experiment 64
3.7 Conclusion 69
Acknowledgements 69
References 70

Chapter 4 73

Sensor System including Silicon Nanowire Ion Sensitive FET Arrays and CMOS readout 73

4.1 Abstract 74
4.2 Introduction 75
4.3 System description 78
4.4 Silicon nanowire FET sensor array 79
4.5 CMOS readout chip 81
4.6 FPGA and software 83
4.7 Results and discussions 84
4.7.1 Electrical characterization 84
4.7.2 Experimental results 85
4.8 Conclusions 93
Acknowledgments 93
References 94

Chapter 5 ..... **97**

Adding the 'heart' to hanging drop networks for microphysiological multi-tissue experiments **97**

5.1 Abstract 98

5.2 Introduction 99

5.3 Results and discussion 101

5.3.1 Device architecture 101

5.3.2 Pump concept 101

5.3.3 Integration of flow buffers 102

5.3.4 Pump characteristics 103

5.3.5 Influence of the applied pressure 103

5.3.6 Influence of drop volume 104

5.3.7 Pulsatile flow profiles 104

5.3.8 Parallelization 105

5.3.9 Large networks 107

5.3.10 Real-time closed-loop pump regulation using a beating cardiac microtissue 108

5.4 Methods 112

5.4.1 Microfluidic device fabrication 112

5.4.2 Experimental setup 112

5.4.3 Particle tracing 113

5.4.4 Closed-loop actuation 113

5.4.5 Spheroid cultures 113

5.5 Conclusion 114

Acknowledgements 115

References 116

Chapter 6 ..... **119**

Conclusions and Outlook **119**

6.1 Conclusion: Multifunctional high-density MEA 119

6.2 Conclusion: Experimental setup design 120

6.3 Outlook 121

Appendix A **123**

Hanging-drop pump setup design **123**

A. Hardware design 123

B. Software design 124

C. Closed loop operation 125

References....125

Appendix B.... 127

Publications 127

Journal Papers....127

Conference Contributions....128

Appendix C.... 131

Personal conclusions 131

Appendix D.... 135

Acknowledgments.... 135

ABSTRACT

This thesis includes mixed-signal circuit design, system integration, and experimental results of a multi-functional complementary-metal-oxide-semiconductor (CMOS) high-density (HD) microelectrode array (MEA) chip for in-vitro studies of electrogenic cells. Additionally, concepts and designs of experimental setups for nanowire-based sensor arrays and microfluidic hanging-drop networks will be presented.

To study the functional characteristics and the coordinated activity of electrogenic cells (e.g., neurons, heart cells, retinal cells), systems are required that can interact with these cells while the cells form part of cellular networks. MEA systems that feature arrays of densely packed microelectrodes constitute an ideal platform for bidirectional interaction with cultured cells or tissue-slices and are extensely used in neuroscience research and pharmaceutical applications. By integrating electronic circuits with the microelectrodes on the same CMOS substrate, larger number of electrodes and readout channels can be realized. This dissertation reports on a novel multifunctional CMOS platform featuring 2048 action-potential (AP, bandwidth: 300 Hz to 6 kHz) recording units, 32 local-field-potential (LFP, bandwidth: 1 Hz to 300 Hz) recording units, 32 current recording units, 32 impedance measurement units, 28 neurotransmitter detection units and 16 current/voltage stimulation units.

While the generation of electrical stimuli is commonly done by voltage-controlled or current-controlled stimulation circuits, a new circuit was designed for this multifunctional HD-MEA chip, which can simultaneously control the current and voltage levels of stimulation pulses. This circuit was implemented around a positive current conveyor of type II (CCII+) with added feedback path for controlling the output voltage levels. By employing these on-chip stimulation units, the evoked responses of neurons to stimulation pulses with different current and voltage values were examined. Since the amplitudes of stimulation pulses were orders of magnitude higher than the amplitude of neural signals, the stimulation pulses caused artifacts in the recording channel circuits that were connected to electrodes neighboring the stimulation site. To suppress stimulation artifacts, a novel technique was implemented, which decreases the blank-out time of the recording channels by increasing their high-pass cut-off frequency during the stimulation period. This technique allows the recording channels to return to the operating state within 200 µs after the stimulation pulse.

To detect neurotransmitter compounds, released by the neuronal cells, a fast-scan cyclic voltammetry technique was employed. To minimize disturbances to other measurement modalities on the chip, the scan voltage for voltammetry was directly applied to the working electrodes. Two structures of transimpedance amplifiers (TIA) were designed, which were suitable for different measurement scenarios. The TIAs enabled the scanning of the electrode voltage within a 2 V window and enabled to

record currents in the nA to µA range. The recorded data were digitized on the chip by using 10-bit SAR ADCs. Different concentrations of dopamine in phosphate-buffered saline (PBS) were detected using these on-chip neurotransmitter detection circuits.

To operate the multifunctional HD-MEA chip, a setup was developed including a custom-designed printed-circuit board (PCB) to provide reference voltages and a data-acquisition (DAQ) card for transmitting and saving the output data on a PC. An FPGA was incorporated on the PCB to buffer configuration commands and for providing the proper timing between transmitted and received data streams. In a collaborative project, a complete setup was developed for operating a CMOS nanowire sensor array chip. The nanowire sensor chip comprised 8 current-to-frequency converters and 8 sigma-delta modulators to detect the small current changes (100 pA range) in nanowires. Data communication and power delivery to the setup was implemented by using a universal serial bus (USB) interface. To decrease the data load, signal processing and filtering was performed in the hardware by means of an FPGA. In a separate project, a feedback-controlled setup was developed for hanging-drop networks (HDN). Microfluidic HDNs enable culturing and analysis of 3D microtissue spheroids, derived from different cell types, in an environment providing inter-tissue communication. In this project, a real-time feedback control loop was developed to detect the beating motion of cardiac microtissue cultured in a hanging-drop loop and to drive and synchronize pulsatile pump actuation with the beating of the cardiac microtissue. The developed experimental setups provided robust and user-friendly solutions to perform characterization and experimental measurements with the CMOS or microfludic chips.

ZUSAMMENFASSUNG

Diese Arbeit beschreibt die Entwicklung von Mixed-Signal Schaltkreisen, Systemintegration eines multifunktionellen dicht gepackten CMOS (Complementary Metal Oxide Semiconductor) Mikroelektrodenarrays (HD-MEA) und experimentelle Messungen zur Untersuchung von elektrogenen Zellen. Ausserdem werden verschiedene Konzepte und Methoden zur Verwendung von Nanodraht-Sensor-Arrays und mikrofluidischen Netzwerken aus hängenden Tropfen beschrieben.

Zum besseren Verständnis der koordinierten Aktivität von elektrogenen Zellen (z.B. Neuronen, Herzzellen, Netzhautzellen) benötigt man Systeme, welche die Untersuchung dieser Zellen in ihrem natürlichen Zellverband ermöglichen. MEAs mit dicht-gepackten Mikroelektroden sind ideal zur bidirektionalen Messung und Manipulation von Zellkulturen und Gewebeschnitten und sind weit verbreitet in der neurowissenschaftlichen und pharmazeutischen Forschung. Die Integration von elektrischen Schaltkreisen und Mikroelektroden in einer CMOS-Einheit erlaubt die Anordnung vieler Mikroelektroden und Messkanäle auf einem CMOS-Chip. Die vorliegende Arbeit beschreibt die Entwicklung eines multi-funktionellen CMOS-Chips mit 2048 Kanälen zur Messung von Aktionspotentialen (AP, Bandbreite: 300 Hz - 6 kHz), sowie jeweils 32 Kanälen zur Messung lokaler Feldpotentiale (LFP, Bandbreite: 1 Hz - 300 Hz), elektrischer Ströme oder Impedanz. Weiterhin wurden 28 Kanäle zur Detektion von Neurotransmittern und 16 Kanäle zur Strom-Spannungs-Stimulation in das System integriert.

Zur elektrischen Stimulation von Zellen werden üblicherweise spannungs- oder strom-gesteuerte Schaltkreise verwendet. In dieser Arbeit beschreibe ich die Entwicklung und Integration eines neuartigen Schaltkreises, welcher es erlaubt, elektrische Stimulationspulse mit definierter Spannung und Stromstärke zu erzeugen. Hierbei wird die gewünschte Spannung mit Hilfe einer positiven Stromlieferschaltung des Typs II (CCII+) mit zusätzlicher positiver Rückkopplung erzeugt. Unter Nutzung des beschriebenen Schaltkreises wurde die Reaktion von Neuronen auf Stimulationspulse unterschiedlicher Spannung und Stromstärke untersucht. Die benötigten Stimulationspulse waren um Größenordnungen stärker als die gemessenen neuronalen Signale und erzeugten daher Artefakte in den Messkanälen der umgebenden Elektroden. Durch eine Senkung der Totzeit der Messkanäle über eine Erhöhung der Hochpassgrenzfrequenz konnten die genannten Artefakte unterdrückt werden. Durch diese Maßnahme wurden die Messkanäle schon 200 µs nach dem Stimulationspuls wieder nutzbar.

Die Entwicklung einer schnellen Methode zur zyklischen Voltametrie an Mikroelektroden erlaubte die Detektion von abgesonderten Neurotransmittern neuronaler Zellen. Um mögliche Interferenzen mit anderen Messkanälen des Chips

auszuschließen, wurde die für die zyklische Voltametrie benötigte Spannung direkt an die Arbeitselektroden geliefert. Die Entwicklung zweier Transimpedanzverstärker (TIAs) erlaubt eine Vielfalt an möglichen Messmodalitäten. Mit Hilfe der TIAs konnten Stromstärken im nA - µA Bereich über einen Spannungsbereich von 2V gemessen werden. Die Digitalisierung der gemessenen Ströme erfolgte durch im Chip integrierte 10bit analog-digital Wandler mit sukzessiver Annäherung. Die integrierten Messkanäle für zyklische Voltametrie ermöglichten die Bestimmung verschiedener Dopaminkonzentrationen in phosphatgepufferter Salzlösung (PBS).

Der Betrieb des multifunktionellen HD-MEA Chips wurde durch die Entwicklung einer Leiterplatte zur Bereitstellung der entsprechenden Referenzspannungen und einer Datenerfassungskarte zur Kommunikation des Chips mit dem PC ermöglicht. Die Integration eines FPGA auf der Leiterplatte erlaubt die Pufferung der Konfigurationsbefehle und Synchronisation der ein- und ausgehenden Datenströme. Eine vollständige Messanordnung zum Betrieb eines CMOS Mikrochips mit mehreren Nanodraht-Sensoren wurde in einer engen Kollaboration entwickelt. Der Mikrochip erlaubt die Detektion von kleinen Änderungen der Stromstärke (100 pA) in Nanodrähten, und besteht aus 8 Strom-Frequenz-Wandlern und 8 Sigma-Delta-Modulatoren. Die Stromversorgung und Datenkommunikation der Messanordnung wurde über eine USB-Schnittstelle realisiert. Durch die Nutzung eines FPGAs zur hardware-basierten Filterung und Verarbeitung der Signale konnte die kommunizierte Datenmenge begrenzt werden. In einem separaten Projekt wurde eine Feedback-kontrollierte Methode zum Betrieb von mikrofluidischen Netzwerken aus hängenden Tropfen (HDN) entwickelt. Mikrofluidische HDNs erlauben die gleichzeitige Kultivierung mehrerer dreidimensionaler Mikrogewebe unterschiedlicher Zelltypen, unter Austausch von Stoffen zwischen den einzelnen Geweben. Durch die Implementierung einer Feedbackschleife konnte die Frequenz des pulsierenden Pumpsystems des mikrofluidischen Systems mit der Schlagfrequenz eines Mikrogewebes aus Herzzellen synchronisiert werden.

CHAPTER 1

Introduction

The thesis consists of two parts. The first and main part of the thesis reports on the design of a novel multifunctional MEA chip for interfacing to electrogenic cells. The second part of the thesis reports on the design of experimental setups for interfacing to CMOS and microfluidic chips.

1.1 Multifunctional MEA

The human brain is the most sophisticated information-processing system. This processing capability is due to the large number of neuronal cells in the brain and their complex interconnections. In order to better understand neuronal communication and functions, systems to extract information from neurons in a network are required.

MEA systems provide the means for extracellular interfacing with the neurons. These systems, in contrast to patch-clamp systems, are non-invasive and allow for long-term monitoring and measurements of the cells within an intact neuronal network [1]. Moreover, MEA systems provide a unique platform for pharmacological research on brain diseases and for drug discovery [2-5].

Passive microelectrodes have been realized as metal electrodes on glass substrates with the electrodes connected to external low-noise amplifiers for electrophysiological recordings [6, 7]. However, these systems are limited in the number of recording electrodes and in their spatial resolution. Moreover, the amplitude of extracellular signals is on the order of tens of microvolts, and the use of rather long leads and wires between MEA electrodes and the recording electronics exposes the signals to electromagnetic interference (EMI) and degrades signal-to-noise ratio (SNR). By utilizing CMOS technology, sensitive readout and measurement electronics can be integrated on the same substrate as MEA electrodes, minimizing the EMI noise and increasing the number of parallel measurement channels by multiplexing and digitizing the readout data on the chip [8-12].

Achieving high spatial resolution requires low pitches between the MEA electrodes, however achieving low-noise CMOS readout requires larger chip area for the circuits to decrease flicker noise, which is the dominant noise in the low-frequency range of electrophysiological signals. The introduction of the switch-matrix concept enabled the connection of the MEA electrodes to arbitrary CMOS recording units located

outside of the array through configuring switches implemented underneath each electrode. This technique provides more flexibility and helps to overcome the area constraints for the readout circuits [13, 14].

The use of a switch-matrix not only relaxes the area requirements for the electrophysiological recording units, but also enables the connection of electrodes to various types of integrated sensing or stimulating units. By integrating other sensing units, such as impedance measurement, neurotransmitter detection, current readout, etc. on the MEA chip, multiple information can be gained from the very same cell culture or tissue on top of the electrodes. Therefore, developing a multifunctional CMOS MEA system has been in the focus of this PhD project.

1.2 Experimental setup design considerations

Increased data rates of mixed-signal CMOS circuits have posed new challenges for experimental setup design in terms of signal integrity and power delivery to the chip. A careless design of the experimental setup can degrade the performance of the chips used for measurements.

The setup design was divided into hardware and software parts. The hardware part provided supply of power and reference voltages to the chips and provided the communication platform for interfacing the chips to a host PC for online visualization and data storage. For designing setups for mixed-signal ICs, special care has to be taken to limit the induced noise (EMI, cross talk, ground bounce) on the sensitive analog signals. The hardware was planned according to specifications, such as data-rate, noise-budget, supply voltages, portability etc. For designing the software, robustness, flexibility and user-friendliness needed to be considered. If the timing constraint was not very tight, the processing tasks could be shifted to the software. However for precise timing tasks (e.g. accurate stimulation pulse periods), the use of an FPGA and hardware processing was required.

1.3 Scope and structure of the thesis

The main focus of this thesis is on the development and application of a multifunctional switch-matrix-based HD-MEA chip presented in the first chapters. This HD-MEA chip has been designed and fabricated in industrial 0.18 µm CMOS-technology. To facilitate neuron stimulation and neurotransmitter detection, high voltage (3.3 V) MOS transistors were employed. The CMOS system was suitable for extracting electrical and electrochemical information from a variety of biological preparations, such as cell-cultures, brain-slices and retinae. While each functional unit on the chip was designed to work as a stand-alone unit, the interaction and integration of these units into an overall system also needed to be considered.

In the following chapters, the design details of two collaborative projects concerning a nanowire sensor array and a feedback-controlled hanging-drop network are provided. Experimental setups have been designed for these projects, which provided a robust interface to facilitate device characterization and biological measurements.

This thesis includes four publications:

1. A Multi-Functional Microelectrode Array Featuring 59760 Electrodes, 2048 Electrophysiology Channels, Stimulation, Impedance Measurement and Neurotransmitter Detection Channels

Jelena Dragas*, Vijay Viswam*, Amir Shadmani*, Yihui Chen*, Raziyeh Bounik, Alexander Stettler, Milos Radivojevic, Sydney Geissler, Marie Obien, Jan Müller, Andreas Hierlemann

*Authors contributed equally.

IEEE Journal of Solid-State Circuits, Accepted

2. Integrated current/voltage controlled stimulation buffer and artifact suppression circuits for microelectrode arrays

Amir Shadmani, Vijay Viswam, Yihui Chen, Raziyeh Bounik, Jelena Dragas, Milos Radivojevic, Sydney Geissler, Sergey Sitnikov, Jan Mueller, Andreas Hierlemann

IEEE Transactions on Biomedical Circuits and Systems, Submitted

3. Sensor System including Silicon Nanowire Ion Sensitive FET Arrays and CMOS readout

Paolo Livi, Amir Shadmani, Mathias Wipf, Ralph L. Stoop, Jörg Rothe, Yihui Chen, Michel Calame, Christian Schönenberger and Andreas Hierlemann

Sensors and Actuators B: Chemical, 2014

4. Adding the 'heart' to hanging drop networks for microphysiological multi-tissue experiments

Saeed Rismani Yazdi, Amir Shadmani, Sebastian C. Bürgel, Patrick M. Misun, Andreas Hierlemann and Olivier Frey

Lab on a Chip, 2015

The first paper describes the design of NeuroCMOS chip with explanation on different implemented units such as electrophysiological recording channels, impedance measurements units, neurotransmitter detection units, stimulation units, analog-to-digital converters (ADCs), digital control unit and the system integration. The results of electrical characterization and biological measurements are reported showing the performance of these units and the integrated chip. The second paper describes in detail, the design and characterization of stimulation units and their performance in combination with the artifact suppression circuit. The result of biological results in excitability of neurons to different current/voltage levels of biphasic stimulation pulse is reported. Moreover the stimulation artifact phenomena and the effect of implemented artifact suppression technique are examined. The third paper describes the design and measurement results of the nanowire sensor array, for which a portable setup has been developed and the last paper provides the information on the feedback controlled hanging-drop networks and the application of this system.

1.4 Author contributions

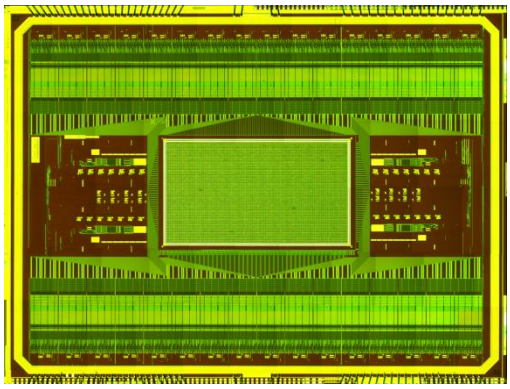
- Amir Shadmani: Design, characterization and measurement of current/voltage controlled stimulation units, artifact suppression unit and neurotransmitter detection unit integrated in the multifunctional CMOS MEA (NeuroCMOS) chip. Conception and design of experimental setups for NeuroCMOS and nanowire sensor array chips, as well as of a setup for microfluidic hanging-drop network chips. Data analysis, C++/VHDL programming and writing of scientific manuscripts.
- Yihui Chen: Design of recording channels, artifact suppression features and full chip integration of NeuroCMOS. Supervising of NeuroCMOS project. Guidance and correction of manuscripts.
- Vijay Visvam: Design and verification of impedance measurement units and electrophysiology recording channels of the NeuroCMOS chip. Writing scientific manuscripts and support with measurements.
- Jan Müller: Support and guidance for digital signal processing (VHDL, Matlab, C++, etc.) and troubleshooting throughout the NeuroCMOS project.
- Jelena Dragas: Design and characterization of the switch-matrix array and the digital core and control unit of the NeuroCMOS chip.
- Paolo Livi: Design, characterization and measurement of the nanowire sensor array, as well as support and guidance for the stimulation-unit design for the NeuroCMOS project. Writing of scientific manuscripts.
- Milos Radivojevic: Support and advice with cell culture handling and stimulation experiments
- Olivier Frey: Support, guidance and advice for neurotransmitter-detection measurements. Support, guidance and advice for the hanging-drop network project.

- Raziyeh Bounik: Support for electrophysiology recording-channel measurement
- Marie Obien: Help with acute slice measurements
- Saeed Rismani Yazdi: Design, fabrication and measurements of microfluidic hanging-drop networks and writing of scientific manuscripts.
- Sebastian C. Bürgel, Patrick M. Misun: technical support with hanging-drop network fabrication.
- Mathias Wipf, Ralph L. Stoop, Michael Calame, Christian Schönenberger: support for nanowire fabrication.
- Jörg Rothe: Sigma-delta design in nanowire sensor array project.
- Sydney Geissler: Support with cell culturing. Advice with biological measurements.
- Sergey Sitnikov: Support with cell imaging.
- Alexander Stettler: CMOS post-processing and platinum microelectrode fabrication.

1.5 Major results

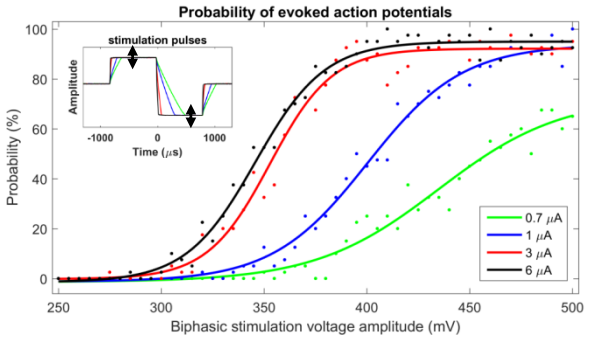
CMOS-based multifunctional HD-MEA

A fully integrated CMOS-based multifunctional HD-MEA was successfully designed and fabricated. The system included: 2048 action-potential (bandwidth: 300 Hz to 6 kHz) recording units, 32 local-field-potential (bandwidth: 1 Hz to 300 Hz) recording units, 32 current-recording units, 32 impedance-measurement units, 28 neurotransmitter-detection units and 16 stimulation units.



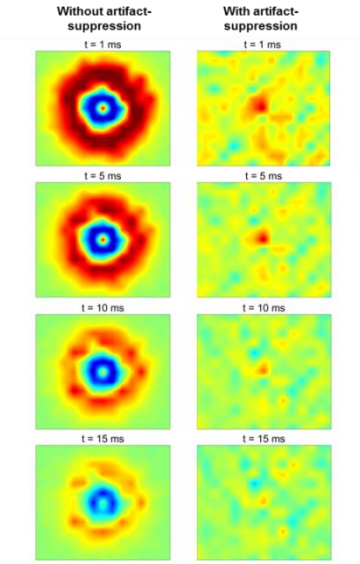
Current/voltage controlled stimulation

The stimulation units of NeuroCMOS chip were designed to operate in two operation regimes including voltage-only or current/voltage-control. For current/voltage-controlled mode, a positive current conveyor of type II (CCII+) was deployed and a feedback path was added for limiting the output voltage. Initial experiments were performed to investigate the excitability of neurons by using biphasic pulses with different current and voltage values.



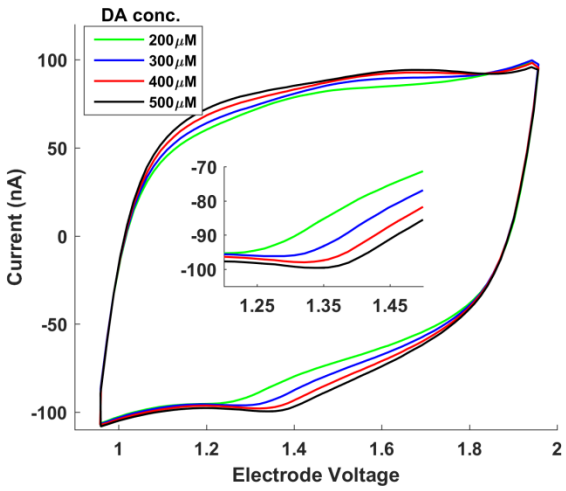
Artifact suppression

The artifact suppression circuit was integrated in the electrophysiological recording units. This circuit increases the high-pass frequency of the first- and second-stage amplifiers during the stimulation phase. The duration of stimulation artifacts upon useage of the artifact removal technique was decreased to 200 µs in comparison to 13 ms without applying artifact suppression.



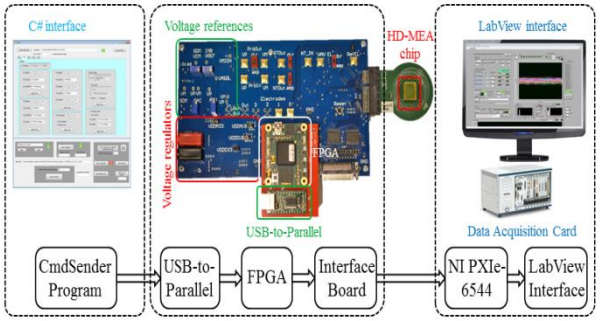
Neurotransmitter detection

Circuitry units to realize voltammetry measurements were implemented with the aim to electrochemically detect neurotransmitter compounds. Transimpedance amplifiers (TIA) were designed to directly apply the scanning voltage to the working electrodes and to then convert the resulting currents to digital output data. Different concentrations of dopamine in phosphate-buffered saline were measured using on-chip circuits.



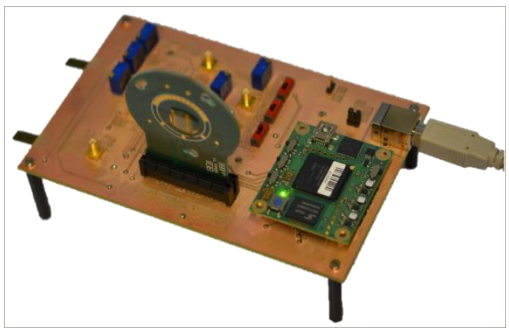
Low-noise setup for the NeuroCMOS project

In order to achieve the best performance and capitalize on the low-noise features of the NeuroCMOS chip, a new setup was developed with due consideration of signal integrity and EMC issues. An FPGA was included in the setup to (i) buffer the commands for precise timing of the commands (e.g., during stimulation pulse generation and artifact suppression) and to (ii) introduce a phase shift between transmitter and receiver clocks in order to decrease fluctuations on digital supply rail.



Portable setup for nanowire sensor array

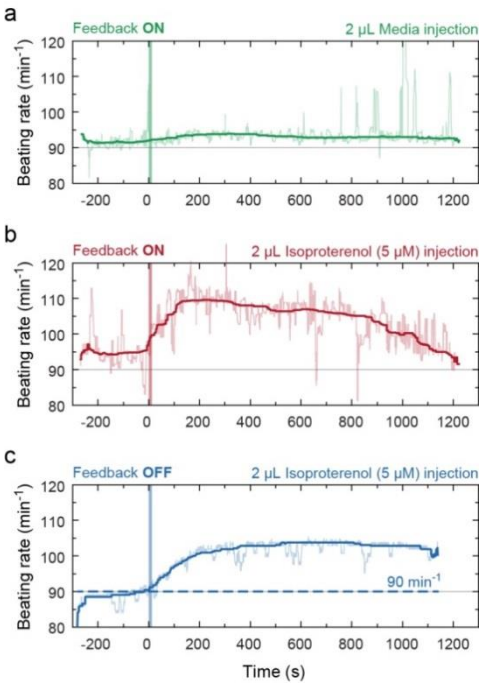
A nanowire sensor array had been developed to detect pA levels of current changes upon absorption of target molecules on the surface of the nanowires. A setup including FPGA and USB controller was developed to perform the initial signal processing and to transmit the recorded data to a PC. Software was



developed for real-time visualization and data storage using the C# programming language.

Closed-loop feedback for "adding the heart" to hanging-drop networks

In order to study the effects of cardiac beating on the function of other body tissues that have been cultured in hanging-drop networks, a real-time feedback control loop was developed. This feedback mechanism was designed to detect the motion of cardiac microtissues, cultured in a hanging-drop loop, and tosynchronize pulsatile pump actuation with the beating of the cardiac microtissue.



References

[1] M. E. J. Obien, K. Deligkaris, T. Bullmann, D. J. Bakkum, and U. Frey, “Revealing neuronal function through microelectrode array recordings,” *Front. Neurosci*., vol. 9, no. JAN, p. 423, 2015.

[2] A. Stett et al., “Biological application of microelectrode arrays in drug discovery and basic research,” *Anal. Bioanal. Chem*., vol. 377, no. 3, pp. 486–95, Oct. 2003.

[3] S. I. Morefield, E. W. Keefer,K. D. Chapman, and G. W. Gross, “Drug evaluations using neuronal networks cultured on microelectrode arrays,” *Biosens. Bioelectron*., vol. 15, pp. 383–396,Oct. 2000.

[4] G. W. Gross, B. K. Rhoades, H. M. E. Azzazy, and M.-C. Wu, “The use of neuronal networks on multielectrode arrays as biosensors,” *Biosens. Bioelectron*., vol. 10, no. 6, pp. 553–567, 1995

[5] Y. Jimbo, H. P. Robinson, and A. Kawana, “Strengthening of synchronized activity by tetanic stimulation in cortical cultures: application of planar electrode arrays,” *IEEE Trans. Biomed. Eng.*, vol. 45, no. 11, pp. 1297–304, Nov. 1998.

[6] B. Ghane-Motlagh, M. Sawan A review of microelectrode array technologies: design and implementation challenges *Second International Conference on Advances in Biomedical Engineering (ICABME),* Tripoli, Lebanon (2013), pp. 38–41

[7] R. Kim et al., "Recent trends in microelectrode array technology for in vitro neural interface platform", *Biomed. Eng. Lett.*, vol. 4, pp. 129-141, 2014.

[8] A. Hierlemann, U. Frey, S. Hafizovic, F. Heer, "Growing cells atop microelectronic chips: Interfacing electrogenic cells in vitro with CMOS-based microelectrode arrays", *Proc. IEEE*, vol. 99, no. 2, pp. 252-284, Feb. 2011.

[9] L. Berdondini, K. Imfeld, A. Maccione, M. Tedesco, S. Neukom, M. Koudelka-Hep, and S. Martinoia, “Active pixel sensor array for high spatio-temporal resolution electrophysiological recordings from single cell to large scale neuronal networks.,” *Lab Chip*, vol. 9, no. 18, pp. 2644–51, 2009.

[10] B. Eversmann, A. Lambacher, T. Gerling, A. Kunze, P. Fromherz, and R. Thewes, “A neural tissue interfacing chip for in-vitro applications with 32k recording / stimulation channels on an active area of 2.6 mm2,” *Eur. Solid-State Circuits Conf*., pp. 211–214, 2011.

[11] G. Bertotti, et al., "A CMOS-based Sensor Array for in-vitro Neural Tissue Interfacing with 4225 Recording Sites and 1024 Stimulation Sites," *IEEE BioCAS,* pp. 304-307, Oct. 2014.

[12] F. Heer, S. Hafizovic, T. Ugniwenko, U. Frey, W. Franks, E. Perriard, J.-C. Perriard, A. Blau, C. Ziegler, A. Hierlemann, "Single-chip microelectronic system to interface with living cells", *Biosens. Bioelectron*., vol. 22, pp. 2546-2553, May 2007.

[13] U. Frey, J. Sedivy, F. Heer, R. Pedron, M. Ballini, J. Mueller, D. Bakkum, S. Hafizovic, F. D. Faraci, F. Greve, K.-U. Kirstein, A. Hierlemann, "Switch-matrix-based high-density microelectrode array in CMOS technology", *IEEE J. Solid-State Circuits*, vol. 45, no. 2, pp. 467-482, Feb. 2010.

[14] M. Ballini, J. Muller, P. Livi, et al., “A 1024-channel CMOS microelectrode array with 26,400 electrodes for recording and stimulation of electrogenic cells in vitro,” *IEEE J. Solid-State Circuits*, vol. 49, no. 11, pp. 2705–2719, 2014.

CHAPTER 2

In-Vitro Multi-Functional Microelectrode Array Featuring 59760 Electrodes, 2048 Electrophysiology Channels, Stimulation, Impedance Measurement and Neurotransmitter Detection Channels

Jelena Dragas*, Vijay Viswam*, **Amir Shadmani***, Yihui Chen*, Raziyeh Bounik, Alexander Stettler, Milos Radivojevic, Sydney Geissler, Marie Obien, Jan Müller, Andreas Hierlemann

*Authors contributed equally.

Accepted to:

IEEE Journal of Solid-State Circuits (JSSC)

Keywords

high-density microelectrode array (HD-MEA), neural interface, multi-functionality, high channel count, low noise, low power, switch matrix, extracellular recording and stimulation, neurotransmitter detection, impedance spectroscopy, pre-charging, pseudo-resistor

2.1 Abstract

Biological cells are characterized by highly complex phenomena and processes that are, to a great extent, interdependent. To gain detailed insights, devices designed to study cellular phenomena need to enable tracking and manipulation of multiple cell parameters in parallel; they have to provide high signal quality and high spatiotemporal resolution. To this end, we have developed a CMOS-based microelectrode array system for in-vitro applications that integrates six measurement and stimulation functions, the largest number to date. Moreover, the system features the largest active electrode array area to date (4.48×2.43 mm^2) to accommodate 59,760 electrodes, while its power consumption, noise characteristics, and spatial resolution (13.5 µm electrode pitch) are comparable to the best state-of-the-art devices. The system includes: 2,048 action-potential (AP, bandwidth: 300 Hz to 10 kHz) recording units, 32 local-field-potential (LFP, bandwidth: 1 Hz to 300 Hz) recording units, 32 current recording units, 32 impedance measurement units, and 28 neurotransmitter detection units, in addition to the 16 dual-mode voltage-only or current/voltage-controlled stimulation units. The electrode array architecture is based on a switch matrix, which allows for connecting any measurement/stimulation unit to any electrode in the array and for performing different measurement/stimulation functions in parallel.

2.2 Introduction

Electrogenic cells, such as neuronal, cardiac, and pancreatic cells are capable of generating and transmitting electrical signals. The flow of ions through the cell membrane generates changes in electrical potentials that can be measured using standard electronic circuitry. Apart from electrical signals, cells also use chemical compounds for signaling, as it is the case in neuronal synapses (contacts between neuronal cells). The chemicals that transmit signals across the synaptic cleft, so-called neurotransmitters, play a significant role in brain diseases, such as schizophrenia, Alzheimer's, and Parkinson's disease [1]; electrochemical methods, such as cyclic voltammetry are commonly used to detect their presence [2]-[4]. To study cell network dynamics, a bidirectional interaction, which also includes electrical stimulation of cells, is desired. To ensure precise and selective stimulation of individual neurons, characterization of the electrodes [5], the cell-electrode attachment, and the cell morphology [6] is first performed, typically by means of impedance spectroscopy [7]. This method can also be used for investigation of anisotropic and inhomogeneous electrical conductivity of tissue and culture medium, label-free detection of cancer [8], [9] and studies of neurodegenerative diseases [10]. Simultaneous examination of electrical and chemical cell signaling noninvasively over extended periods of time, performing targeted electrical stimulation, and tracking cellular movements or changes in the electrical interface via changes of impedance, will provide valuable insights in the behavior of individual cells and cell ensembles and help diagnose and treat different disorders.

Microelectrode arrays (MEAs) are well-established platforms for *in vitro*, extracellular investigation of electrogenic cells that feature multiple electrodes for parallel electrical recording and stimulation [11]-[21]. They are widely used for investigations of cell cultures and cell tissues, such as organotypic brain slices and retinae [14]-[15]. Various MEA systems have been developed to date, featuring different substrates, numbers of electrodes, and electrode pitches [22]. Owing to the small feature size of CMOS technology, recent CMOS-based MEA systems comprise thousands of electrodes at pitches of tens of µm, down to a few µm [17]-[21], which is crucial to obtain detailed insights into cellular and sub-cellular phenomena [23]. The two most common CMOS-MEA architectures base either on the in-pixel front-end amplifiers [18]-[20], or on the switch matrix [17], [21], where the amplifiers reside outside of the electrode array and are connected to the electrodes via a programmable matrix of switches. Both approaches have been detailed and compared in [22].

Contemporary MEAs feature up to three different types of measurement/stimulation functions. The HD-MEAs in [17]-[21] feature only voltage recording and/or voltage/current stimulation; they are limited either in the noise performance [18]-[20], spatial resolution [19], or number of readout channels [17]-[21]. The systems in [2]-[4], [6] feature a combination of neurotransmitter, impedance and/or electrophysiology measurement functions, however, they have a very low number of channels, low spatial resolution and no stimulation capabilities.

In this paper, we present a high-density MEA system that allows performing multiple measurement/stimulation functions in parallel (Fig.1). The implemented switch-matrix approach [17], [21] allows connecting any electrode to any of the measurement/stimulation channels. Moreover, the system significantly advances some of the characteristic MEA features.

The paper is organized as follows. The system architecture, its functional units, and their integration into a single system are described in Section II, III, and IV, the electrical and biological measurements are described in Sections V and VI, while Section VII gives a detailed comparison to the state-of-the-art and concludes the paper.

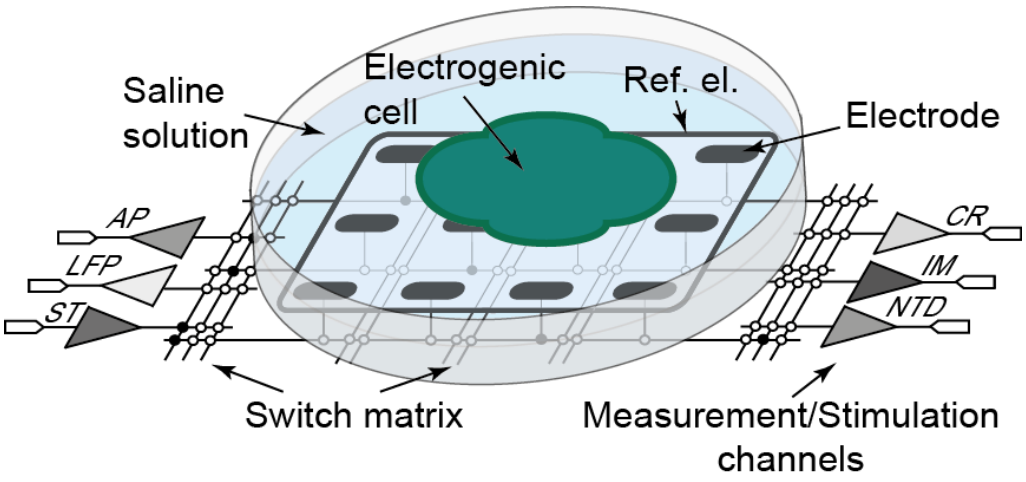


Fig. 1. Illustration of the multi-functional, switch-matrix-based HD-MEA system featuring action potential (AP) readout, local field potential (LFP) readout, current readout (CR), impedance measurement (IM), neurotransmitter detection (NTD) and stimulation (ST) channels.

2.3 System architecture

A list of the different functional units integrated in the system as well as of the respective design requirements is given in Table I. The number of measurement/stimulation channels is limited by the chip area and overall power consumption. The processes of neuronal cells can extend over a few hundred µm (Purkinje neurons) and more, hence, to explore functional connections in fairly distant regions of tissue slices (mm distance), we have opted for at least 4×2 mm^2 of active electrode array area. The data sampling frequency of 20 kS/s has been chosen based on the frequency range of extracellular signals of different cells, including neuronal action potentials (APs), 300 Hz-10 kHz, neuronal local field potentials (LFPs), 1 Hz-300 Hz, and cardiac APs, 1 Hz-1 kHz. To prevent cells heating by more than 2 °C, the targeted power consumption was below 100 mW.

TABLE I

Overview of design requirements for different measurement/stimulation units

Function	Application	Spatial Resolution	Signal Level	Signal Frequency	Noise & Linearity	Number of Channels
AP Readout	Electrical: Action-potential recording	Sub-cellular[a, b]	Neuronal AP: <1 mV	Neuronal AP: 300 Hz-6 kHz	<5 μV_{RMS}[c]	Thousands
LFP Readout	Electrical: Local-field potential recording	Few hundred µm[d]	Neuronal LFP: <5 mV. Cardiac/ pancreatic AP/LFP: tens of mV	Neuronal LFP: ~1 Hz - 300 Hz, Cardiac AP: ~1Hz-1 kHz.	<10 μV_{RMS}	Few tens
Neurotransmitter Detection	Electrochem: Neurotrans-mitter detection	Sub-cellular/ cellular[a]	nA - µA	~10 kHz for FSCV	<0.5 nA_{RMS}	Few tens
Impedance Meas.	Electrical/ Electrochem: Cell-electr. interface characteri-zation	Sub-cellular[a]	10 kΩ-10 GΩ	1 Hz-1 MHz	<1.0 pA_{RMS}	Few tens
Stimulation	Electrical: Evoking activity by applying voltage/ current pulse	Sub-cellular[a]	Voltage:±1.3 V Current: ±30µA in low range, ±300 µA in high range[e]	10 kHz – 100 kHz for single electrode	>9 bit linearity	Few tens
Current Readout	Electrical: Current recording	Sub-cellular[a]	pA	-	<100 fA_{RMS}	Few tens

[a] Cell diameters: few µm to few tens of µm.
[b] To closely approach axons (weak signals, ~20 µV, [25]) and track cellular morphological changes (e.g., axonal initial segment [24]).
[c] Level comparable to neuronal background activity and electrode noise (e.g., ~80 $\mathrm{nV}/\sqrt{\mathrm{Hz}}$ at 1 kHz, for Pt electrodes of 25 µm diameter [28]). To reveal weak axonal signals [25]
[d] LFPs spread up to 2-3 mm, measured *in vivo* in [26], [27]
[e] [50], [51]

The block diagram of the overall system is depicted in Fig. 2. The details on each of the functional units are given in the next section, with the exception of the current recording unit, which will not be covered here due to space limitations.

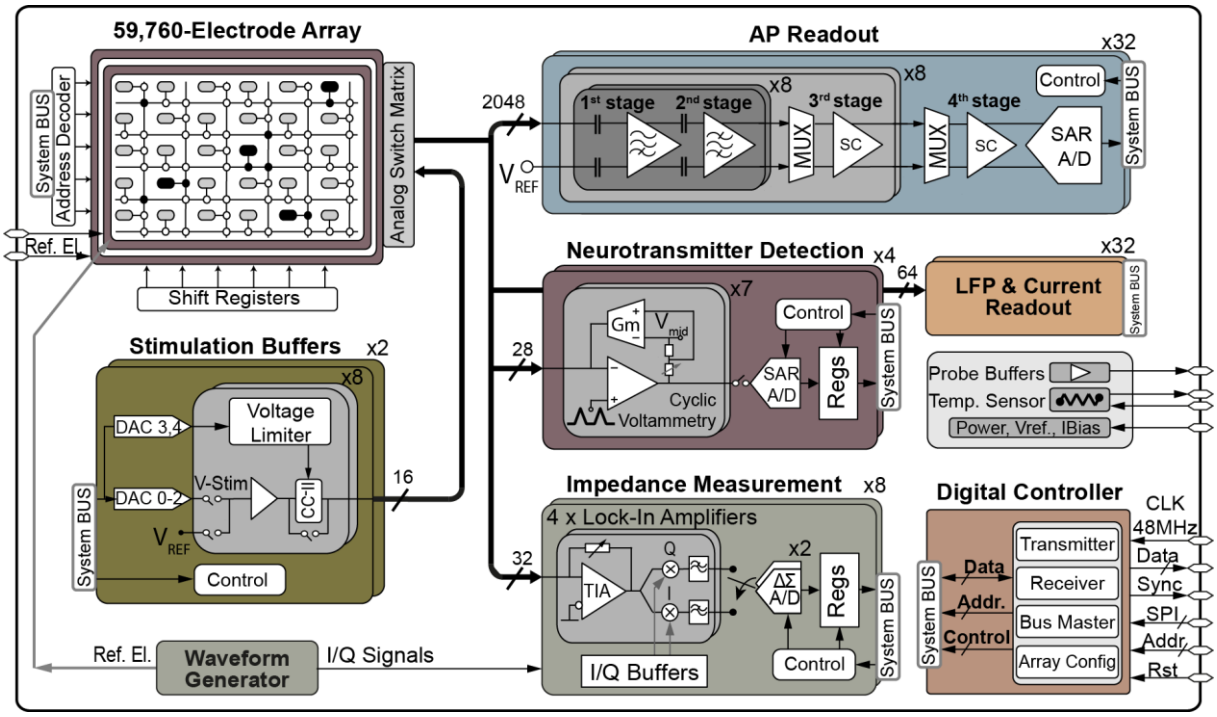


Fig. 2. Block diagram of the multi-functional MEA system.

2.4 Functional units

2.4.1 Electrode array

The electrode array features 59,760 microelectrodes (3.0×7.5 µm^2), distributed over 332×180 pixels, at a 13.5 µm pitch (5,487 electrodes/mm^2), covering an area of 4.48×2.43 mm^2. The electrode array was organized in pixels (Fig. 3), and the underlying electrode routing was switch-matrix-based [17], [21]. Each pixel contained 4 transmission gates with approximately 600 Ω on-resistance, and 3 SRAM cells. The SRAM cells were used to set the state of three switches (SW0-SW2) that connected either the electrode to a signal wire or pairs of wires. The fourth switch (SW3) was used to connect the electrode to a stimulation unit using a low-impedance routing path, which contained only two switches. The largest electrode block providing maximum spatial resolution consisted of 45×45 electrodes or pixels (607.5×607.5 µm^2, 2025 pixels), containing 4 times more pixels and covering twice the area of the largest HD block in [21]. The more advanced features of this switch matrix came from the more advanced CMOS technology used (0.18 µm, instead of 0.35 µm), and from the particular switch arrangement implemented.

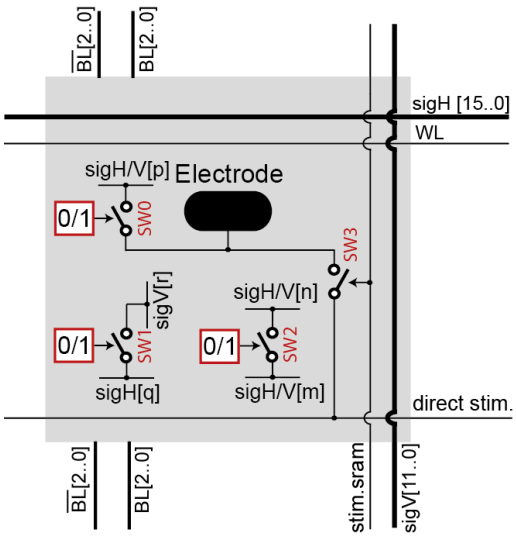


Fig. 3. Block diagram of an array pixel. Switch SW0 connects the electrode to either a horizontal (sigH) or a vertical (sigV) wire. SW1 connects two intersecting wires, SW2 connects a pair of parallel wires. $m, n, p, q \in [0,15]$, $r \in [0,11]$.

The arrangement of the switches in the electrode array achieved a maximal routing flexibility, so that any electrode could be connected to any measurement/stimulation channel. Additionally, different electrodes can be simultaneously routed to different measurement/stimulation units, which enable conducting complex experiments, targeting different cell features in parallel. The array switch pattern also allows for locally connecting multiple electrodes (e.g., 4, 9, or 16 electrodes) to obtain a sensing area of rectangular shape and arbitrary size. These pseudo-large electrodes enable charge integration over a large area, which can be used in recording LFPs and may prove beneficial in neurotransmitter detection.

The physical layout of the array was generated using a custom C++ application that described the switch matrix as a mathematical graph. An electrode routing configuration is generated by means of Integer Linear Programming, where the routing is mapped into a max-flow, min-cost problem, similar to [17], [21]. The reconfiguration of the entire array takes up to 4.5 ms.

2.4.2 Voltage recording channels

2048 AP recording channels were organized in 32 recording blocks (Fig. 4(a)). Each block consisted of 64 AP channels, including 64 continuous-time (CT) first-stage and second-stage amplifiers, 8 switched-capacitor (SC) third-stage amplifiers, one SC fourth-stage amplifier, and one successive-approximation-register (SAR) ADC. Inside each block, bias circuitry and digital control logic were shared among 16/64 channels. Weak extracellular signals were amplified by the four amplification stages, with a programmable gain from 29 to 77 dB (steps of 6 dB), after which they were sampled at 20 kS/s and digitized by the 10-bit 1.28 MS/s SAR ADC. To suppress interferences from power supplies and substrate, the whole signal chain of the AP recording channel featured fully differential structure.

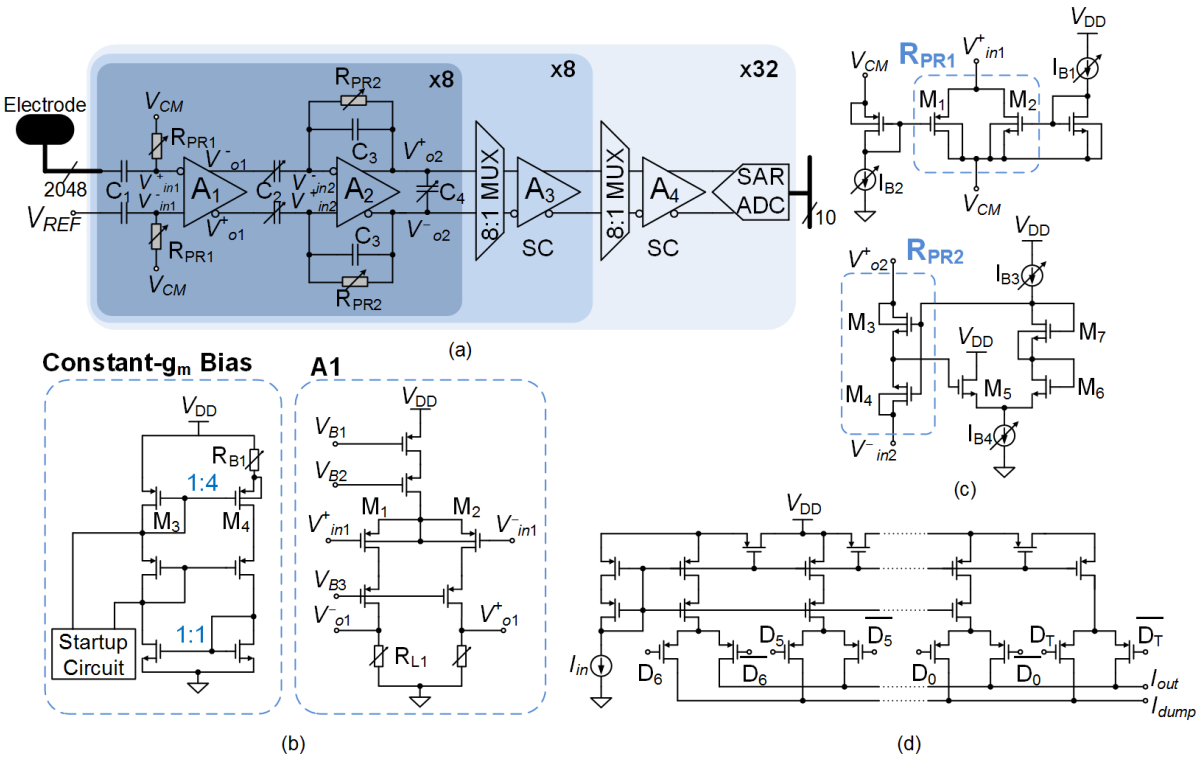


Fig. 4. AP recording channels. (a) Block diagram of the 2048 AP recording channels. (b) Open-loop amplifier (A_1) used in the first stage and its constant-g_m bias circuit. (c) Two types of pseudo resistors and their bias circuits. (d) MOSFET-only R-2R current DAC for tuning high-pass corner frequencies.

As the noise performance of the signal chain was mostly determined by the first stage, about half of the total recording channel power (~8.6 µW) was allocated to this stage to achieve a low noise level. To further reduce the noise, a resistively loaded open-loop amplifier topology has been adopted [28], [29], [30] (Fig. 4(b)). Traditional CMOS neural amplifiers employ closed-loop topologies with high-gain OTAs, [31], which use PMOS input differential pairs and NMOS active loads. NMOS transistors normally have a much higher flicker noise level than PMOS transistors, and thus, add substantial noise in the AP band. In contrast, the resistive loads do not generate flicker noise and contribute less thermal noise, if the voltage drop across the two types of loads is equal. Therefore, the proposed open-loop topology achieved a lower noise level (<3 μV_{RMS}) than traditional topologies with similar power consumption. However, one drawback of this topology is that its gain can vary significantly due to process and temperature (PT) variations. To keep such variations low, a constant-g_m biasing circuit was used to generate the tail current of the amplifier A_1 (Fig. 4(b)). By matching the PMOS transistors (M_1, M_2 and M_4) and the resistors (R_{L1}, R_{L2} and R_{B1}) in both A_1 and the biasing circuit, a uniform gain was achieved.

The second stage (Fig. 4(a)) employed a closed-loop structure, which provided an accurate gain of $A_{V2} = C_2/C_3$ (programmed by tuning C_2) and could handle a large swing. Owing to the relaxed noise requirements, the g_m of its input transistors was reduced to cut power consumption, and thus less load capacitance was required to realize the anti-aliasing filtering. The load capacitor, C_4, can be tuned to set the low-pass corner frequency to around 6 kHz.

In addition to achieving low noise, the two CT stages had to remove low-frequency (<1 Hz) electrode potential drifts and control their offsets to avoid saturating succeeding amplifiers in large-gain configurations. Therefore, a high-pass filter (HPF, C_1 and R_{PR1}) has been implemented in front of the first stage to remove the electrode potential drifts, and the HPF (C_3 and R_{PR2}) in the second stage to null the first-stage offset. Due to the DC feedback through R_{PR2}, the offset of the amplifier A_2 was divided by A_{V2} when referred to the input of the second stage, and further divided by the first-stage gain when referred to the electrode.

Both HPFs had to have very low corner frequencies (f_{HPF} < 10 Hz) to avoid filtering out AP signals and adding excessive noise into the AP band. Nonetheless, C_1 and C_2 had to be around a few pF due to area constraints, and, thus, R_{PR1} and R_{PR2} had to be in the TΩ range. Such large resistances are usually realized with so-called “pseudo-resistors”, i.e., MOS transistors biased in the weak inversion region. However, in conventional structures, the transistor gates are biased with fixed voltages, so that the threshold voltage (V_T) may significantly vary due to PT variations, which yields a large spread in resistances. To guarantee a good uniformity of f_{HPF}, two pseudo-resistor structures have been devised, [29], [30] (Fig. 4(c)). R_{PR1} consisted of a PMOS (M_1) and an NMOS (M_2) transistor connected like a transmission gate. Their gates were biased by a PMOS and an NMOS current mirror, respectively. Hence, the first-order dependence of their overdrive voltages (V_{OV}) on V_T has been cancelled. For R_{PR2} a similar structure as in [32] was adopted, which

consists of two NMOS transistors (M_3 and M_4) connected in series. Here, an NMOS-only level shifter (M_5, M_6, and M_7) was used to bias the gates of M_3 and M_4, instead of a PMOS level shifter in [32], to obtain a better process tracking. V_{OV3} was approximately equal to $(V_{OV6} + V_{OV7} - V_{OV5}) + (V_{T6} + V_{T7} - V_{T5} - V_{T3})$, and had a weak dependence on V_T. Thus, resistance variations of both R_{PR1} and R_{PR2} were substantially smaller than those of the conventional structures. In addition, both R_{PR1} and R_{PR2} were designed symmetrically around their common-mode voltages, which entailed a good linearity. Furthermore, the two resistances were inversely proportional to the bias currents (I_{B1}~I_{B4}), which can be tuned to adjust f_{HPF} for compensating PT variations and to accommodate different experimental scenarios. This adjustability was realized by using two 7-bit current DACs (shared by 16 channels, Fig. 4(d)) for two stages. The DACs relied on the current-division principle of MOS transistors and were realized using a compact MOSFET-only R-2R ladder structure [33].

To reduce area and power consumed by the last two stages and the ADCs, 8-to-1 multiplexing has been realized in front of both stages. To ease the implementation of multiplexing and handle large signal swings, the SC topology has been chosen. With a large gain (~47 dB) provided by the two CT stages, the sampling noise of the two SC amplifiers is negligible when referred to the electrode. However, when the CT output of the second-stage amplifier,V_{O2_i}, was connected to an input capacitor of the SC third stage (e.g., C_{1_i} in Fig. 5(a)), V_{O2_i} got disturbed and could not settle to a 10-bit accuracy within the tracking phase (~47 µs), as the second-stage bandwidth was limited to around 6 kHz. To solve this issue a CT buffer could have been added after each second-stage amplifier, but this would have entailed additional power consumption. In this work, a pre-charging amplifier, $A_{Pre,}$ was connected to V_{O2_i} at the beginning of the tracking phase and drove C_{1_i} (Fig. 5(a)). Thus, V_{O2_i} was disturbed only by a small capacitor (input capacitance of A_{Pre} << C_{1_i}). At the end of pre-charging, the voltage of C_{1_i} was already quite close to V_{O2_i} (Fig. 5(b)), so that V_{O2_i} could settle to the required accuracy in the remaining tracking phase. As A_{Pre} was shared among 8 channels, the power overhead was negligible.

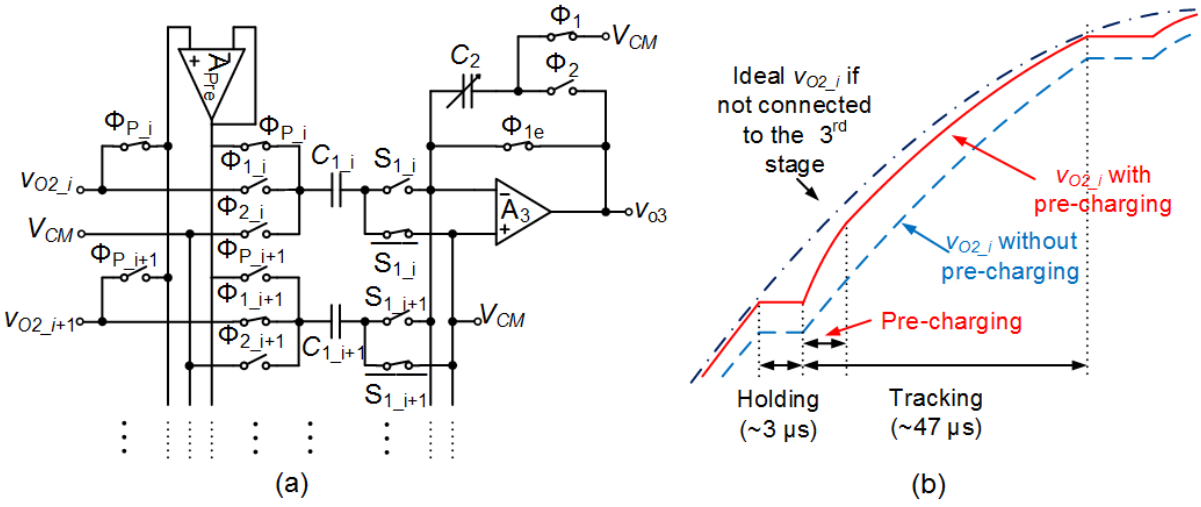


Fig. 5. The pre-charging technique. (a) Schematic of a portion of the SC third-stage amplifier, drawn as single-ended circuit for brevity. (b) A sketch of V_{o2_i} with/without using the pre-charging technique.

The LFP and AP channels shared a similar topology. The 32 LFP channels have been divided into 4 blocks, each containing 8 CT first- and second-stage amplifiers, one SC third-stage amplifier, and one SAR ADC. The two CT stages featured the same topologies as the AP channels, with high-pass corner frequencies lower than 1 Hz. The third stage and the ADC were the same as those in the AP channels with a reduced clock rate.

2.4.3 Neurotransmitter detection channels

The neurotransmitter detection (NTD) circuit is based on the fast-scan cyclic voltammetry (FSCV) technique, [34]-[38]. This technique relies on applying a triangular-shape voltage signal with gradients of more than 100 V/s between a working and a reference electrode, and measuring the resulting current flowing through the working electrode. The trace of the measured current versus the applied voltage, i.e., cyclic voltammogram (CV), provides information about the electrochemical properties of the analytes in the cell medium. In this realization, the reference electrode potential was kept constant while the working electrode potential was swept to minimize disturbances to the other measurement units on the chip.

The NTD channels were grouped into 4 blocks, each block consisting of 7 trans-impedance amplifiers (TIAs), which were designed to copy the CV scan voltage to the electrode and measure the electrode current. To maintain the timing information, the scan voltage and the 7 current signals were multiplexed and then sampled at 20 kS/s by a 10-bit SAR ADC.

Two novel TIA circuit topologies were developed: TIA_A was designed to record from large electrodes, and $TIA_{B,}$ which featured low noise levels, was tailored to record from small electrodes. Both topologies could scan the working electrode potential within a 2 V window and record currents on the order of µA, at a 10-bit resolution (Fig. 6). TIA_A included a local common-mode feedback (LCMFB) OTA, configured in a unity-gain feedback to buffer the input scan voltage, and a resistive TIA. The cascode transistors in the current mirror ensured that the mirrored current closely followed the electrode current (I_{el}), irrespective of the voltage difference between the electrode and V_{MID}. The TIA_A output voltage (V_{out}) was equal to $R_f I_{el}$, where R_f could be configured as 1 MΩ or 3 MΩ. The OTA inside the TIA_A had its dominant pole at the output stage and remained stable even for very large capacitances of the aforementioned pseudo-large electrodes, and rendered slow-rate CV applications possible.

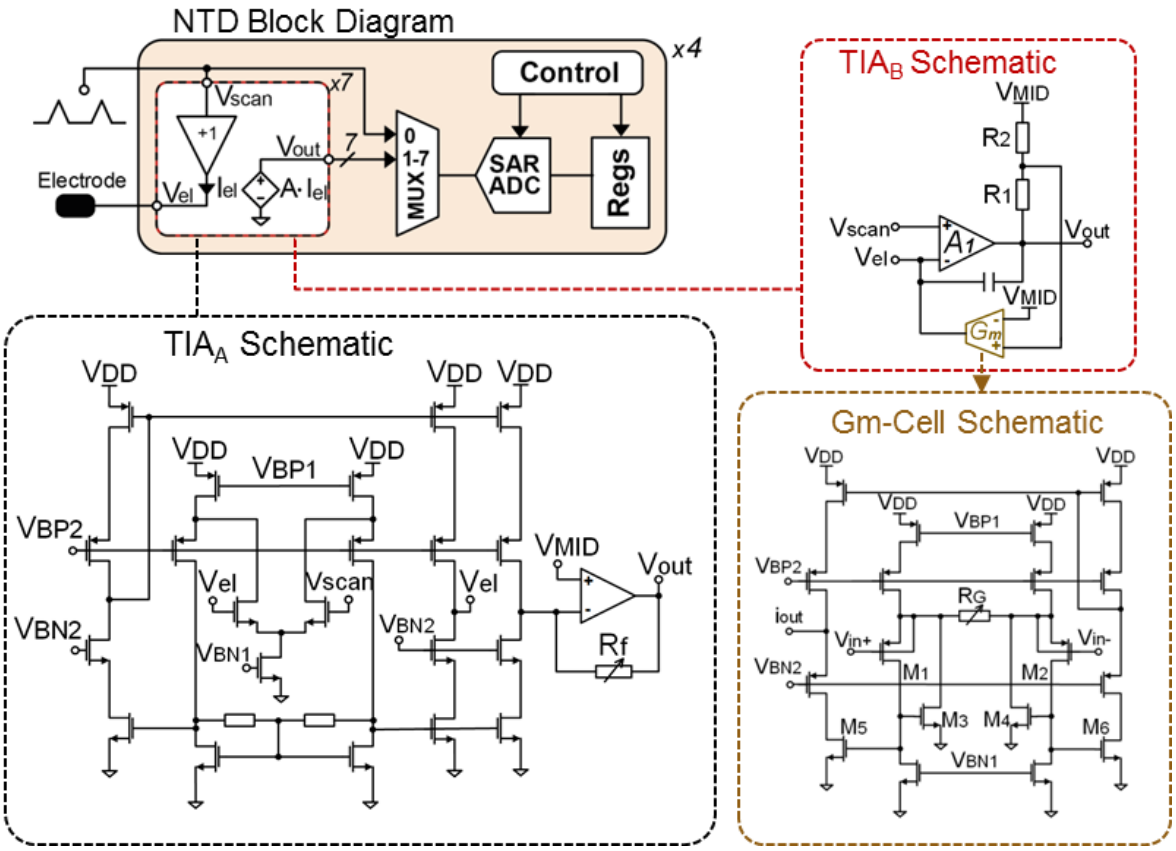


Fig. 6. Block diagram of the neurotransmitter detection channel and schematics of TIA_A, TIA_B, and the Gm cell.

TIA_B was composed of a two-stage Miller-compensated opamp (A_1) and a G_m cell. This topology resembled a resistive TIA, but the feedback resistor was replaced with a linear Gm cell. The Gm cell eliminated the feedforward path passing through the feedback resistor and thus prevented the scan voltage appearing at the output. The linearity of the Gm cells (Fig. 6) was significantly improved by avoiding the body effect of M_1 and M_2, and keeping V_{GS1} and V_{GS2} constant by a local feedback around M_1, M_3 and M_2, M_4, [39]. Therefore, the input differential voltage was almost entirely applied across R_1 and converted to a current. This current flowed through M_3 and M_4 and then got mirrored to the output branch. The cascode transistors at the output branch reduced the output current variation due to the scan voltage. The output current of the Gm cell can be calculated as:

$$i_{out} = 2 \cdot \frac{(W/L)_5}{(W/L)_3} \cdot \frac{1}{R_G} \cdot \frac{R_2}{R_1 + R_2} \cdot v_{out} = \frac{2}{R_G} \cdot \frac{R_2}{R_1 + R_2} \cdot v_{out},$$

where $(W/L)_5$ was equal to $(W/L)_3$ for this design. Since the input impedance of A_1 was very high, i_{out} had to exactly match the electrode current. Consequently, the transimpedance of TIA_B, $R_{TIA,}$ can be calculated as $R_{TIA} = \frac{v_{out}}{i_{in}} = \frac{R_G}{2} \cdot \frac{R_1+R_2}{R_2}$, which can be tuned according to the experimental conditions.

2.4.4 Impedance measurement channels

The main challenge in integrating many impedance measurement (IM) channels with limited silicon area is to achieve simultaneously low noise level, high dynamic range, and low power consumption. The state-of-the-art IM systems typically trade off large area per channel and power for low noise [40]. Most systems feature exclusively IM channels [10], and only few systems combine both electrophysiology and IM [41], [6]. While [41] reports electrode IM only, [6] reports IM of cardiac cells, however, with a poor spatial resolution.

To run IM in real time, a waveform generator with a programmable frequency (1 Hz-1 MHz) and amplitude (set by a 10-bit DAC) was integrated on chip (Fig. 7). The generated sinusoidal voltage (V_{Stim}) was applied between the reference electrode and the selected working electrodes. The resulting current signal from each working electrode ($I_{el} = V_{Stim}/Z_{el}$) was converted into a voltage, and the impedance magnitude and phase were reconstructed with a lock-in amplifier, realized by a low-noise TIA and two quadrature phase mixers, followed by a low-pass filter (LPF). The implemented TIA can be programmed with either resistive feedback for measuring large electrodes (impedance < 100 MΩ) or capacitive feedback for measuring electrodes smaller than 10 µm diameter (impedance > 100 MΩ). For the capacitive feedback TIA, a periodic reset [42] was used to avoid the amplifier going to saturation and keep the electrode potential at the reference voltage, V_{REF}. To mitigate the drift of the electrode-electrolyte interface capacitor C_e, and the amplifier offset, the TIA circuitry included an auto zeroing (AZ) function. The amplifier offset was first stored in C_{AZ} (200 fF) during the reset phase, so that during the measurement phase the voltage stored in C_{AZ} canceled the offset and low-frequency drift from the electrode (Fig. 7). The TIA's output signal was mixed with the in- and quadrature-phase (I/Q) signals, which were generated by the waveform generator and phase-locked with V_{Stim} to produce I/Q demodulated impedance signals. A double-balanced passive mixer architecture was selected, since passive mixers normally show higher linearity and negligible 1/f noise compared to their active counterparts [43]. An active differential LPF was used to filter out higher order harmonics and intermodulation products before the A/D conversion. The LPF can be bypassed or its gain set to 1 or 5 by tuning the feedback resistor. The cut-off frequency of the LPF was set to16 kHz, which was about 1/46 of the sampling frequency of the ΔΣ ADC. The I/Q output of the lock-in amplifier was digitized by the incremental single-bit SC ΔΣ ADC, consisting of an on-chip ΔΣ modulator and an off-chip decimation filter. To handle large input signal swing, while relaxing the slewing requirements of the integrators, an input feed-forward (CIFF) topology was employed [44]. Setting the sampling frequency to 750 kHz relaxed the anti-aliasing requirement, and the oversampling ratio was high enough to guarantee a resolution of 14 bits and a temporal resolution of 10 ms upon using proper filtering [40]. To reduce kT/C noise, a large sampling capacitor (1.6 pF) was used. The flicker noise and the offset contributed by the first-stage integrator opamp were suppressed by the AZ technique [45].

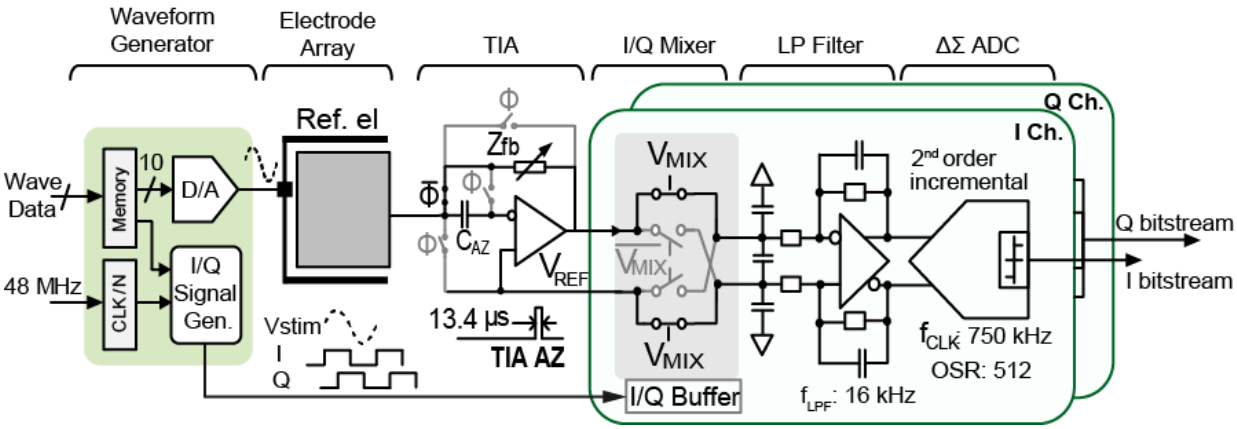


Fig. 7. Schematic of the lock-in-amplifier-based impedance measurement channel.

2.4.5 Stimulation channels

16 stimulation buffers were integrated on the chip, grouped into two blocks of 8. Additionally, each block contained 3 DACs for producing the stimulation waveforms and 2 DACs for setting the limiting voltages, all 5 DACs shared by the 8 stimulation buffers. Each buffer can be configured as voltage-only or current/voltage-controlled buffer. Most reported stimulation buffers provide either voltage- or current-controlled stimulation pulses [46]. Here, the current/voltage-controlled buffers can simultaneously control both the output current and the output voltage. Current-controlled stimulation is desirable for precise charge delivery, independent of the electrode impedance, whereas controlling the stimulation voltage is needed to avoid tissue damage upon exceeding safe stimulation range. Fig. 8 shows the schematic of the stimulation buffer configured in the current/voltage-controlled mode. The core of this circuit is a positive current conveyor of type II (CCII+), an improved version of [47], which converts DAC voltages into electrode currents. The default current range can be increased by closing S1 and S2, which increases the gain of the current mirror in the output stage from 1.0 A/A to about 11.0 A/A. R_{IN} had a value of 50 kΩ, therefore ±1.5 V DAC signal produced ±30 µA in the low-current mode and ±330 µA in the high-current mode. The voltage controllability was realized by OTA_{UP} and $OTA_{DN,}$ which turned off either the positive or the negative output current, when the output voltage reached the predefined values of V_{UP} or V_{DN}, respectively. For the voltage-only controlled stimulation, OTA_{VolBuf} was configured in the unity-gain feedback and directly connected to the output. OTA_{VolBuf} used the LCMFB technique as well, and had the dominant pole located at the output node, suited to drive the mainly capacitive electrodes.

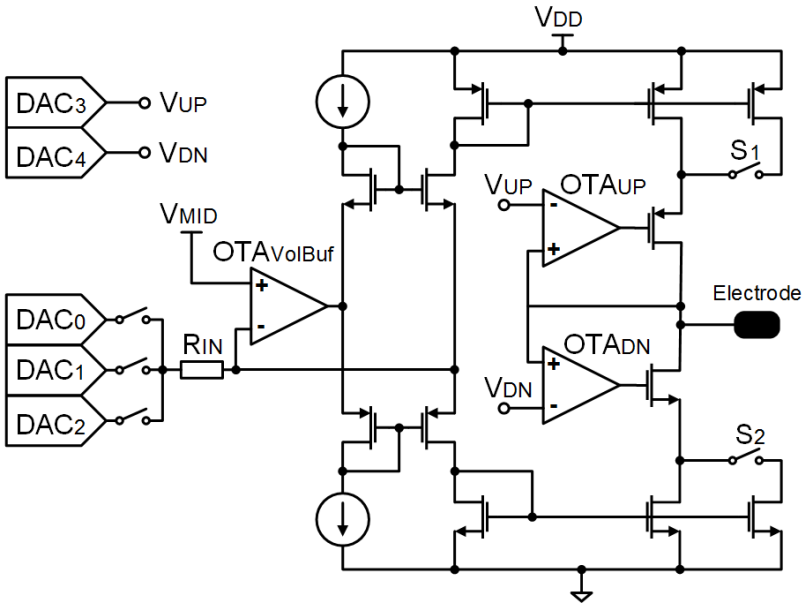


Fig. 8. Schematic of the current/voltage-controlled stimulation buffer.

2.5 System integration

2.5.1 Digital control

A 48 MHz-SPI interface was used to stream the configuration data to the chip, followed by an on-chip cyclic redundancy check (CRC). The measurement/stimulation units and the electrode array were grouped into 26 independently addressable blocks that were interfaced by the on-chip controller using an addressing scheme. As the chip extended over 12.0×8.9 mm^2, to guaranty the settling times of the 10-bit-word data generated by the different measurement blocks, the system Read Data Bus was divided into two busses, both running at 24 MHz. The data from the busses were fetched by the controller in an alternating manner at 48 MHz and sent off chip. The order in which the data generated by the measurement units were allocated on the two data busses was defined by the Bus Master module of the digital controller. To avoid data collision, an OR-tree was inserted between the busses and the measurement units. The chip output data was organized in packages, aligned with the sampling cycles, each featuring 2400 10-bit words, ending with a 20-bit CRC checksum.

2.5.2 Chip layout and fabrication

The chip was fabricated in a 0.18 µm CMOS technology (2P6M) and has a size of 12.0×8.9 mm^2. The platinum electrodes were post-processed at wafer level by means of ion beam deposition and etching. In the same step, two Pt-resistors used as on-chip temperature sensors were fabricated and implemented in 4-terminal sensing configurations. A multilayer SiO_2/Si_3N_4 passivation stack was deposited by plasma-enhanced chemical vapor deposition (PECVD), on the one hand, to protect the CMOS circuits against the saline solution that was used as cell culture medium, and, on the other hand, to prevent the release of toxic aluminum and copper from the CMOS metal layers into the cell culture. Openings in the passivation, defining the electrode areas and wire bonding contacts, were fabricated in a reactive-ion etching (RIE) step. A shifted-electrode layout [48] was employed to ensure a tight seal between the electrodes in the liquid phase and the CMOS metal layers and circuitry in the chip. Fig. 9 shows the micrograph of the chip, bio-compatible chip packaging, and an SEM image of the post-processed array surface with rat cortical neurons.

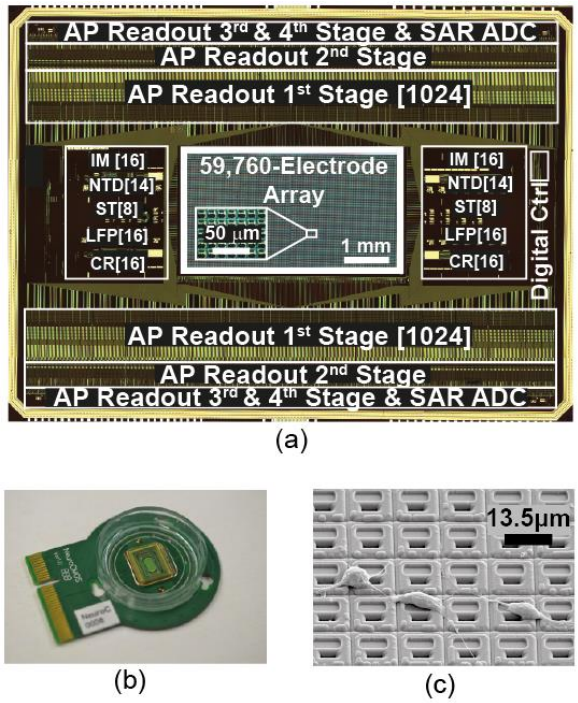


Fig. 9. (a) Chip micrograph; the die size is 12.0×8.9 mm^2. (b) Bio-compatible chip packaging and PCB. (c) SEM image of the chip surface, showing in-house post-processed Pt-electrodes and dissociated primary rat cortical neurons, cultured on top.

2.6 Electrical characterization

2.6.1 Voltage recording channels

The integrated input-referred noise of an AP recoding channel, including the ADC, was 2.4 μV_{RMS} in the AP band (300 Hz−10 kHz), and 5.4 μV_{RMS} over the full band (1 Hz−10 kHz) (Fig. 10(a)). The noise of an LFP recording channel was 3.6 μV_{RMS} in the LFP band (1 Hz−300 Hz), and 4.3 μV_{RMS} over the full band (1 Hz−10 kHz). The measured maximum input-referred offset ranged from -110 µV to 70 µV, which was achieved without dedicated offset compensation/calibration circuitry [29], [30] (Fig. 10(b)). The measured maximum gain of the recording channel was 76.4 dB. An excellent gain uniformity of σ = 0.14 dB (Fig. 10(c)) was measured by applying a common sinusoidal signal to all inputs. The two tuning curves of the first and second stage HPF corner frequency (f_{HPF}) are illustrated in Fig. 10(d). Fig. 10(e) shows the overall transfer function of all 2048 AP recording channels. The histogram inset demonstrates a good uniformity of f_{HPF} (σ = 0.97 Hz), which indicates a good matching of the pseudo resistors across all channels. Moreover, the SAR ADC achieved an SNDR of 56.2 dB at a sampling rate of 1.28 MS/s and with an input signal frequency of 1.1 kHz. Finally, the measured power consumption of each AP channel was around 16 µW.

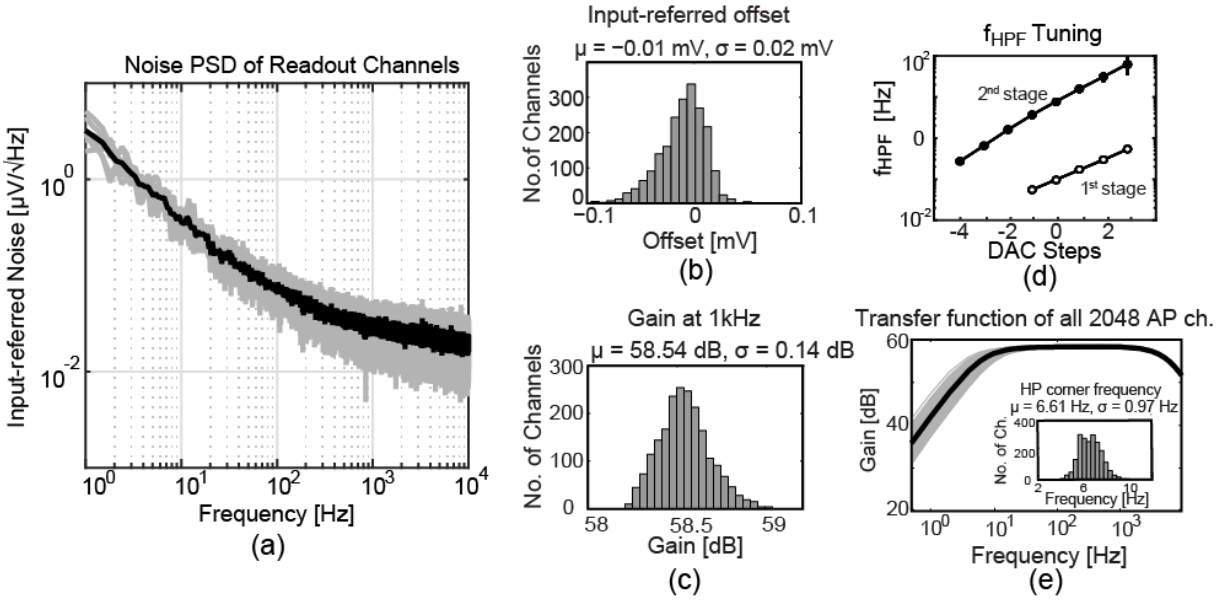


Fig. 10. (a) Noise power spectral density of the AP recording channel including the ADC. (b) Input-referred offset distribution of the 2048 AP channels. (c) Gain distribution of the 2048 AP channels. (d) The first- and second-stage HPF corner frequency (f_{HPF}) tuned by the current DAC. (e) Transfer function of 2048 AP channels and f_{HPF} distribution.

2.6.2 Neurotransmitter detection channels

The area per neurotransmitter channel was 0.04 mm^2 (TIA_A) and 0.047 mm^2 (TIA_B), whereas the power consumption amounted to 222 µW (TIA_A) and 178 µW (TIA_B). The integrated input-referred noise of TIA_A and TIA_B, including the multiplexer and the ADC, was 200 pA_{rms} and 120 pA_{rms}, respectively (Fig. 11(a)). The gain distribution is depicted in Fig. 11(b). In Fig. 11(c), FSCV at 300 V/s using the TIA_B was conducted to measure 4 different concentrations of dopamine (DA) in phosphate-buffered saline (PBS, a saline which has similar electrical properties as physiological solution).

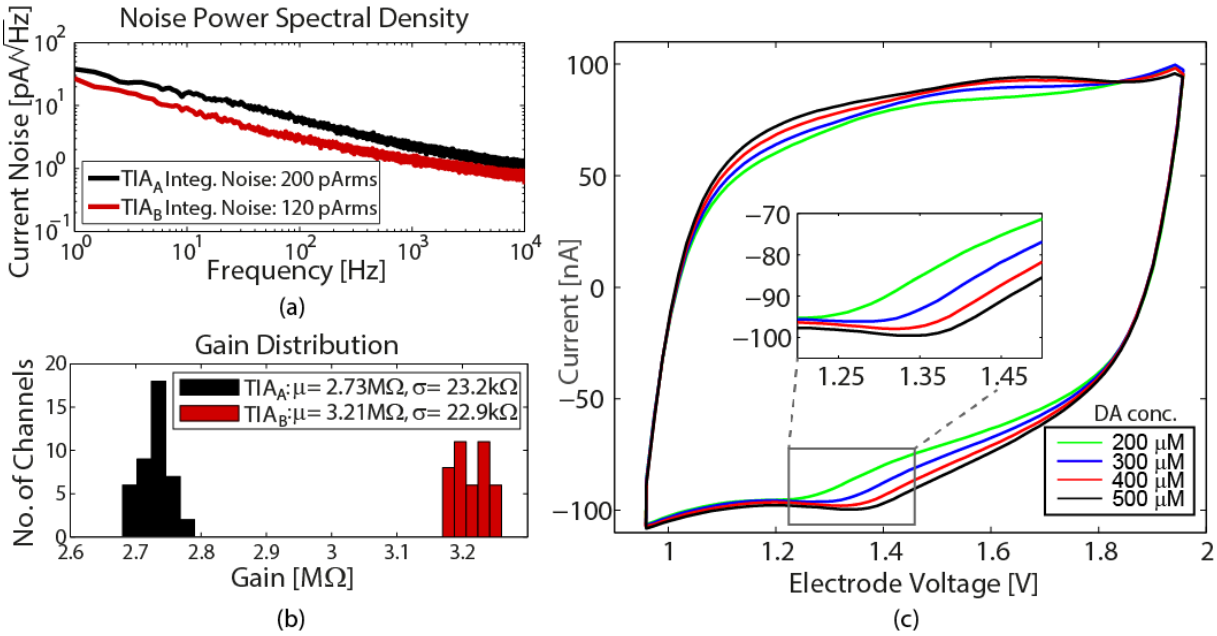


Fig. 11. (a) Noise power spectral density of TIA_A and TIA_B including multiplexer and ADC. b) Gain distribution across 3 chips. (c) Recorded FSCV results for concentrations of 200 µM, 300 µM, 400 µM and 500 µM dopamine in 0.1 M PBS. TIA_B connected to a single electrode has been used. Each of the curves is an average over 98 scans applied within 1 s.

2.6.3 Impedance measurement channels

The integrated input-referred noise (1 Hz-100 kHz) of the TIA with a 10 MΩ feedback resistor was 6.4 pA_{RMS} (Fig. 12(a)). The area and power consumption of the impedance measurement channel amounted to 0.1 mm^2 and 412 μW, respectively, whereas the waveform generator occupies an area of 0.012 mm^2 and consumes 1.1 mW at 3.3 V supply voltage for driving a load of 10 nF at 1 MHz frequency. Pt-electrode-electrolyte interface was characterized in PBS, in a frequency range between 1 Hz and 1 MHz (Fig. 12(b)). The same measurement was repeated after Pt-black electroplating, which was done by driving the current from an external Pt electrode (immersed into the Pt working solution) to the array electrodes, which were shorted to the chip's probe pad (current sink). As expected, in the frequency range from 10 Hz to 10 kHz the impedance showed a 1/f roll-off behavior, which evidenced the dominance of the double-layer capacitance. The predominantly capacitive behavior was also indicated by the phase angle, which ranged between 60-80 degrees. At frequencies over 10 kHz, spreading resistance started dominating the impedance, as the phase angle moved towards zero [48]. The effect of Pt-black deposition was 30-40 fold reduction of impedance magnitude.

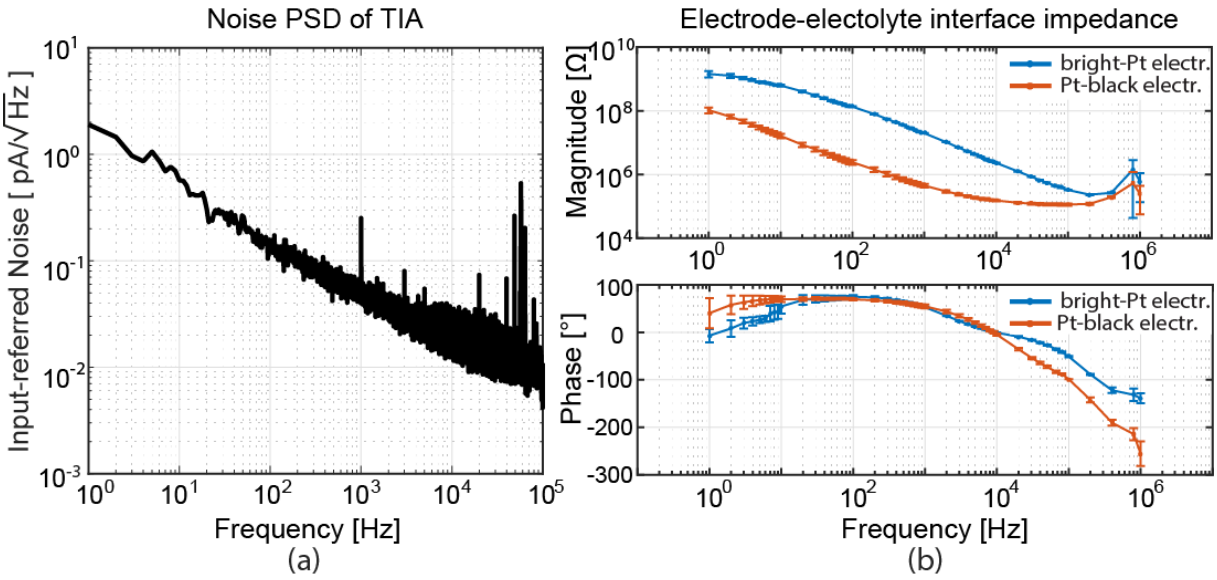


Fig. 12. (a) Noise power spectral density of the TIA with R_f =10 MΩ. (b) Magnitude and phase of the bright Pt and Pt-black electrode impedance in a frequency range between 1 Hz and 1 MHz, averaged over 6 and 8 electrodes for bright Pt and Pt-black, respectively.

2.6.4 Stimulation channels

The area of a stimulation channel was 0.046 mm^2. The power consumption was measured to be 251 µW and 306 µW per current/voltage-controlled stimulation buffer in the low- and high-current mode, respectively. Fig. 13 shows generated current/voltage-controlled pulses using the on-chip stimulation buffers with a load of 610 pF. The output current was switched off when the output voltage reached the predefined limits, in this case 2.65 V for V_{UP} and 1.65 V for V_{DN}. A ±1 V DAC signal resulted in ±20 µA output current for the low-current mode and a ±0.55 V DAC signal resulted in ±100 µA output current for the high-current mode (Fig. 13(a) and (b)). The linearity of the circuit was measured by running a DC sweep, which evidenced a 10-bit linearity for the low-current mode in the range of ±30 µA and a 9-bit linearity for the high-current mode in the range of ±300 µA. The linearity of the voltage-only controlled stimulation, was measured to be 10-bit in the range of ±1.45 V.

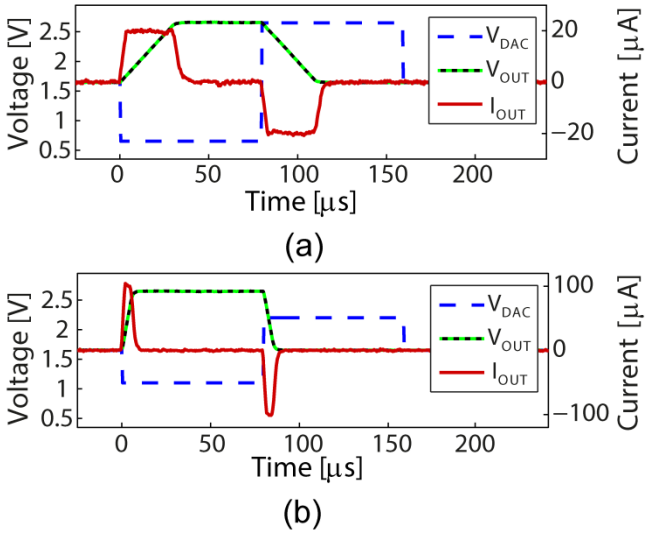


Fig. 13. Voltage (black-green dotted line) and current (red bold line) waveforms, generated by the stimulation buffer operated in current/voltage-controlled mode and connected to a 610 pF external load. The blue dashed line indicates the input waveform. (a) Low-current mode configuration, (b) High-current mode configuration.

2.7 Biological measurement results

The presented HD-MEA system was verified in *in-vitro* measurements. Fig. 14(a) depicts average spike amplitudes of dissociated rat cortical neurons (Wistar) recorded by an HD 45×45 pixel electrode block connected to 2025 AP channels. A close-up of the spatial distribution of average AP waveforms, which most likely belong to a single neuron, is plotted in Fig. 14(b). Fig. 14(c) depicts the average activity in a mouse cerebellar slice (C57Bl6J), recorded over the entire array by using a sparse electrode configuration (2000 AP channels, in a honey-comb configuration with an average of ~5.5 pixel spacing between the recording electrodes). Fig. 14(d) shows a superposition of the recorded electrical activity map with a microscopy image of the cerebellar slice. The large number of AP channels featured on the chip enabled a quick overview of the tissue activity by using a single, sparse electrode configuration, which revealed the locations of the electrically active Purkinje neurons [49].

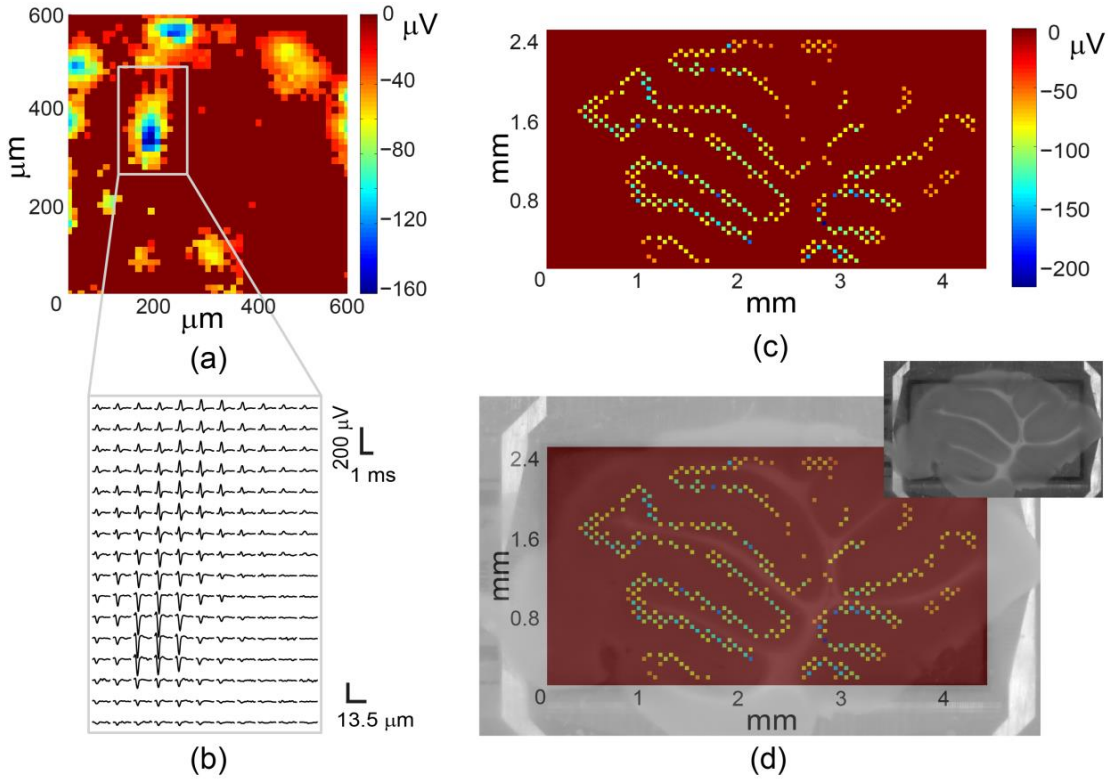


Fig. 14. (a) Average negative peak amplitudes, recorded by an HD electrode configuration (45×45 pixels, 607.5×607.5 µm^2) from dissociated primary rat cortical neurons. Each pixel in the image corresponds to an electrode. (b) AP waveforms, recorded in a sub-region of (a), averaged 20 times. (c) Mouse cerebellar slice electrical activity, recorded with a single sparse configuration of 2000 electrodes. The recording electrodes are displayed 9 times larger than their original size for visual clarity. (d) Overlap of (c) and a microscopy image of the slice (inset).

Stimulus–evoked average activity of dissociated rat cortical neurons, triggered by a biphasic voltage pulse of 300 mV amplitude, was recorded over the entire electrode array area at full spatial resolution and depicted in Fig. 15.

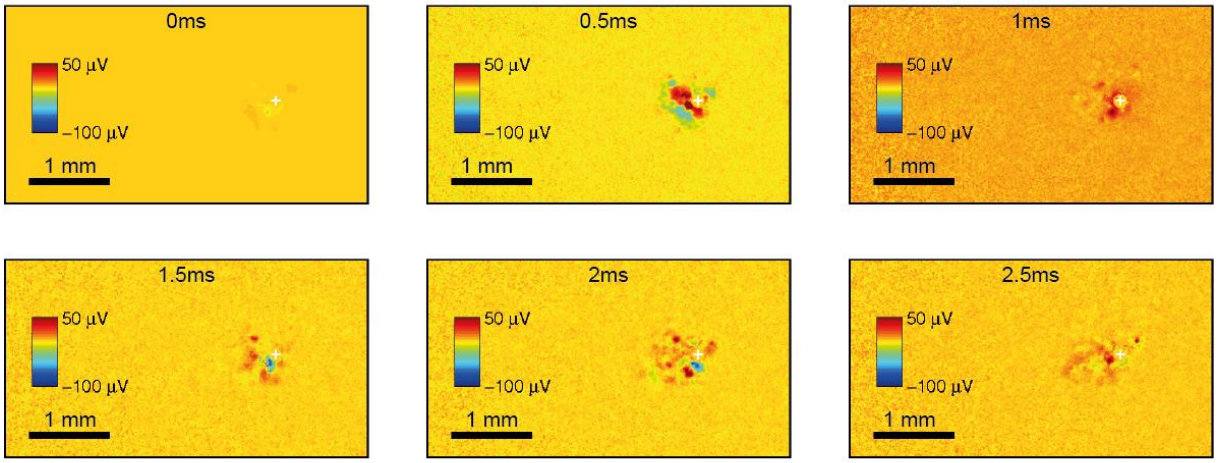


Fig. 15. Six frames showing 2.5 ms of evoked neuronal activity from a culture of dissociated rat cortical neurons. A biphasic voltage-controlled pulse (300 mV phase amplitude and 200 µs phase duration) was applied to a single electrode (white cross), while the other electrodes were scanned for recording from the entire array (4.48×2.43 mm^2) at full spatial resolution (13.5 µm electrode pitch). 100 averages were computed for each electrode and temporally aligned with respect to the stimulation pulses.

2.8 Conclusion

The presented HD-MEA system includes the largest number of measurement/stimulation functions to date, comprising electrophysiology readout, impedance measurement, and neurotransmitter detection, as well as stimulation capabilities (Table II). At the same time, it features the largest active electrode array area (10.9 mm^2) to date, whereas the number of voltage recording channels (2048 AP + 32 LFP), the spatial resolution (13.5 µm electrode pitch), the noise characteristics, and the power consumption are comparable to those of the best state-of-the-art MEA systems. Any measurement/stimulation unit can be connected to any electrode on the array owing to the flexible electrode routing via the switch matrix. The possibility to perform different measurement/stimulation functions in parallel will pave the way to complex experiments that target multiple cellular parameters at once.

TABLE II
Comparison to the state-of-the-art in-vitro MEA systems

Reference	[18]	[6]	[20]	[21]	**This work**
Type	In-pixel Amp	In-pixel Amp	In-pixel Amp	Switch Matrix	**Switch Matrix**
Technology	0.5 µm	0.13 µm	0.18 µm	0.35 µm	**0.18 µm**
Active Area	2.28×1.14 mm^2	9 pixel groups of 0.128 mm^2	1×1 mm^2/ 2×2 mm^2	3.85×2.10 mm^2	**4.48×2.43 mm^2**
No. Transducers	32768	144	4225 / 1024	26400	**59760**
No. Different Measurement Modes	1	4	1	1	**5**
Pixel Pitch	8.775 µm	89.4 / 357.8 µm[d]	16 / 32 µm	17.5 µm	**13.5 µm**
Transducers/ mm^2	12987	125 / 7.8	977, 3906	3265	**5487**
No. Voltage Readout Channels	32768	144	4225	1024	**2048 AP + 32 LFP**
A/D Conversion	128 pipeline	Off chip	Off chip	1024 single-slope	**32 + 4 SAR**
ADC Resolution	9 bit	-	-	10 bit	**10 bit**
Sampling Rate	2.4 kS/s[a]	-	77 (25) kS/s[a]	20 kS/s	**20 kS/s**
Input-referred Noise	-	12.7 µVrms[e]	44 µVrms[b]	2.4 µVrms[b,c]	**2.4 µVrms[b,c]** **3.6 µVrms[c,f]**
Total Power	4 W	-	-	75 mW	**86 mW**
Die Size	9.5×9.5 mm^2	2.2×2 mm^2	-	10.1×7.6 mm^2	**12×8.9 mm^2**

[a] When reading out all channels.
[b] Noise integration over 300Hz-10kHz,for AP recording channels.
[c] Including ADC.
[d] Equivalent square-shaped pixel dimension, for the same spatial density.
[e] Noise integration over 0.5Hz-10kHz.
[f] Noise integration over 1 Hz-300 Hz, for LFP recording channels.

Acknowledgments

This work was supported by the ERC Advanced Grants 267351 “NeuroCMOS” and 694829 “neuroXscales”. A. Shadmani received individual support through FP7-MTN “EngCaBra” (Contract 264417). The authors thank M. Ballini, P. Livi, U. Frey, W. Gong, and D. Bakkum, all at ETH Zurich, for discussions and help with neuronal cultures. The authors also thank Evi Bieler, University of Basel, for providing the SEM image.

References

[1] P. J. Uhlhaas and W. Singer, “Review Neural Synchrony in Brain Disorders: Relevance for Cognitive Dysfunctions and Pathophysiology,” *Neuron J*., vol. 52, pp. 155–168, 2006.

[2] M. Roham, D. P. Covey, D. P. Daberkow *et al.*, “A wireless IC for time-share chemical and electrical neural recording,” *IEEE J. Solid-State Circuits*, vol. 44, no. 12, pp. 3645–3658, 2009.

[3] M. H. Nazari, H. Mazhab-Jafari, L. Leng, A. Guenther, and R. Genov, “CMOS neurotransmitter microarray: 96-channel integrated potentiostat with on-die microsensors,” *IEEE Trans. Biomed. Circuits Syst.,* vol. 7, no. 3, pp. 338–348, 2013.

[4] J. Guo, W. Ng, J. Yuan, S. Li, and M. Chan, “A 200-Channel Area-Power-Efficient Chemical and Electrical Dual-Mode Acquisition IC for the Study of Neurodegenerative Diseases,” *IEEE Trans. Biomed. Circuits Syst.*, vol. 10, no. 3, pp. 567–578, 2016.

[5] J. D. Ross, S. M. O’Connor, R. a Blum, E. a Brown, and S. P. DeWeerth, “Multielectrode impedance tuning: reducing noise and improving stimulation efficacy.,” *Conf. Proc. IEEE Eng. Med. Biol. Soc*., vol. 6, pp. 4115–4117, 2004.

[6] T. Chi, J. S. Park, J. C. Butts, T. A. Hookway, A. Su, C. Zhu, M. P. Styczynski, T. C. McDevitt, and H. Wang, “A Multi-Modality CMOS Sensor Array for Cell-Based Assay and Drug Screening*,” IEEE Trans. Biomed. Circuits Syst.,* vol. 9, no. 6, pp. 801–814, 2015.

[7] J. R. Buitenweg, W. L. C. Rutten, W. P. A. Willems, and J. W. Van Nieuwkasteele, “Measurement of sealing resistance of cell-electrode interfaces in neuronal cultures using impedance spectroscopy,” *Med. Biol. Eng. Comput.,* vol. 36, no. 5, pp. 630–637, 1998.

[8] Y. Chen, C. C. Wong, T. S. Pui, *et al.*, “CMOS high density electrical impedance biosensor array for tumor cell detection,” *Sensors and Actuators B Chem,* vol. 173, pp. 903-907, 2012.

[9] R. Elshafey, C. Tlili, A. Abulrob, A. C. Tavares, and M. Zourob, “Label-free impedimetric immunosensor for ultrasensitive detection of cancer marker Murine double minute 2 in brain tissue.,” *Biosens. Bioelectron.,* vol. 39, no. 1, pp. 220–5, Jan. 2013.

[10] A. Manickam, A. Chevalier, M. McDermott, A. D. Ellington, and A. Hassibi, “A CMOS electrochemical impedance spectroscopy (EIS) biosensor array*,” IEEE Trans. Biomed. Circuits Syst*., vol. 4, no. 6 PART 1, pp. 379–390, 2010.

[11] C. A. Thomas, P. A. Springer, G. E. Loeb, Y. Berwald-Netter, and L. M. Okun, “A miniature microelectrode array to monitor the bioelectric activity of cultured cells,” *Exp. Cell Res.,* vol. 74, no. 1, pp. 61–66, 1972.

[12] P. Fromherz, “Electrical interfacing of nerve cells and semiconductor chips,” *ChemPhysChem*, vol. 3, no. 3, pp. 276–284, 2002.

[13] S. I. Morefield, E. W. Keefer, K. D. Chapman, and G. W. Gross, “Drug evaluations using neuronal networks cultured on microelectrode arrays*,” Biosens. Bioelectron.,* vol. 15, no. 7–8, pp. 383–396, 2000.

[14] G. W. Gross, B. K. Rhoades, H. M. E. Azzazy, and Ming-Chi Wu, “The use of neuronal networks on multielectrode arrays as biosensors*,” Biosens. Bioelectron*., vol. 10, no. 6, pp. 553–567, 1995.

[15] A. Stett, U. Egert, E. Guenther, F. Hofmann, T. Meyer, W. Nisch, and H. Haemmerle, “Biological application of microelectrode arrays in drug discovery and basic research.,” *Anal. Bioanal. Chem*., vol. 377, no. 3, pp. 486–95, Oct. 2003.

[16] M. Litke, N. Bezayiff, E. J. Chichilnisky, *et al.*, “What does the eye tell the brain?: Development of a system for the large-scale recording of retinal output activity,” *IEEE Trans. Nucl. Sci.*, vol. 51, no. 4, pp. 1434–1440, Aug. 2004.

[17] U. Frey, J. Sedivy, F. Heer, *et al.,* “Switch-matrix-based high-density microelectrode array in CMOS technology,” *IEEE J. Solid-State Circuits*, vol. 45, no. 2, pp. 467–482, 2010.

[18] B. Eversmann, A. Lambacher, T. Gerling, A. Kunze, P. Fromherz, and R. Thewes, “A neural tissue interfacing chip for in-vitro applications with 32k recording / stimulation channels on an active area of 2.6 mm 2*,” Eur. Solid-State Circuits Conf*., pp. 211–214, 2011.

[19] L. Berdondini, K. Imfeld, A. Maccione, M. Tedesco, S. Neukom, M. Koudelka-Hep, and S. Martinoia, “Active pixel sensor array for high spatio-temporal resolution electrophysiological recordings from single cell to large scale neuronal networks.,” *Lab Chip*, vol. 9, no. 18, pp. 2644–51, 2009.

[20] G. Bertotti, D. Velychko, N. Dodel, *et al.*, “A CMOS-based sensor array for in-vitro neural tissue interfacing with 4225 recording sites and 1024 stimulation sites,” *IEEE Biomedical Circuits and Systems Conference (BioCAS) Proceedings*, 2014, pp. 304–307.

[21] M. Ballini, J. Muller, P. Livi, *et al.*, “A 1024-channel CMOS microelectrode array with 26,400 electrodes for recording and stimulation of electrogenic cells in vitro,” *IEEE J. Solid-State Circuits*, vol. 49, no. 11, pp. 2705–2719, 2014.

[22] M. E. J. Obien, K. Deligkaris, T. Bullmann, D. J. Bakkum, and U. Frey, “Revealing neuronal function through microelectrode array recordings,” *Front. Neurosci*., vol. 9, no. JAN, p. 423, 2015.

[23] C. M. Gray, P. E. Maldonado, M. Wilson, and B. McNaughton, “Tetrodes markedly improve the reliability and yield of multiple single-unit isolation from multi-unit recordings in cat striate cortex.,” *J. Neurosci. Methods*, vol. 63, no. 1–2, pp. 43–54, Dec. 1995.

[24] M. S. Grubb and J. Burrone, “Activity-dependent relocation of the axon initial segment fine-tunes neuronal excitability,” *Nature*, vol. 465, no. 7301, pp. 1070–1074, Jun. 2010.

[25] J. Müller, M. Ballini, P. Livi, *et al.* “High-resolution CMOS MEA platform to study neurons at subcellular, cellular, and network levels,” *Lab Chip*, vol. 15, no. 13, pp. 2767–2780, 2015.

[26] E. Juergens, A. Guettler, and R. Eckhorn, “Visual stimulation elicits locked and induced gamma oscillations in monkey intracortical- and EEG-potentials, but not in human EEG*,” Exp. Brain Res*., vol. 129, no. 2, pp. 247–259, Nov. 1999.

[27] U. Mitzdorf, “Properties of the evoked potential generators: current source-density analysis of visually evoked potentials in the cat cortex.,” *Int. J. Neurosci*., vol. 33, no. 1–2, pp. 33–59, Mar. 1987.

[28] J. Guo, J. Yuan, and M. Chan, “Modeling of the cell-electrode interface noise for microelectrode arrays.,” *IEEE Trans. Biomed. Circuits Syst.,* vol. 6, no. 6, pp. 605–13, Dec. 2012.

[29] V. Viswam, J. Dragas, A. Shadmani, Y. Chen, A. Stettler, J. Muller, and A. Hierlemann, “Multi-functional microelectrode array system featuring 59,760 electrodes, 2048 electrophysiology channels, impedance and neurotransmitter measurement units,” *IEEE International Solid-State Circuits Conference (ISSCC),* pp. 394–396, 2016.

[30] V. Viswam, Y. Chen, A. Shadmani, J. Dragas, R. Bounik, R. Milos, J. Müller, A. Hierlemann, “2048 action potential recording channels with 2.4μVrms noise and stimulation artefact suppression”, *Biomedical Circuits and Systems Conference (BioCAS)*, Oct. 2016.

[31] B. Gosselin and Benoit, “Recent Advances in Neural Recording Microsystems,” *Sensors*, vol. 11, no. 12, pp. 4572–4597, Apr. 2011.

[32] A. Tajalli, Y. Leblebici, *Extreme Low-Power Mixed Signal IC Design - Subthreshold Source-Coupled Circuits*, Springer, 2010.

[33] C. M. Hammerschmied and Q. Huang, “Design and implementation of an untrimmed MOSFET-only 10-bit A/D converter with -79-dB THD,” *IEEE J. Solid-State Circuits,* vol. 33, no. 8, pp. 1148–1157, 1998.

[34] E. S. Bucher and R. M. Wightman, “Electrochemical Analysis of Neurotransmitters,” *Annu. Rev. Anal. Chem.,* vol. 8, no. 1, pp. 239–261, Jul. 2015.

[35] W. P. Kang, S. Raina, J. L. Davidson, and J. H. Huang, “High temporal resolution electrochemical biosensor using nitrogen-incorporated nanodiamond ultra-microelectrode array,” *IEEE Sensors*, pp. 1–4, 2012.

[36] L. M. Borland and A. C. Michael, *An Introduction to Electrochemical Methods in Neuroscience*. CRC Press/Taylor & Francis, 2007.

[37] A. J. Bard and L. R. Faulkner, *Electrochemical methods : fundamentals and applications*. Wiley, 2001.

[38] D. L. Robinson, A. Hermans, A. T. Seipel, and R. M. Wightman, “Monitoring rapid chemical communication in the brain,” *Chem. Rev.*, vol. 108, no. 7, pp. 2554–84, Jul. 2008.

[39] F. Maloberti, *Data Converters*, Springer 2007.

[40] F. Gozzini, G. Ferrari, and M. Sampietro, “An instrument-on-chip for impedance measurements on nanobiosensors with attofarad resolution,” *Dig. Tech. Pap. - IEEE Int. Solid-State Circuits Conf.*, pp. 346–348, 2009.

[41] J. Guo, W. Ng, J. Yuan, and M. Chan, “A 51fA/Hz0.5 low power heterodyne impedance analyzer for electrochemical impedance spectroscopy,” *VLSI Circuits (VLSIC)*, *Symp.*, no. 1, pp. 56–57, 2013.

[42] M. Crescentini, M. Bennati, M. Carminati, and M. Tartagni, “Noise limits of CMOS current interfaces for biosensors: a review.,” *IEEE Trans. Biomed. Circuits Syst.*, vol. 8, no. 2, pp. 278–92, Apr. 2014.

[43] S. Zhou and M. C. F. Chang, “A CMOS passive mixer with low flicker noise for low-power direct-conversion receiver,” *IEEE J. Solid-State Circuits*, vol. 40, no. 5, pp. 1084–1093, 2005.

[44] V. Quiquempoix, P. Deval, A. Barreto, G. Bellini, J. Márkus, J. Silva, and G. C. Temes, “A low-power 22-bit incremental ADC,” *IEEE J. Solid-State Circuits*, vol. 41, no. 7, pp. 1562–1571, 2006.

[45] R. Wu, Y. Chae, J. H. Huijsing, and K. A. A. Makinwa, “A 20-b ±40-mV range read-out IC with 50-nV offset and 0.04% gain error for bridge transducers,” *IEEE J. Solid-State Circuits*, vol. 47, no. 9, pp. 2152–2163, 2012.

[46] S. Joucla and B. Yvert, “Modeling extracellular electrical neural stimulation: from basic understanding to MEA-based applications.,” *J. Physiol. Paris*, vol. 106, no. 3–4, pp. 146–58.

[47] P. Livi, F. Heer, U. Frey, D. J. Bakkum, and A. Hierlemann, “Compact voltage and current stimulation buffer for high-density microelectrode arrays,” *IEEE Trans. Biomed. Circuits Syst.*, vol. 4, no. 6, pp. 372–8, Dec. 2010.

[48] F. Heer, S. Hafizovic, W. Franks, T. Ugniwenko, A. Blau, C. Ziegler, and A. Hierlemann, “CMOS microelectrode array for bidirectional interaction with neuronal networks,” *Proc. ESSCIRC 2005 31st Eur. Solid-State Circuits Conf.*, vol. 41, no. 7, pp. 335–338, 2005.

[49] H. Gray, Anatomy of the Human Body, Lea & Febiger, 1918.

[50] D.A. Wagenaar, J. Pine, S.M. Potter, “Effective parameters for stimulation of dissociated cultures using multi-electrode arrays,” J. Neurosci. Methods 138 (1-2), 27–37, 2004.

[51] J.B. Zimmermann and A. Jackson, “Closed-loop control of spinal cord stimulation to restore hand function after paralysis,” Front. Neurosci. 8:87, 2014.

CHAPTER 3

Integrated Current/Voltage Controlled Stimulation Buffer and Artifact Suppression Circuits for Multifunctional High-density Microelectrode Arrays

Amir Shadmani, Vijay Viswam, Yihui Chen, Raziyeh Bounik, Jelena Dragas, Milos Radivojevic, Sydney Geissler, Sergey Sitnikov, Jan Müller, Andreas Hierlemann

Submitted to:

IEEE Transactions on Biomedical Circuits and Systems (TBioCAS)

Keywords

CMOS, high-density microelectrode array (HD-MEA), extracellular recording and stimulation, current/voltage controlled stimulation, pseudo-resistor, artifact suppression, soft-reset

3.1 Abstract

We present novel stimulation buffers to generate controlled current and voltage stimulation pulses, along with recording circuits featuring adjustable high-pass cut-off filters to suppress stimulation artifacts, all of which have been integrated into a multifunctional high-density microelectrode array. The proposed stimulation buffers can operate in two modes: either control of only the stimulation voltage, or control of current and voltage during stimulation. The voltage-only controlled mode employs a local common mode feedback operational transconductance amplifier with a near rail-to-rail input/output range, suitable for driving high capacitive loads. The current/voltage controlled mode is based on a positive current conveyor generating adjustable output currents, while its upper and lower output voltages are limited by two feedback loops. For traditional electrophysiological recording channels, high amplitude stimulation pulses can produce large artifacts that easily saturate recording amplifiers. To suppress such artifacts, the high-pass corner frequencies of the recording channels reported here can be raised from several Hz to several kHz by applying a “soft reset” technique. This technique prevents recording amplifiers from being saturated and allows for prompt recovery of the recording circuits. The current/voltage controlled circuit can generate stimulation pulses up to 30 µA with less than ±0.1% linearity error in the low-current mode, and up to 300 µA with less than ±0.2% linearity error in the high-current mode. In addition, with the implemented artifact suppression technique, the saturation time of the recording circuits connected to electrodes adjacent to the stimulation site could be reduced to 200 µs.

3.2 Introduction

Microelectrode arrays (MEAs) have been widely used in recent years for in-vitro examination of neuronal cells and neural networks. These MEAs enable long-term bi-directional interfacing with networks of living cells by providing extracellular electrical stimulation and electrophysiological recording features [1, 2]. Recent CMOS-based active MEA devices, especially high-density (HD) MEAs, provide the capability to simultaneously perform electrophysiological recordings from thousands of electrodes at cellular/subcellular spatial resolution [3-6]. The high spatio-temporal resolution of these devices enables better separation and assignment of neural activities for closely spaced neurons as well as very localized stimulation of single individual neurons [7].

Extracellular electrical stimulation of neurons using MEAs is achieved by, for example, applying potentials through electrodes in the vicinity of targeted neurons, thus altering their membrane potentials. Stimulus with proper amplitude and duration will cause depolarization of the neuron membrane and will evoke an action potential (AP) response [8]. Faradaic and non-Faradaic processes may occur during stimulation. In the case of a non-Faradaic process, a charge redistribution in the double-layer capacitor formed at the electrode/electrolyte interface occurs. In the case of a Faradaic process, a charge transfer and redox reactions may occur due to electron transfer at the electrode surface [9]. In contrast to non-Faradaic processes, Faradaic processes may entail unwanted electrochemical reactions and formation of reactive species. These reactions may be irreversible, lead to electrode degradation, and may produce compounds that are toxic to the cells [10]. Upon exceeding a safe potential window, determined also by the electrode material, electrolysis may occur, and the Faradaic current may then increase exponentially [9]. Therefore, to avoid the occurrence of Faradaic processes, the electrode voltage should always be limited to the safe potential window.

The electrical stimulus is, in most cases, generated by either voltage-controlled or current-controlled circuits. The choice of the stimulation technique depends on the preferences for controlling either electrode voltage or injected current [11]. The voltage-controlled technique exploits the electrode/electrolyte double layer capacitor and issues a voltage pulse across this capacitor. This technique has the advantage of being able to obviate Faradaic processes by precisely controlling the electrode voltage and maintaining it within a safe range.

The voltage-controlled stimulation circuit has been widely used in MEAs, as it is efficient and safe [12]. The current generated while using such a circuit, however, depends on the electrode impedance, which may vary considerably due to fabrication variation or aging. This may cause variations in the volume of the tissue that is stimulated [13, 14]. Therefore, current-controlled stimulation circuits have been developed that provide direct control over the electrode current [15].

Current-controlled stimulation does not offer control over the electrode voltage, and excessive electrode voltage can entail accumulation of charges on the electrode, leading to unwanted electrochemistry, tissue damage, or electrode degradation, as discussed before. In the current-controlled stimulation, mostly biphasic current pulses with opposite polarity in each phase are used, which minimizes charge accumulation. However, imperfect device matching may lead to unequal positive and negative currents, so charge-balanced techniques are necessary to assure zero net charge transfer in the current-controlled stimulation [16]. Charge balance can be achieved using a large blocking capacitor (in the range of several hundred nF) placed at the output of the stimulation circuit to block the transfer of any DC currents [17, 18]. It is unfeasible to integrate such a large blocking capacitor on a chip, therefore, active charge balancing techniques have been proposed, where either the offset current is regulated or a compensating charge is transferred to the electrode whenever a residual voltage is detected after stimulation [19-21].

The high impedance of small MEA electrodes entails that the electrode voltage rapidly reaches the supply voltage, which prevents the generation of balanced positive and negative current pulses upon using current-controlled stimulation. To address this issue, high-voltage compliance circuits have been developed to detect and regulate the stimulation in case high voltages appear at the electrodes [22-25].

An undesirable issue accompanying stimulation, either voltage-controlled or current-controlled, is the influence of the stimulation signals on the recording channels. Typically, recording channels are designed to detect electrophysiological signals with amplitudes from tens of µV to a few mV and frequencies from 1 Hz to 10 kHz, whereas the stimulation circuits generate signals with amplitudes of a few hundred mV. Such large amplitude signals can easily drive the recording circuits or channels that are connected to electrodes near the stimulation site into saturation and are known as “stimulation artifact”. The recording channels are normally designed with a very low high-pass cutoff frequency for filtering out low-frequency electrode potential drifts, and thus the saturation state of the channels may last up to tens of milliseconds [26]. When saturated, the recording channels cannot record any electrophysiological signals, so techniques are needed to decrease this dead time and suppress the stimulation artifacts. Analog techniques have been previously developed for artifact suppression in MEAs, such as disconnecting the recording channels or resetting them in a unity-gain feedback during the stimulation phase [27-29]. Their application entails additional artifacts because such techniques impose a rapid topological change in the recording channels during disconnection or reset.

In this paper, we present a new stimulation circuit, which can work either in voltage-only controlled or current/voltage controlled mode. The current/voltage controlled mode provides continuous control over the current and voltage of the generated stimulation pulse by exploiting a similar technique as high-voltage compliance stimulation circuits [22-25]. To suppress stimulation artifacts on the electrodes neighboring the stimulation site, a novel “soft reset” technique to temporarily increase the high-pass cut-off frequency of the amplification stages was devised and included

in the design of the recording channels; this technique will be explained and characterized in detail in later sections.

This paper is organized as follows. Section II shows the proposed stimulation buffers embedded in a novel multifunctional MEA system with an emphasis on the stimulation buffers, the recording channels, and their interconnection with the electrode array. Section III elaborates on the stimulation buffers in the voltage-only controlled and the current/voltage controlled mode, as well as the circuit implementation of the artifact suppression technique. In Section IV, the electrical characterization results are summarized. Section V shows biological measurement results acquired by exploiting the implemented stimulation circuit and artifact suppression technique, and Section VI concludes the paper.

3.3 System overview

The proposed stimulation buffers and recording channels have been integrated in a recently developed multifunctional HD-MEA microsystem. The system chip includes a large sensing area of 4.48×2.43 mm2 comprising 59’760 electrodes at a pitch of 13.5 µm. The system incorporates several types of sensing units with the aim of extracting a wide spectrum of information from the same neuronal culture or brain slice. The overall HD-MEA system includes 16 dual-mode stimulation buffers (can be configured in the voltage-only or current/voltage controlled mode), 2048 action-potential (AP, bandwidth: 300 Hz to 10 kHz) recording channels, 32 local-field-potential (LFP, bandwidth: 1 Hz to 300 Hz) recording channels, 32 current recording units, 32 impedance measurement units and 28 neurotransmitter detection units [6]. Both AP and LFP recording channels are equipped with stimulation artifact suppression capability. A switch-matrix approach has been used to route different measurement/stimulation buffers to arbitrarily selectable electrodes or subsets of electrodes. Fig. 1 shows the block diagram of the whole system.

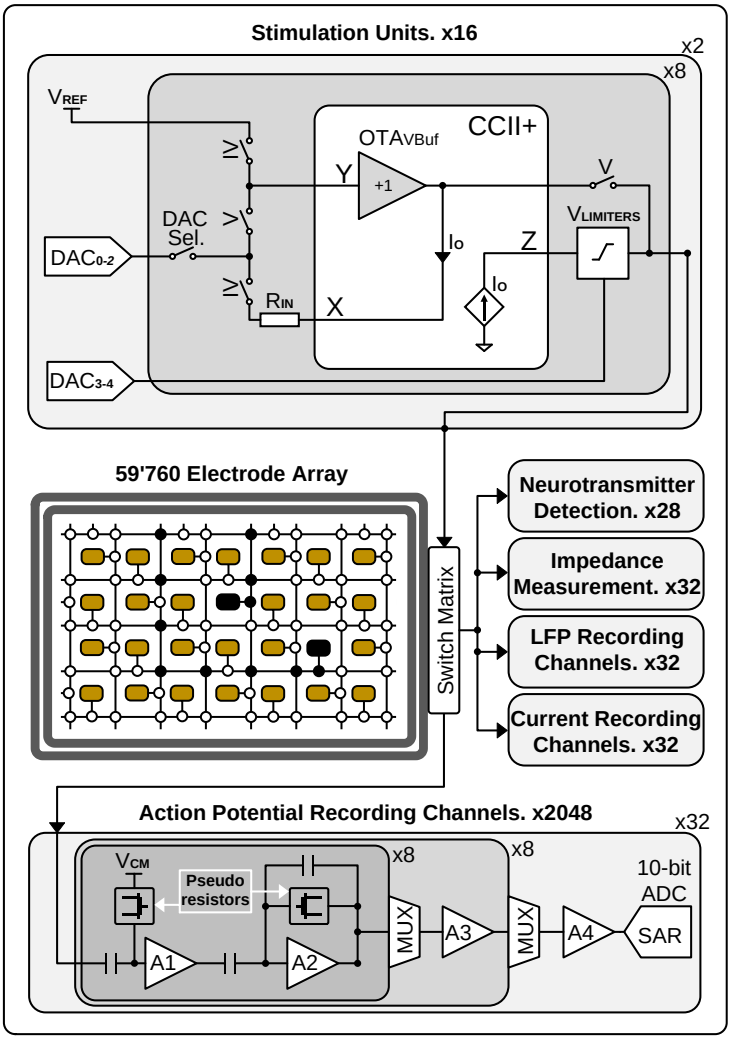


Fig. 1. Block diagram of the stimulation buffers and AP recording channels integrated in the multi-functional HD-MEA system.

The 16 dual-mode stimulation buffers are grouped in two blocks of 8 buffers. Three digital/analog converters, DAC_0 to DAC_2, are shared among all buffers in each block to generate arbitrary stimulation pulses. As shown in Fig. 1, all stimulation buffers can work in two modes: (1) by closing the switches labeled with “V”, voltage-only controlled stimulation is achieved; (2) by closing the switches labeled with “IV”, current/voltage controlled stimulation is achieved. OTA_{VBuf} is a near rail-to-rail operational transconductance amplifier (OTA), which is used in both operating modes. In the voltage-only controlled mode, OTA_{VBuf} is configured in unity-gain feedback, whereas in the current/voltage controlled mode, it works as the buffer part of a positive current conveyor of type II (CCII+). During stimulation, this current conveyor generates tunable output current pulses, which are delivered to the stimulation electrode. The upper and lower bounds of the electrode voltage are controlled by two voltage limiters, whose threshold voltage are set by two digital/analog converters, DAC_3 and DAC_4, respectively.

2048 AP recording channels have been integrated on the chip. Each channel consists of 4 fully-differential amplification stages. The first two stages employ MOS transistors, biased in the weak inversion region, called pseudo-resistors, to realize very high resistances in the range of TΩ. Using these pseudo-resistors together with the AC-coupling input capacitors, a very low high-pass cut-off frequency (tens of mHz to a few Hz) has been achieved to filter out low-frequency electrode potential drifts. Meanwhile, during stimulation, the gate voltages of these MOS transistors can be adjusted in a way that their resistances are two to three orders of magnitude smaller than their normal value. Thus, the recording channels can recover very quickly after stimulation, a feature that will be elaborated in more detail in the next section.

3.4 Circuit implementation

3.4.1 Near rail-to-rail OTA for the voltage-only controlled stimulation mode

The implementation of the stimulation buffer included the design of a near rail-to-rail OTA for the voltage-only controlled mode. For this stimulation mode, the buffer has to drive a large capacitive load of one or several electrodes (up to nF range) while consuming low quiescent power.

Given the large electrode capacitance, the two-stage Miller-compensated opamp would not be a suitable choice for this application due to the necessity of using a large Miller capacitor to keep the opamp stable. Therefore, an OTA topology with a dominant pole at the output was chosen, as such topology is more stable for large electrode capacitances. For OTAs, a large gain is achieved by having a high output resistance, so that resistive loads can degrade the gain substantially. For the voltage-only controlled stimulation mode, the OTA is configured as unity-gain feedback, eliminating the necessity to use feedback resistors. To have the ability to generate high-amplitude stimulation pulses (the supply voltage is limited to 3.3 V), the OTA needs to have a near rail-to-rail output swing, which, in turn, requires the same common-mode input range due to the unity-gain feedback.

Fig. 2 shows the schematic of the proposed OTA. Its near rail-to-rail input/output range is achieved as follows; the common-mode input voltage is sensed by M_7 and M_8. M_7 and M_8 then set the gate voltages of M_{12} and M_{13}, which, in turn, control the drain voltages of the input transistors M_1-M_4. When the input common-mode voltage is close to V_{DD}, M_3 and M_4 are turned off, but the drain voltages of M_1 and M_2 remain high to keep them in saturation. In contrast, upon applying an input common-mode voltage close to ground, the drain voltages of M_3 and M_4 are sufficiently low to keep them in saturation while M_1 and M_2 are turned off. The local common-mode feedback technique [30] has been adopted by adding the resistors R_1 and R_2 in the OTA to provide a large output current with a low quiescent current. The use of this technique also increases the gain-bandwidth product (GBW) significantly.

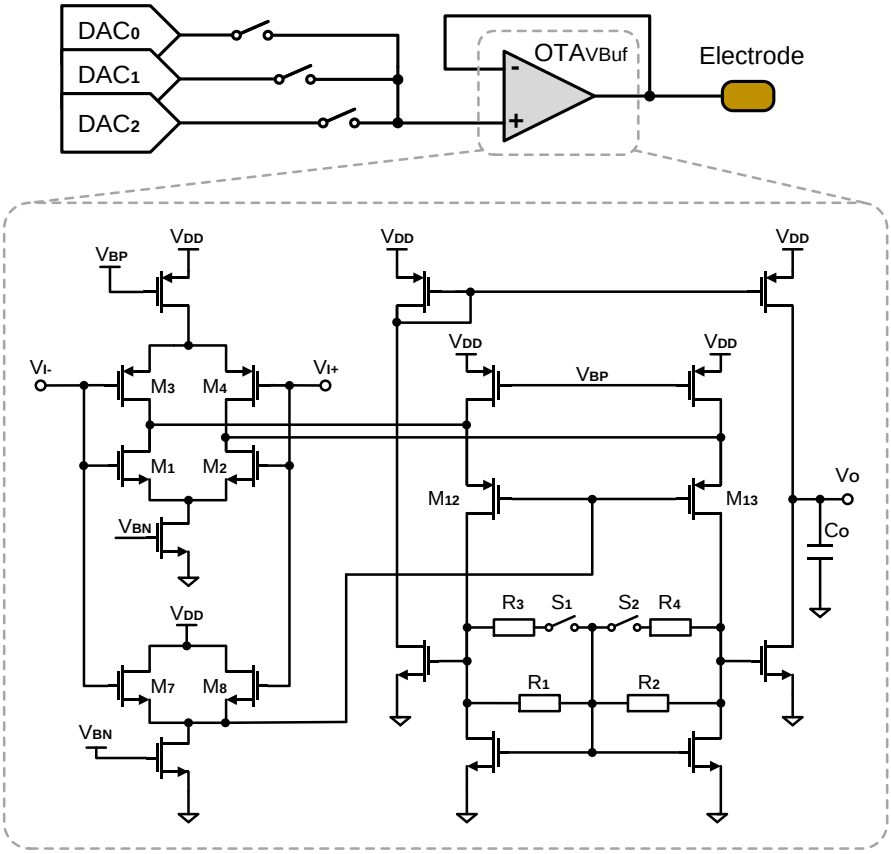


Fig. 2. Schematic of the proposed near rail-to-rail OTA for the voltage-only controlled stimulation mode.

3.4.2 Current conveyor with output voltage limiters for the current/voltage controlled stimulation mode

For the voltage-only controlled mode, the voltage level of the stimulation pulse is precisely controlled, but the output current delivered to the electrode is not controlled. The maximum output current of the OTA for the voltage-only mode can be controlled by adjusting its bias current [26], however, the relationship between maximum output current and the bias current is nonlinear, so the control would be poor. To achieve better control over both output voltage and current, we improved the current conveyor reported in [31] and developed a new design, illustrated in Fig. 3.

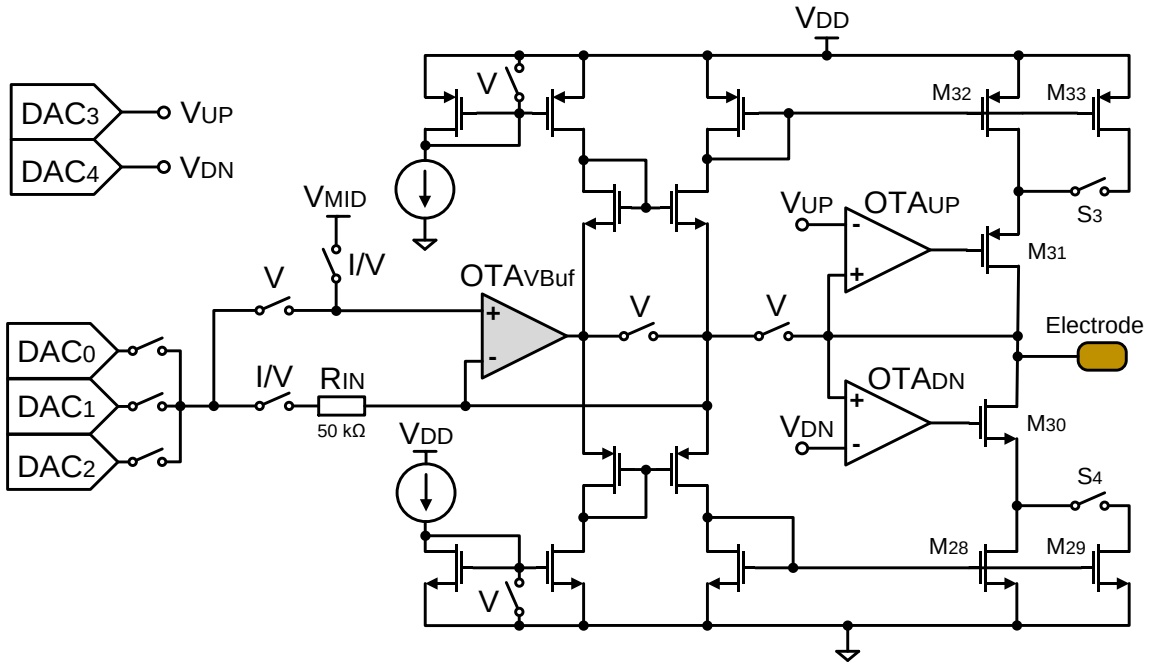


Fig. 3. Schematic of the proposed stimulation buffer. The switches marked with “V” are used for configuring the buffer for operation in the voltage-only mode and those marked with “IV” are used for configuring the buffer for operation in the current/voltage controlled mode.

The proposed stimulation buffer incorporates a CCII+ that converts the input DAC voltage into a corresponding output current. The same OTA, OTA_{VBUF}, was reused for this mode as well, and the resistors R_3 and R_4 have been added in parallel to R_1 and R_2 to decrease current boosting and the OTA's gain. According to simulations, R_3 and R_4 decrease the gain by 13 dB and increase the phase margin from 15º to 70º. Meanwhile, R_{IN} has a value of 50 kΩ, so that a ±30 µA current is generated upon applying ±1.5 V through the DACs. This current flows into the current conveyor where it is then mirrored to the output branch and delivered to the electrode. The circuit features a low- and high-current mode. In the low-current mode, the generated current is directly mirrored to the output, whereas, in the high-current mode, M_{29} and M_{33} are connected to the output branch by closing the switches S_3 and S_4. M_{29} and M_{33} are ten times larger than M_{28} and M_{32}, and therefore, the output current can become as large as ±330 µA without increasing the quiescent current in the first two branches of the current conveyor.

OTA_{UP}, M_{31} and OTA_{DN}, M_{30} form two independent feedback loops to control the electrode voltage (V_{EL}). The OTA_{UP}, M_{31} feedback loop is triggered when $V_{EL} > V_{UP}$, and the OTA_{DN}, M_{30} feedback loop is triggered when $V_{EL} < V_{DN}$. Whenever one of the two feedback loops is triggered, the output current is nearly turned off, and only a small bias current is still flowing. This bias current keeps the feedback loop active and continuously regulates V_{EL} to be fixed either at V_{UP} or V_{DN}.

3.4.3 Pseudo resistors and stimulation artifact suppression

A simplified schematic of the AP recording channel is shown in Fig. 4. A more elaborated description of its overall design can be found in [6, 32], and here we will focus mainly on the design of the pseudo resistors and the stimulation artifact suppression features. As already mentioned, each AP recording channel includes four fully-differential gain stages. The first stage employs an open-loop topology with resistive loads to achieve low noise. The second stage includes a closed-loop structure to have an accurate gain and to provide anti-aliasing filtering. Both the third and fourth stages have been realized with switched-capacitor amplifiers to handle a large signal swing and implement 8-to-1 multiplexing. Finally, every 64 neural signals are sampled at 20 kS/s and digitized by a 10-bit successive-approximation register (SAR) analog/digital converter (ADC) capable of converting at 1.28 MS/s. The digital data is sent off chip and processed by a field-programmable gate array (FPGA).

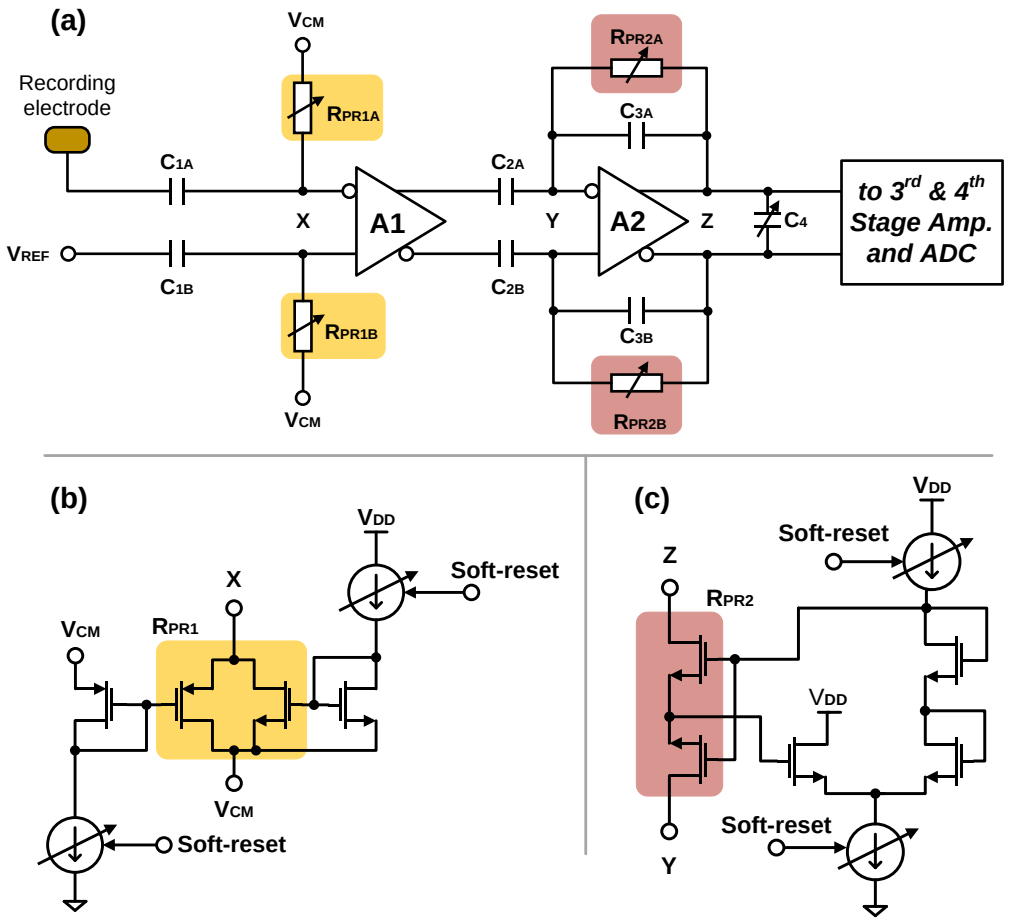


Fig. 4. Schematic of the pseudo-resistors for implementing the soft reset feature in the first and second stage amplifiers of the AP recording channels.

Pseudo resistors, normally realized with MOS transistors biased in the weak inversion region, play an important role in realizing the two first-order high-pass filters (HPFs) in front of the first and the second stage, as shown in Fig. 4. In the first stage, the HPF (C_1 and R_{PR1}) eliminates the low frequency drifts of the electrode potentials. In the second stage, the HPF (C_3 and R_{PR2}) removes the offset of the first stage, avoiding saturation of the whole amplification chain. Both HPFs should have very low cutoff frequencies (<10 Hz) in order not to distort AP signals. As the capacitors are limited to around a few pF due to area constraints, the pseudo resistors (R_{PR1} and R_{PR2}) need to feature large enough resistances (in the TΩ range).

Although conventional pseudo-resistor structures can achieve such large resistances, the resistance spreads are large due to process and temperature variations. To overcome this drawback, we devised two novel structures [6, 32], shown in Fig. 4. R_{PR1} consists of a PMOS and an NMOS transistor, connected like a transmission gate and biased by a PMOS and an NMOS current mirror. R_{PR2} includes two NMOS transistors connected in series, whose gate voltages are biased with an NMOS-only level shifter, improved from a structure using a PMOS level shifter [33] for better process tracking. In both structures, the first-order dependence of the overdrive voltages of the transistors in R_{PR1} and R_{PR2} on their threshold voltages is cancelled out, which results in a much smaller resistance spread compared to conventional structures.

In addition, the resistances of R_{PR1} and R_{PR2} can be easily adjusted through their bias currents. Therefore, the high-pass cutoff frequencies can be tuned to compensate for process and temperature variations and to accommodate different experimental scenarios. This feature is further exploited in this work to suppress stimulation artifacts. Shortly before stimulation, the bias currents of the pseudo resistors are set to much higher values than during normal operation, so that the high-pass cutoff frequencies are increased to the kHz range. As a result, the gains of the first two stages are reduced to an extent that these two stages are not saturated during stimulation and their operating points are preserved. In addition, the time constants associated with the two HPFs became small (in the 100 µs range), so that, right after stimulation, the recording channels quickly recover and record neuronal signals. We call this technique “soft reset”, as the amplifier is gently reset.

3.5 Electrical characterization

The multifunctional MEA chip, including the stimulation buffers and AP recording channels, has been fabricated in an industrial 0.18-µm 1P6M CMOS technology followed by in house post-processing to deposit the platinum electrodes. A shifted-electrode design and the application of a silicon nitride/silicon oxide passivation stack is used to protect the circuitry in the chip against liquids and metabolic products of the cell cultures and to prevent toxic materials from being released by the chip into the cell medium [2].

Fig. 5 shows a chip micrograph with the floorplan of the different measurement blocks. The electrode array is in the center of the chip. The 2048 AP recording channels are placed on the top and bottom side of the chip, while the rest of the measurement units are placed on the left and right side in a symmetrical arrangement. The chip was wire-bonded to a four-layer packaging printed circuit board (PCB), which is partitioned into analog and digital sections to prevent noisy digital signals from contaminating sensitive analog signals. A glass ring was glued on the packaging PCB to contain the cell culture medium. Finally, epoxy was used to cover the bonding wires while leaving the electrode array area open to interact with the cells and liquid phase.

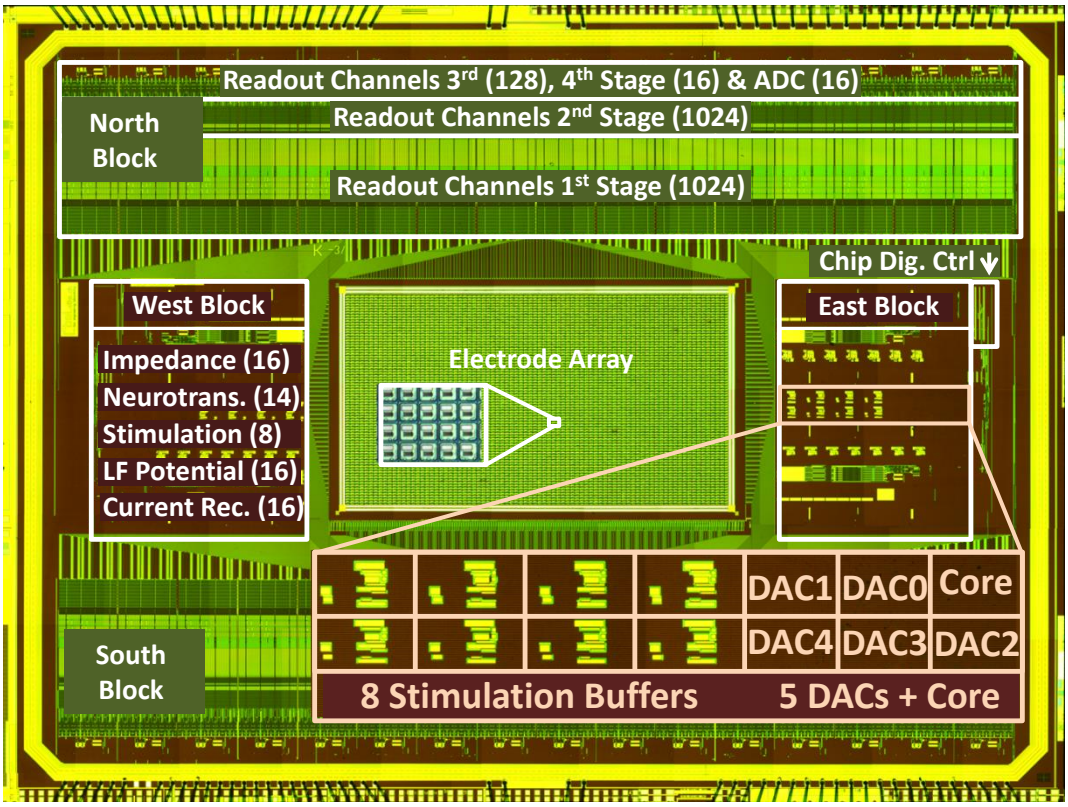


Fig. 5. Multifunctional HD-MEA chip micrograph.

To evaluate the performance of the chip, a measurement setup has been developed, including: (1) A custom-designed supporting PCB to generate bias voltages and clock signals, and to provide the test input/output connections. On the supporting PCB, a USB controller (FT245RL) and an FPGA (Spartan-3E, Xilinx Inc., San Jose, USA) were mounted to fetch and buffer the configuration commands and to transmit them to the chip with precise timing; (2) A data acquisition (DAQ) Card (NI PXIe-6544, National Instruments, Austin, USA) to acquire the digital data from the chip and send them to a PC for online visualization and data storage.

For characterization of the stimulation buffer in the voltage-only controlled mode, a 500 pF external capacitor was used as load, which emulated several electrodes connected in parallel. A biphasic ±1 V voltage pulse, centered around 1.65V, was generated using the on-chip DACs and applied to the buffer. The output voltage and output current of the buffer are shown in Fig. 6.a and 6.b, respectively. The maximum output currents obtained with the 4 µA, 6 µA, and 8 µA bias currents are approximately 85 µA, 190 µA, and 315 µA. Fig. 6.c shows the linearity analysis results. The buffer features a 10-bit linearity for a voltage range of ±1.45 V relative to the common-mode voltage of 1.65 V, which is 200 mV away from the power supply and ground rails. The open-loop characteristics of the buffer included a DC gain of 74 dB, a unity gain bandwidth of 75 kHz and a phase margin of 89º for a bias current of 4 µA; and a DC gain of 77 dB, a unity gain bandwidth of 200 kHz and a phase margin of 82º for a bias current of 8 µA, simulated with a capacitive load of 500 pF. The measured quiescent power consumption of the OTA and the biasing circuitry was 105 µW for a bias current of 4 µA and 225 µW for a bias current of 8 µA. The buffer occupies an area of 0.011 mm^2.

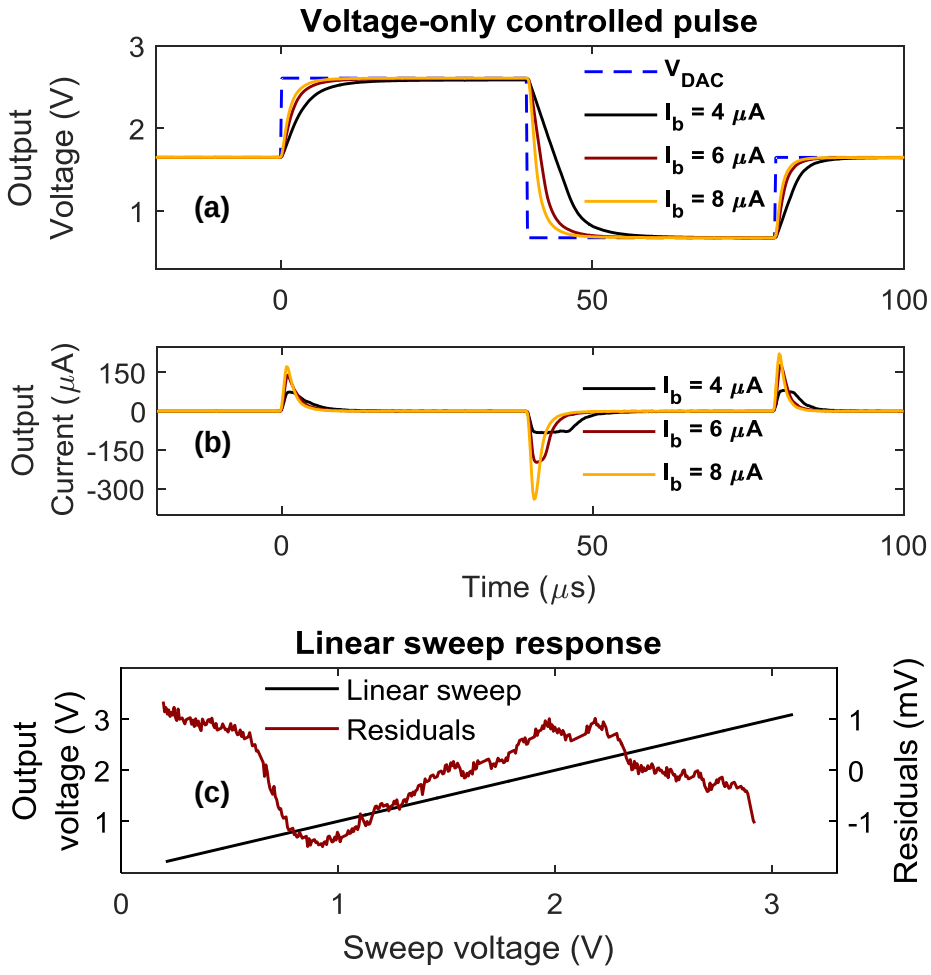


Fig. 6. Characterization of the stimulation buffer working in the voltage-only controlled mode.

Fig. 7 shows the characterization results of the stimulation buffer configured in the current/voltage controlled mode. The input voltage to the current conveyor was generated using the on-chip DACs, and its output current was adjusted to ±20 µA and ±100 µA using the low- and high-current modes, respectively. The current pulses were programmed to include 80 µs of positive currents and 80 µs of negative currents. The lower voltage limit (V_{DN}) was programmed to be 1.65 V, while the upper voltage limit (V_{UP}) was set to 2.65 V so the output voltage swing was constrained to 1.0 V. The load was again an external 500 pF capacitor. The DAC's output voltage, the output current, and the output voltage of the generated pulses for all these configurations are shown in Figs. 7.a and 7.b. It can be seen that the output currents were automatically switched off by the two feedback loops, when the output voltage reached the two programmed voltage limits. It is evident by comparing Fig. 7 with Fig. 6 that the current/voltage controlled stimulation mode provides more accurate control over the output current than the voltage-only mode. Figs. 7.c and 7.d show the linearity measurement results of the current conveyor in the low- and high-current mode. The low-current mode features less than ±0.1% linearity error for a ±30 µA output current, and the high-current mode has less than ±0.2% linearity error for a current range of ±300 µA. The measured quiescent power consumption of the stimulation buffer was 251 µW and 306 µW in the low- and high-current mode, respectively, including the corresponding biasing circuitry. A single buffer occupies an area of 0.046 mm^2.

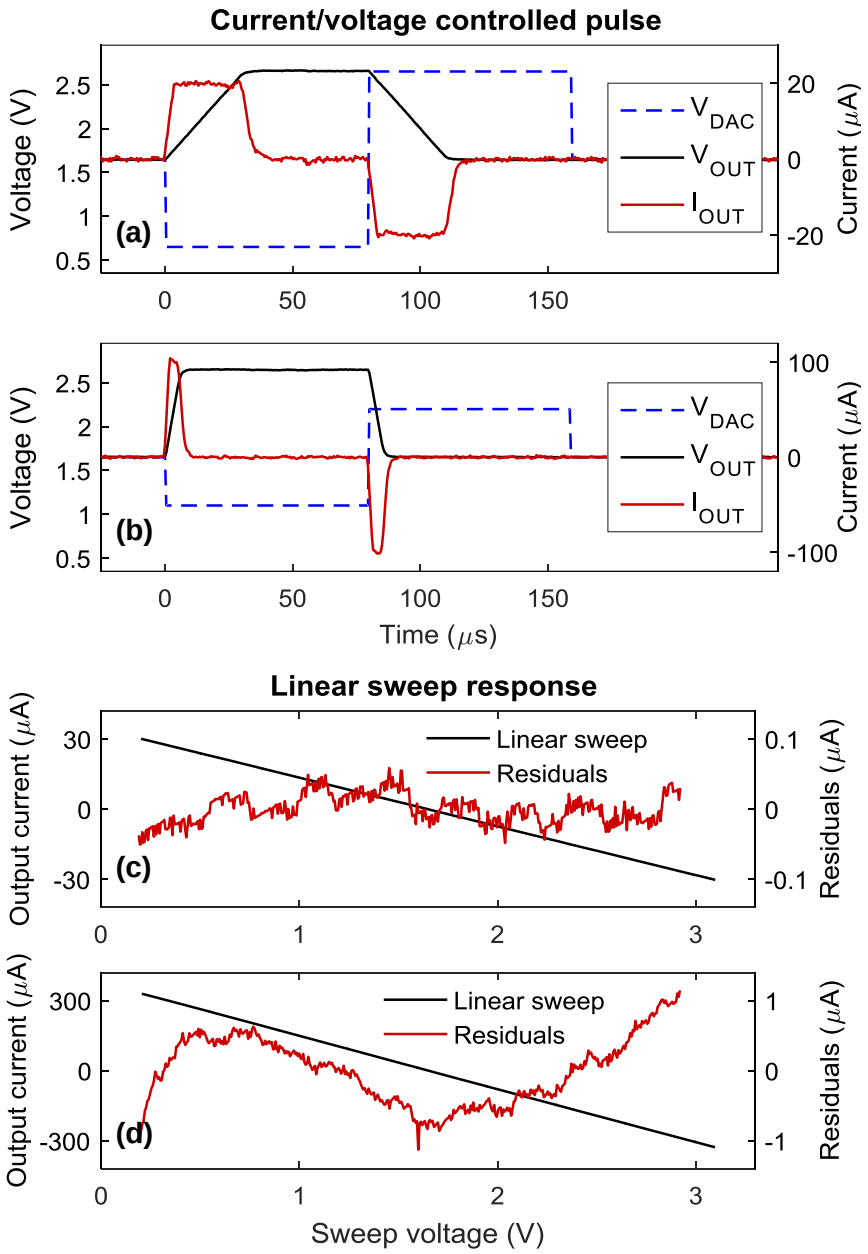


Fig. 7. Characterization of the stimulation buffer in the current/voltage controlled mode. Output response for the low-current mode (a) and the high-current mode (b) with the voltage limiters set to 2.65 V for V_{UP} and 1.65 V for V_{DN}. Linearity analysis for the low-current mode (c) and the high-current mode (d).

To evaluate the influence of the soft reset technique on the high-pass cutoff frequencies (f_{HPF}) of the first- and second-stage amplifiers, a 10 mV sinusoidal chirp signal from 0.1 Hz to 100 kHz was applied to the action potential (AP) recording channels, and the frequency responses of the first two amplification stages were recorded. Fig. 8 shows the measured transfer functions of the first two amplification stages, with and without activating the soft reset technique. The overall high-pass cutoff frequency was increased from 15 Hz to 4.23 kHz (Fig. 8) by the application of soft reset.

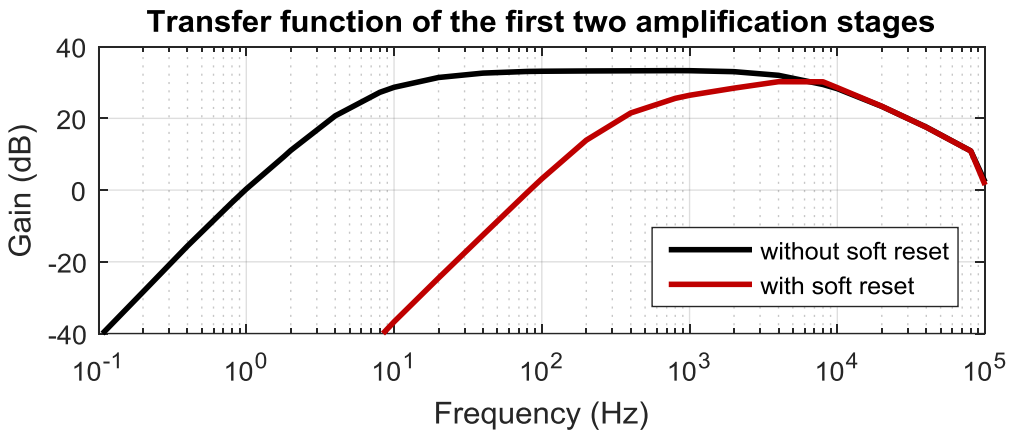


Fig. 8. Characterization of the transfer functions of the first two stages with/without applying the soft reset. Applying the soft reset increased the overall high-pass cutoff frequency from 15 Hz to 4.23 kHz.

3.6 Result of biological experiment

To elicit electrical activity of a single cell, the stimulation site should be close to the axon of the targeted cell [7]. For our experiments, a stimulation electrode at a distance of 192 µm (approx. 14 electrodes) from the cell soma of interest was selected. The application of stimulation pulses, with sufficiently large voltage and current amplitudes, to this electrode evoked APs in the targeted neuron. The recorded data are shown in Fig. 9.a and a consistently low jitter between the stimulation pulses and the recorded APs indicates that these APs were indeed evoked by the stimulation pulses. The recording electrode near the cell soma was far away from the stimulation electrode, the recording channel was not driven into saturation, therefore, artifact suppression was not required for this experiment. By decreasing the current or voltage amplitude of the stimulation pulse the probability of evoking an AP decreased as well (Fig. 9.b).

To study the stimulation probability for different current and voltage levels, a set of biphasic stimulation pulses was generated and applied to the stimulation electrode, and the response of the targeted neuron to these pulses was recorded. The current/voltage controlled stimulation buffer, configured in the low-current mode, was used to generate biphasic stimulation pulses at 4 different current settings of 0.7 µA, 1 µA, 3 µA, and 6 µA. The voltage amplitude of the biphasic pulses was configured to fall within a range of ±250 mV to ±500 mV and was increased in steps of 5 mV. For each current/voltage value 10 stimuli were applied and a stimulation probability was calculated as the percentage of successfully evoked action potentials. This stimulation protocol was repeated 5 times to obtain an average stimulation probability for each current/voltage value and a sigmoid curve was fitted to the data (Fig. 9.c).

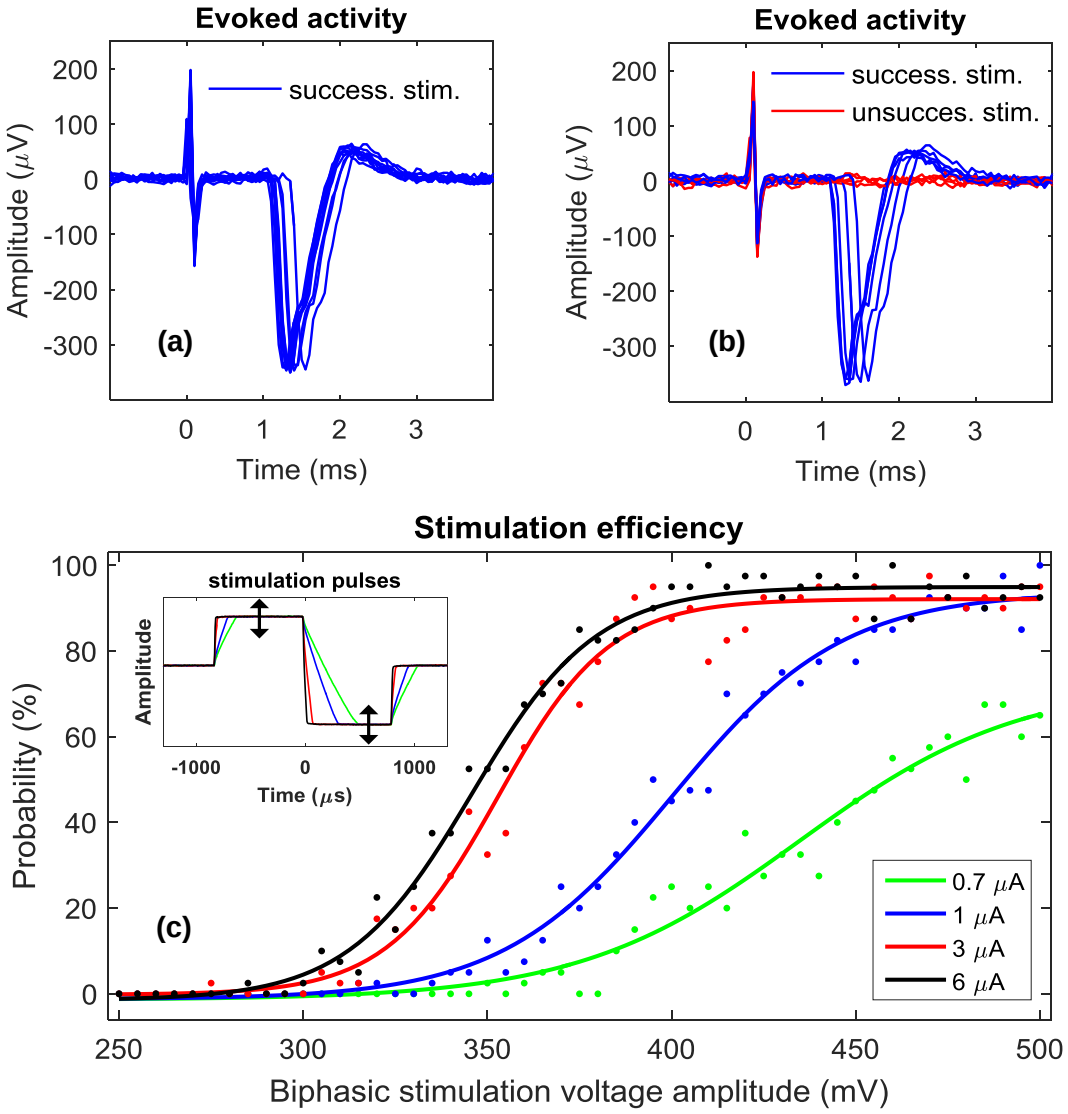


Fig. 9. Probability of evoked spikes for different current and voltage values: successfully evoked activity by applying stimulation pulses with sufficiently large voltage and current amplitudes (a), by decreasing the current or voltage amplitude of the stimulation pulse, the probability of evoked activity drops (b), comparison of stimulation efficiency for the applied biphasic pulses at 4 different current levels and voltage amplitudes from ±250 mV to ±500 mV in 5mV steps (c).

The data indicate that a larger voltage amplitude of the stimulation pulse was required to evoke action potentials at low output current settings. This was likely due to the increase of the rise/fall time of the pulse caused by lower current levels. When the current was large enough to cause a steep slope in the pulse, a further increase of the current did not significantly improve the stimulation efficiency any more. The stimulation efficiency, under these conditions, was solely dependent on the voltage amplitude of the biphasic pulse.

To test the effectiveness of the soft reset technique in suppressing stimulation artifacts, a ±400 mV biphasic stimulation pulse with 200 µs for each phase was applied to an electrode. The stimulation pulse was generated using the on-chip DACs, and the stimulation buffers were configured in the voltage-only controlled mode. The responses were recorded with and without application of the soft reset feature. Fig. 10.a shows that the soft reset technique could substantially reduce the dead time of the recording channels, which were connected to the electrodes surrounding the stimulation site. The close-ups (Fig. 10.b) show the response of one of these recording channels immediately before and after stimulation, with and without application of the soft reset. The soft reset was activated 200 µs before issuing the stimulation pulse and released 200 µs after the pulse. The recording channels were available for recording immediately after the release of soft reset. The exact activation and release period of soft reset depends on several parameters such as electrode impedance, stimulation pulse amplitude and gain/cut-off frequency of recording channels. Fig. 10.c shows two time series of recorded electrical images of the electrodes surrounding the stimulation site and affected by the stimulation artifacts. Responses are shown for a 13×11 electrode configuration with the stimulation electrode in the center. The location of the electrodes is marked with dots in the last image of each row. The top-row images were obtained from the measurements without applying the soft reset, whereas the bottom row shows measurements with the application of the soft reset. The top-row images show that the stimulation pulse affects electrodes within a radius of 67.5 µm (5 electrodes) and lasts for more than 10 ms. The bottom row shows the effectiveness of the soft reset technique in suppressing the stimulation artifacts. Although some offset was superimposed on the recorded response, the channels recovered quickly from saturation upon application of the soft reset. The remaining offset can be further reduced by digital filtering of the recorded data.

To evaluate the performance of the artifact suppression on neuronal recordings, two electrodes near an active neuron were selected for stimulation and recording (Fig. 10.d). Biphasic stimulation pulses (±450 mV) were generated using the stimulation buffers in the voltage-only mode. The responses of the selected recording electrode with and without application of the soft reset are illustrated in Fig. 10.e. The resulting large neuronal spikes are likely to be the post-synaptic action potentials, according to their delay with respect to the stimulation event. The consistency in the occurrence of the small activities (shown in the inset of Fig. 10.e) that could be recorded immediately after the stimulation evidenced the efficiency of the artifact suppression technique. Such small signals immediately after the stimulation would otherwise remain undetectable due to the saturation of recording channels.

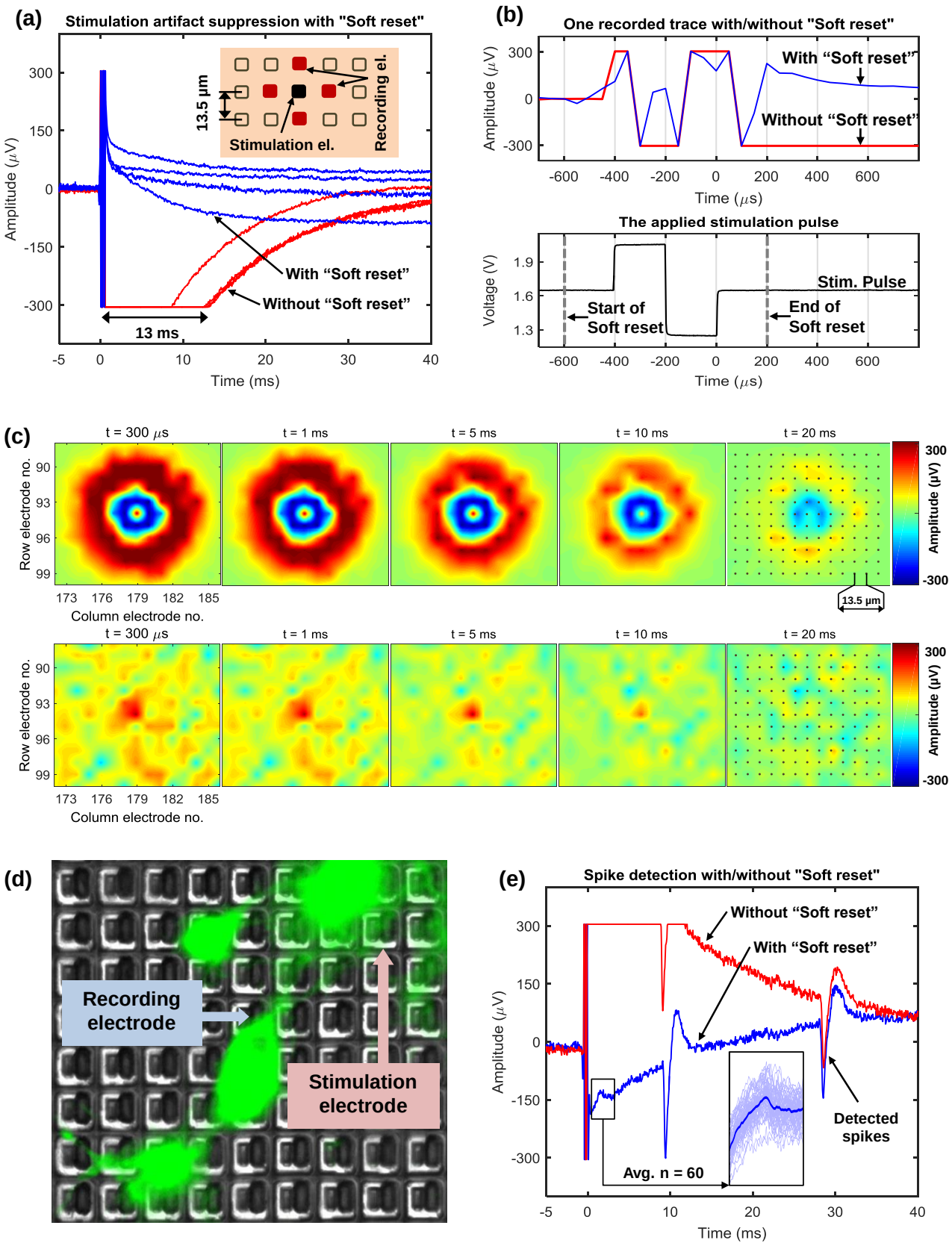


Fig. 10. Artifact suppression using the proposed soft-reset technique: recorded traces around a stimulation site with and without applying the soft reset (a). Zoom-in around the stimulation time (b). The soft reset starts 200 µs before the stimulation pulse and ends 200 µs after the pulse. Two time series of electrical images in response to a ±400 mV stimulation pulse (c). Top row: no artifact suppression was applied. The saturation of recording channels due to stimulation artifact is observed

within a radius of up to 5 electrodes (67.5 µm) away from the stimulation site. Bottom row: The soft reset technique was applied for artifact suppression. The last frame in each row depicts the locations of the recording electrodes (black dots). Location of neurons on top of the electrode array and positions of the selected electrodes for stimulation and recording around an active cell (d). Recorded spikes upon stimulation with and without application of the soft reset (e).

For all bio-measurements, dissociated cortical neurons (E18 Wistar rats) were cultured for 3-4 weeks on top of the MEA chips prior to the recordings. All experimental protocols were approved by the Basel-Stadt veterinary office according to Swiss federal laws on animal welfare and were carried out in accordance with the approved guidelines. The experiments were conducted under controlled environmental conditions inside an incubator at 37 ºC and 5% CO_2.

3.7 Conclusion

The CMOS circuits for electrical stimulation of electrogenic cells along with a stimulation artifact suppression technique have been presented. The circuits have been integrated into a novel multifunctional HD-MEA system. The stimulation buffers provided controlled current and voltage stimulation pulses with more than 9 bit linearity. The stimulation artifact suppression technique implemented in the recording amplifiers decreased the recovery time of the AP recording channels, connected to the electrodes adjacent to the stimulation site, to 200 µs. The performed biological measurements with cultured neurons showed the dependence of neuronal excitability on current and voltage levels of biphasic stimulation pulses. The possibility of applying controlled current and voltage stimulation pulses, along with the possibility to record neuronal responses immediately after stimulation and close to the stimulation site, provide excellent means to gain a better understanding of extracellular electrical stimulation mechanisms.

Acknowledgements

This work was supported by the European Community through the ERC Advanced Grants “NeuroCMOS” under contract number AdG 267351 and “neuroXscales” under contract number AdG 694829 as well as the Swiss National Science Foundation through Grant 205321_157092/1. Amir Shadmani received individual support through the EU Marie Curie Research Training Network “EngCaBra” (grant agreement no. 264417) and Milos Radivojevic received individual support through the Swiss National Science Foundation Ambizione Grant PZ00P3_132245. The authors would like to thank P. Livi for technical support and A. Stettler and A. Martel for post-processing of CMOS chips.

References

[1] M. E. J. Obien, K. Deligkaris, T. Bullmann, D. J. Bakkum, and U. Frey, “Revealing neuronal function through microelectrode array recordings,” Front. Neurosci., vol. 9, p. 423, 2015.

[2] A. Hierlemann, U. Frey, S. Hafizovic, F. Heer, "Growing cells atop microelectronic chips: Interfacing electrogenic cells in vitro with CMOS-based microelectrode arrays", Proc. IEEE, vol. 99, no. 2, pp. 252-284, Feb. 2011.

[3] L. Berdondini, K. Imfeld, A. Maccione, M. Tedesco, S. Neukom, M. Koudelka-Hep, and S. Martinoia, “Active pixel sensor array for high spatio-temporal resolution electrophysiological recordings from single cell to large scale neuronal networks.,” Lab Chip, vol. 9, no. 18, pp. 2644–2651, 2009.

[4] B. Eversmann, A. Lambacher, T. Gerling, A. Kunze, P. Fromherz, and R. Thewes, “A neural tissue interfacing chip for in-vitro applications with 32k recording / stimulation channels on an active area of 2.6 mm2,” Eur. Solid-State Circuits Conf., pp. 211–214, 2011.

[5] M. Ballini, J. Muller, P. Livi, et al., “A 1024-channel CMOS microelectrode array with 26,400 electrodes for recording and stimulation of electrogenic cells in vitro,” IEEE J. Solid-State Circuits, vol. 49, no. 11, pp. 2705–2719, 2014.

[6] V. Viswam, J. Dragas, A. Shadmani, Y. Chen, A. Stettler, J. Muller, and A. Hierlemann, “Multi-functional microelectrode array system featuring 59,760 electrodes, 2048 electrophysiology channels, impedance and neurotransmitter measurement units,” IEEE International Solid-State Circuits Conference (ISSCC), pp. 394–396, 2016.

[7] M. Radivojevic, D. Jäckel, M. Altermatt, J. Müller, V. Viswam, A. Hierlemann, D. Bakkum, "Electrical identification and selective microstimulation of neuronal compartments based on features of extracellular action potentials", Scientific Reports 6, art. 31332, 2016.

[8] H. Lodish, A. Berk, S.L. Zipursky, et al., Molecular Cell Biology, 4th Edition. New York: W. H. Freeman; 2000.

[9] D.R. Merrill, M. Bikson, J.G. Jefferys, “Electrical stimulation of excitable tissue: design of efficacious and safe protocols,” J. Neurosci. Methods, vol. 141, no. 2, pp. 171-198. 2005.

[10] S.F. Cogan, “Neural stimulation and recording electrodes,” Annu.Rev. Biomed. Eng., vol. 10, pp. 275–309, 2008.

[11] S. Joucla and B. Yvert, “Modeling extracellular electrical neural stimulation: from basic understanding to MEA-based applications,” J. Physiol. Paris, vol. 106, pp. 146–58, 2011.

[12] D.A Wagenaar, J. Pine, S.M. Potter, “Effective parameters for stimulation of dissociated cultures using multi-electrode arrays.,” J. Neurosci. Methods, vol. 138, no.1-2, pp. 27–37, 2004.

[13] C.R. Butson, C.C. McIntyre, “Role of electrode design on the volume of tissue activated during deep brain stimulation,” J. Neural. Eng., vol. 3, no. 1, pp. 1-8, 2006.

[14] S.F. Lempka, M.D. Johnson, S. Miocinovic, J.L. Vitek, C.C. McIntyre, “Current-controlled deep brain stimulation reduces in vivo voltage fluctuations observed during voltage-controlled stimulation.,” Clin. Neurophysiol., vol. 121, no. 12, pp. 2128–2133, 2010.

[15] M. van Dongen, S. Wouter, Design of Efficient and Safe Neural Stimulators: Springer, 2016.

[16] K. Sooksood, T. Stieglitz, and M. Ortmanns, "Recent advances in charge balancing for functional electrical stimulation," IEEE Conf. of Eng. in Med. and Bio. Society, pp. 5518-5521, Sept. 2009.

[17] D. Prutchi, M. Norris, Design and development of medical electronic instrumentation: a practical perspective of the design, construction and test of medical devices: Wiley, New Jersey, 2005.

[18] M. Ortmanns, "Charge balancing in functional electrical stimulators: A comparative study", Proc. IEEE Int. Symp. Circuits Syst., pp. 573-576, 2007.

[19] K. Sooksood, T. Stieglitz, M. Ortmanns, "An active approach for charge balancing in functional electrical stimulation", IEEE Trans. Biomed. Circuits Syst., vol. 4, no. 3, pp. 162-170, 2010.

[20] E. K. F. Lee, A. Lam, "A matching technique for biphasic stimulation pulse", Proc. IEEE International Symposium on Circuit and Systems, pp. 817-820, 2007.

[21] J. J. Sit, R. Sarpeshkar, "A low-power blocking-capacitor-free charge-balanced electrode-stimulator chip with less than 6 nA dc error for 1-mA full-scale stimulation", IEEE Trans. Biomed. Circuits Syst., vol. 1, no. 3, pp. 172-183, 2007.

[22] E. Noorsal, K. Sooksood, H. Xu, R. Hornig, J. Becker, M. Ortmanns, "A neural stimulator frontend with high-voltage compliance and programmable pulse shape for epiretinal implants", IEEE J. Solid-State Circuits, vol. 47, no. 1, pp. 244-256, Jan. 2012.

[23] H. Chun, N. Tran, Y. Yang, O. Kavehei, S. Bai, S. Skafidas, "A precise charge balancing and compliance voltage monitoring stimulator front-end for 1024-electrodes retinal prosthesis", Proc. Int. Conf. IEEE Eng. Med. Biol. EMBC 2012, pp. 3001-3004, 2012.

[24] C.S.A. Gong, H.Y. Lai, S.H. Huang, et. al., “A programmable high-voltage compliance neural stimulator for deep brain stimulation in vivo,” J. Sensors, vol. 15, no. 6, pp. 12700-12719, 2015.

[25] M. Ghovanloo, K. Najafi, "A compact large voltage compliance high output impedance programmable current source for biomedical implantable microstimulators", IEEE Trans. Biomed. Eng., vol. 52, no. 1, pp. 97-105, Jan. 2005.

[26] E. A. Brown, J. D. Ross, R. A. Blum, Y. Nam, B. C. Wheeler, S. P. DeWeerth, "Stimulus-artifact elimination in a multi-electrode system", IEEE Trans. Biomed. Circuits Syst., vol. 2, no. 1, pp. 10-21, Mar. 2008.

[27] Hottowy, P., Skoczen, A., Gunning, D. E., Kachiguine, S., Mathieson, K., Sher, A., et al. “Properties and application of a multichannel integrated circuit for low-artifact, patterned electrical stimulation of neural tissue.,” J. Neural Eng., vol. 9, no. 6, 066005. 2012.

[28] F. Heer, S. Hafizovic, T. Ugniwenko, U. Frey, W. Franks, E. Perriard, J.-C. Perriard, A. Blau, C. Ziegler, A. Hierlemann, "Single-chip microelectronic system to interface with living cells", Biosens. Bioelectron., vol. 22, pp. 2546-2553, May 2007.

[29] U. Frey, J. Sedivy, F. Heer, R. Pedron, M. Ballini, J. Mueller, D. Bakkum, S. Hafizovic, F. D. Faraci, F. Greve, K.-U. Kirstein, A. Hierlemann, "Switch-matrix-based high-density microelectrode array in CMOS technology", IEEE J. Solid-State Circuits, vol. 45, no. 2, pp. 467-482, Feb. 2010.

[30] J. A. Galan, A. J. Lopez-Martin, R. G. Carvajal, J. Ramirez-Angulo, C. Rubia-Marcos, "Super class-AB OTAs with adaptive biasing and dynamic output current scaling", IEEE Trans. Circuits Syst. I Reg. Papers, vol. 54, no. 3, pp. 449-457, Mar. 2007.

[31] P. Livi, F. Heer, U. Frey, D. J. Bakkum, A. Hierlemann, "Compact voltage and current stimulation buffer for high-density microelectrode arrays", IEEE Trans. Biomed. Circuits Syst., vol. 4, no. 6, pp. 372-8, Dec. 2010.

[32] V. Viswam, Y. Chen, A. Shadmani, J. Dragas, R. Bounik, R. Milos, J. Müller, A. Hierlemann, “2048 action potential recording channels with 2.4μVrms noise and stimulation artefact suppression”, Proc. IEEE Biomedical Circuits and Systems Conference (BioCAS), pp. 136-139, Oct. 2016.

[33] A. Tajalli and Y. Leblebici, Extreme Low-Power Mixed Signal IC Design: Springer New York, 2010.

CHAPTER 4

Sensor System including Silicon Nanowire Ion Sensitive FET Arrays and CMOS readout

Paolo Livi[1], **Amir Shadmani[1]**, Mathias Wipf[2], Ralph L. Stoop[2], Jörg Rothe[1], Yihui Chen[1], Michel Calame, Christian Schönenberger[2] and Andreas Hierlemann[1]

[1] ETH Zurich, Bio Engineering Laboratory, Department of Biosystems Science and Engineering, Basel, Switzerland

[2] Department of Physics, University of Basel, Basel, Switzerland

Published in:

Journal of Sensors and Actuators B: Chemical

Volume 204, 1 December 2014, Pages 568–577

Keywords

Silicon nanowires, ISFET, chemical sensors, CMOS, integrated current readout circuits

4.1 Abstract

We present a highly sensitive chemical sensor system comprising a chip with an array of silicon nanowire ISFETs and a CMOS chip with custom-designed signal-conditioning circuitry. The CMOS circuitry, comprising 8 sigma-delta (Σ-Δ) modulators and 8 current-to-frequency converters, has been interfaced to each of the nanowires to apply a constant voltage for measuring the respective current through the nanowire. Each nanowire has a dedicated readout channel, so that no multiplexing is required, which helps to avoid leakage current issues. The analog signal has been digitized on chip and transmitted to a host PC via a FPGA. The system has been successfully fabricated and tested and features, depending on the settings, noise values as low as 8.2pA RMS and a resolution of 13.3 bits while covering an input current range from 200 pA to 3 μA. The two readout architectures (Σ-Δ and current to frequency) have been compared, and measurements showing the advantages of combining a CMOS readout with silicon nanowire sensors are presented: 1) simultaneous readout of different silicon nanowires for high-temporal-resolution experiments and parallel sensor experiments (results from pH and KCl concentration sweeps are presented); 2) high speed measurements showing how the CMOS chip can enhance the performance of the nanowire sensors by compensating its non-idealities as a consequence of hysteresis.

4.2 Introduction

Silicon nanowire (SiNW) Field Effect Transistors (FETs) have recently received great attention for their potential with regard to label free detection of biological and chemical species. Owing to their high surface-to-volume ratio and quick response, they are promising candidates for highly sensitive sensors. SiNWs have, for example, been successfully used for the detection of DNA [1], proteins [2], for pH measurements [3], and for recording the activity of electrogenic cells, such as neurons [4].

The detection principle of a SiNW-based sensor is similiar to that of an Ion Sensitive Field Effect Transistor (ISFET), a well-known device first introduced in 1970 by Bergveld [5].

A SiNW FET, as well as an ISFET, can be considered as a normal MOS transistor, whose metallic gate is replaced by a liquid solution and a reference electrode controlling the potential of the liquid. A sketch of such a SiNW FET is shown in Fig. 1a, with its cross section displayed in Fig. 1b (further details will be described in the following sections). The charges coming from, e.g., ions, proteins, DNA, dispersed in the liquid solution accumulate at the surface of the gate oxide on top of the SiNW. This charged layer induces a surface potential change that can modulate the current flowing through the NW, in the same way as a gate voltage modulates the current flowing through the channel of a MOS transistor. The surface potential is therefore depending on the amount of charges in the liquid solution, and thus on the analyte concentration. Since the surface potential is directly affecting the threshold voltage of the transistor, a change in the analyte concentration causes a shift of the threshold voltage: for a p-channel device (having negative threshold voltage), an increase in positive charge on the surface will reduce the threshold voltage and will lead to a decrease in the current flowing through the channel for a defined sensor geometry and defined electrochemical conditions (reference electrode voltage).

An equation showing the relation between surface potential, Ψ, and the threshold voltage, Vth, of an ISFET was first introduced in [6]:

$$V_{th} = V_{\mathrm{lg}} - \Psi + \chi^{sol} - \frac{\Phi_{Si}}{q} - \frac{Q_{ox} + Q_{SS} + Q_B}{C_{ox}} + 2\phi_f \qquad (1)$$

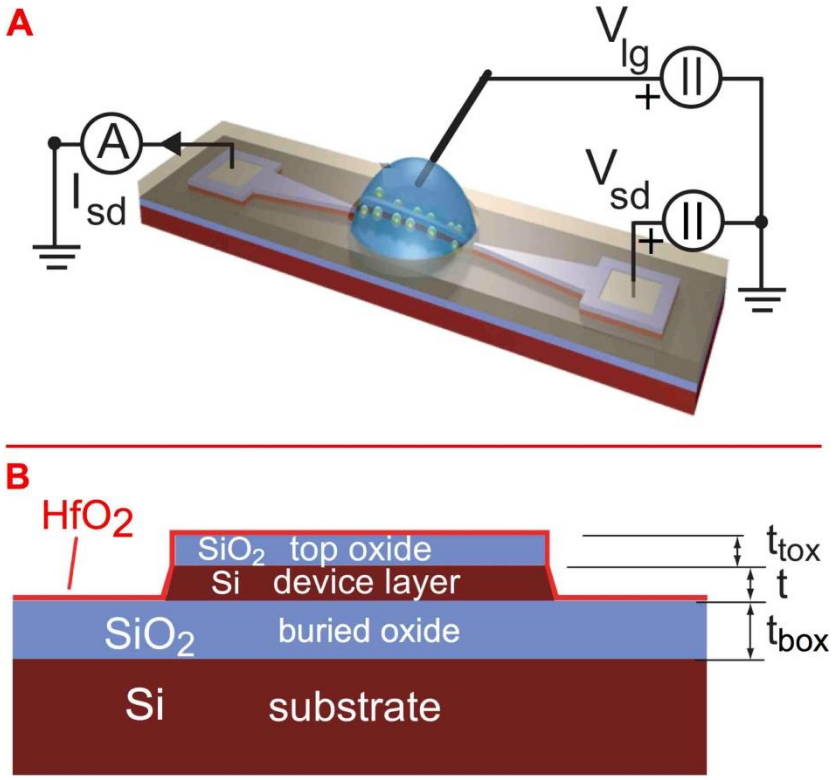


Fig. 1. (a) Sketch of a SiNW FET sensor and its electrical connections: the source-drain voltage (V_{sd}) is clamped, while the current I_{sd} is measured; a voltage V_{lg} is applied to the reference electrode in the liquid solution. (b) Cross section of the SiNW FET sensor including key parameters: buried oxide thickness, t_{box} = 145 nm; silicon device layer, t = 85 nm; top oxide layer, t_{tox} = 15 nm; layer of HfO_2 with thickness of 8 nm.

where Vlg is the voltage applied at the reference electrode, χsol is the surface dipole potential of the solvent, ΦSi is the work function of the silicon; the fifth term represents the charge accumulated in the oxide (Qox), at the oxide-silicon interface (QSS) and the depletion charge (QB); the last term is an indicator of the onset of inversion depending on the dopant concentration. The parameter χsol is a parameter of the liquid solution, whereas all the other terms characterize the device itself. The signal that is transduced from the chemical to the electrical domain is represented by changes in Ψ, the surface potential, which - as already mentioned - will directly be detectable as threshold voltage changes (for a given Vlg). Even though this physical description was originally developed for ISFETs, it can be also used to describe the behavior of SiNW FETs under certain conditions [7].

One important issue in the design of SiNW FET sensors is an efficient readout of the sensor output signal, which is normally rather weak and noisy. Most of the work cited so far [1-4] focus on the fabrication of the NW sensors and on their use. The measurements have been done by using bulky lab equipments, which, however, can limit the sensor performance.

A better solution is to use readout circuits fabricated in CMOS technology. The main advantage is the possibility of having the sensors and readout as physically close together as possible, potentially even on the same substrate. Moreover, the use of CMOS technology brings other advantages: many sensors can be simultaneously read out in parallel; sensor signals can be digitized right after the measurement to take advantage of the high robustness and flexibility of digital circuits; non-idealities of the sensor can be compensated in order to improve its performance. The CMOS approach is very powerful for integrating NW-based sensors into portable point-of-care devices that, e.g., can simultaneously detect different analytes in a solution or body fluid.

At the circuit level, a common mesurement approach for SiNW FET sensors is to keep the voltage across the NW (Vsd) constant, to fix the voltage at the reference electrode (Vlg), and to measure the current flowing through the device (Isd), as sketched in Fig. 1a. In such configuration, changes in the threshold voltage will directly be detected as changes in the current: the resistance or conductance of the NW is measured. Extensive work has been done to design CMOS current readout circuits, many of which are not specifically suitable for NW-based sensors. Here we only briefly review some typical architectures.

A well-known challenge in designing CMOS circuits is to find the best trade-off among noise performance, bandwidth, input current range, power consumption and needed silicon real estate. The last two points become especially important for designing readouts for sensor arrays. Meanwhile, the input sensor current range can be quite large (several orders of magnitude, from few hundreds of pA to few µA), and it is, therefore, not easy to realize circuits with extremely low noise (few pARMS or less) for such wide input current ranges.

A popular current readout circuit architecture is the transimpedance amplifier (TIA), due to its low noise and bandwidth of several kHz that can be achieved (noise of few fARMS have been reported [8, 9]). On the other hand, TIAs can be rather bulky and they also require a dedicated analog-to-digital converter (ADC) to digitize the output voltage resulting from the measured current. Integrated versions of lock-in amplifiers have also been presented [10, 11], which provided good noise performance again at the cost of area and complexity.

Other architectures previously used as ISFET readouts include circuits able to directly measure the threshold voltage changes [6] or circuits that incorporate the sensor itself as part of the readout circuit in so-called sensing circuits [12]. These architectures either use complex circuits or are technologically challenging (in terms of monolithic integration) when applied to SiNWs.

For the work presented in this paper we focused on two architectures also widely used in the chemical and bio-sensor field: sigma-delta (Σ-Δ) modulators and current-to-frequency (ItoF) converters. Both topologies have limited bandwidth (only suitable for slow varying signals) but are fast enough for most applications with SiNW

sensors. Moreover, Σ-Δ and ItoF converters have the advantage of sensing and digitizing the current at the same time. Σ-Δ modulators have been demonstrated to feature low noise levels and high resolution [13, 14], while they have a high degree of complexity and are difficult to design. ItoF converters, on the other hand, are rather simple and scalable circuits and thus suitable for arrays [15]; moreover, they can cover a wide input current range [16, 17], while noise levels and resolution are sometimes not as good as those of Σ-Δ topologies.

Most of the cited work describes the CMOS circuits or their use in combination with micro and nano electrode arrays. Only few publications to date reported the integration of a CMOS readout chip and a SiNW sensor array [18-20]. Most of these systems feature one single readout channel that is multiplexed to several NWs [18, 20]. The multiplexing approach suffers from leakage currents (up to few pA [18, 20]) coming from the switches that have been used to connect each of the NW sensors to the readout circuit; multiplexing also does not allow for a simultaneous readout of different sensors. In addition, previous works [18-20] measure mostly the NW conductance change. Although conductance measurements may be sufficient for most applications, it is important to remember that the characteristic sensing effect in SiNWs is related to the threshold voltage shift. Non-idealities of the SiNW sensor, e.g., hysteresis, may induce unwanted threshold voltage shifts so that measured conductance changes may be caused by irrelevant sensing event.

We present a nanowire sensor system, featuring (i) a CMOS readout chip and a SiNW sensor array; (ii) each NW sensor has a dedicated readout channel; (iii) the CMOS chip includes both Σ-Δ modulators and ItoF converters in order to compare and discuss their performance and potentials as SiNW sensor readout circuits; (iv) simultaneous recordings from several NWs at high temporal resolution and the corresponding statistics are shown; (v) the capability of the CMOS readout chip to compensate sensor non-idealities, such as hysteresis, will be demonstrated.

4.3 System description

A schematic view of the overall system presented in this paper is shown in Fig. 2. The system comprises a chip with the SiNW FET sensor array, three CMOS readout chips to interface all of the SiNWs and an FPGA for data processing and communication to a host PC through a USB connection. All these blocks have been assembled on a custom-designed printed circuit board (PCB), which also provides analog reference voltages and digital settings to configure the readout circuits. Moreover, the PCB includes two circuits realized with off-shelf IC chips that convert the 5V supply from the USB connection into two separate and low-noise 3.3V voltages that are used to power the analog and digital circuits in the CMOS chips. Each block is described in more detail in the following sections.

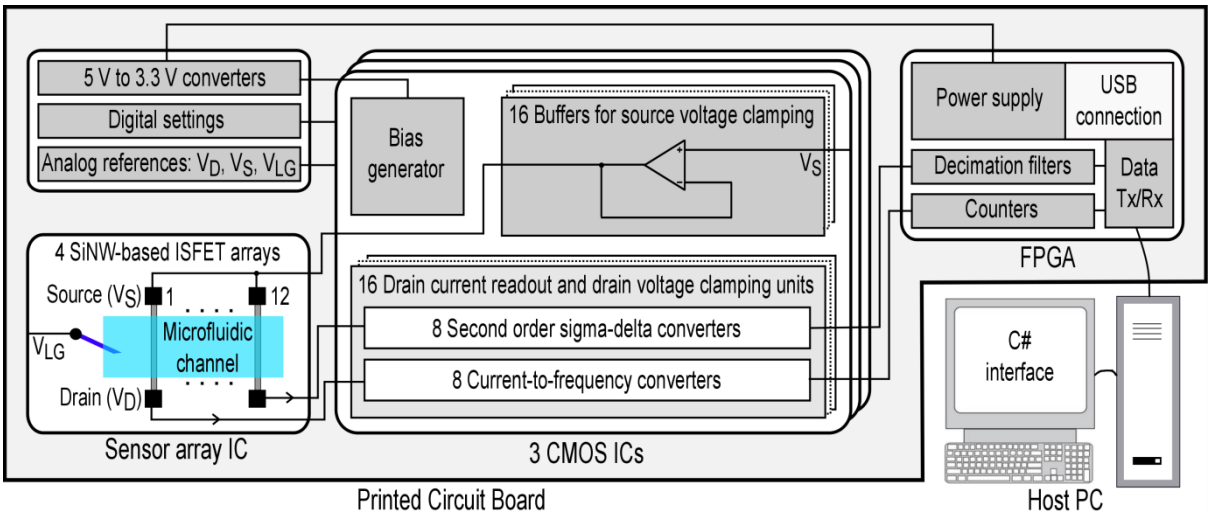


Fig. 2. Building blocks of the overall sensing system and circuitry units of the CMOS readout chip; the nanowire sensor array is interfaced to 3 CMOS chips whose outputs are processed and then transmitted to a host PC by a FPGA. All the components are connected via a printed circuit board (PCB). The acquired data are visualized and stored on the host PC with the help of a C# software interface.

4.4 Silicon nanowire FET sensor array

The SiNW FET sensor array was fabricated according to the procedure previously reported in [21, 22]. A silicon-on-insulator (SOI) wafer from Soitec, France, was structured in a top-down approach. A cross section of such a NW device is shown in Fig. 1b: the thickness of the buried oxide layer, t_{box}, is 145 nm, the thickness of the Si device layer, t, is 85 nm (p-doped with a resistivity of 8.5-11.5 Ωcm). Initially, the Si device layer was thermally oxidized to achieve a SiO_2 layer with a thickness, t_{tox}, of 15 nm. The NW pattern was defined by electron beam lithography and transferred to the SiO_2 layer by reactive ion etching. The structured SiO_2 layer was used as an etching mask for the wet chemical etching of the Si device layer performed in tetra-methyl ammonium hydroxide (TMAH) with 10 vol% isopropanol (IPA) at 45°C. The resulting SiNWs with (111) side faces were 6µm in length, 85nm in height (t) and featured different widths ranging from 100 nm to 1000 nm.

The source and drain contacts were implanted with BF_2^+ ions at a dopant dose of $2.3 \cdot 10^{15}$ cm^{-2} and an energy of 33 keV. Thermal annealing in forming gas (6min at 950°C) was performed to activate the dopants. The top oxide layer was removed in buffered HF. The RCA standard cleaning procedure was used for cleaning the samples from organic and metallic contaminants. Atomic layer deposition (ALD) was employed to cover the sample with a thin layer (8 nm) of HfO_2 at 200°C (Savannah S100, Cambridge NanoTech). The structures of the source and drain contacts were defined by optical lithography, and the ALD layer was then etched away at the contacts in buffered hydrofluoric acid (BHF). The metallization of the source and drain contacts was done with Al-Si (1%). The formation of Ohmic contacts was completed by a thermal annealing step at 450°. To ensure leakage-free operation in

liquid, a 2 µm thick protection layer (SU-8 2002, MicroChem) with 6 µm wide openings (liquid channels) was defined by optical lithography. Finally, the samples were wire-bonded onto a chip carrier (see Fig. 3a), and the bond wires were sealed with epoxy (Epotek 353ND).

Each chip includes four sub-arrays with 12 nanowires. Each sub-array has a shared source contact. A picture of a packaged chip with a zoom-in on the NW area and on some NWs (SEM picture) is shown in Fig. 3a.

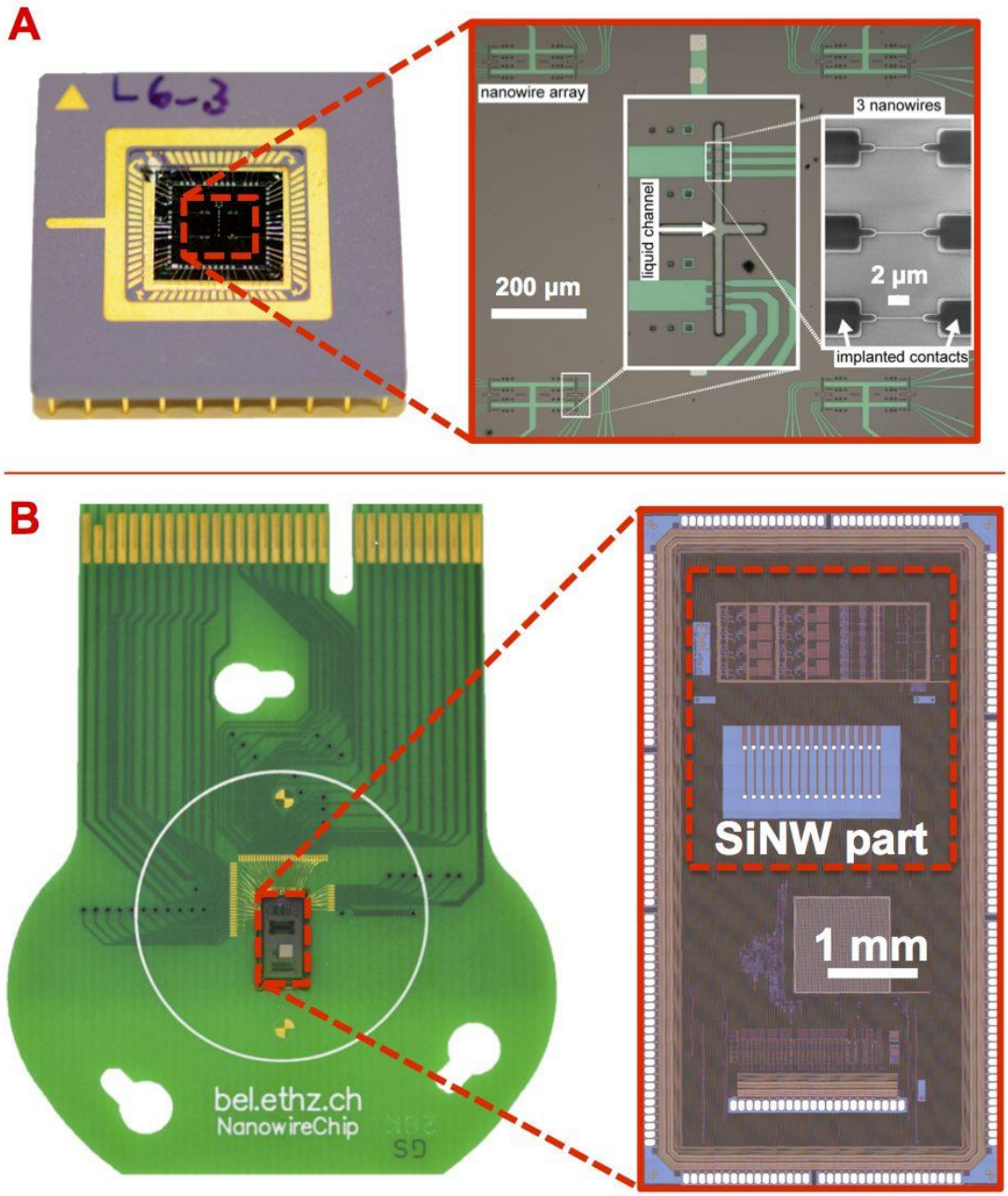


Fig. 3. (a) On the left side, a packaged chip containing the SiNW sensor array; on the right side, a micrograph of the four SiNW sub-arrays of the chip; the insets show (from left to right) a zoom-in on one of the sub-arrays with highlighting of the microfluidic channels, and a SEM picture of three SiNWs. (b) On the left side, a PCB designed to host the CMOS chip and to connect it to the overall system; on the right side, a micrograph of the fabricated CMOS chip (the circuitry for the SiNWs is highlighted at the top; the bottom part features a set of circuits used for another project).

4.5 CMOS readout chip

A schematic view of the CMOS chip is shown in Fig. 2. The circuitry includes 16 voltage buffers to define the voltage at the source terminal of each nanowire, and 16 current readout units (8 Σ-Δ modulators and 8 ItoF converters) that define the voltage at the drain terminal of each nanowire while sensing and digitizing its current values. An on-chip biasing block was also implemented.

Each voltage buffer is an operational amplifier, configured in unity-gain feedback, which can provide voltages between 500 mV and 2.5 V. The used operational amplifier has a class-AB output stage, providing good driving capabilities, i.e., loads as big as 10 nF can be driven, at low power consumption. Stray capacitances resulting from the wire bonding and the PCB tracks that connect the CMOS chip to the SiNW chip will not lead to instability of the voltage buffer.

Each Σ-Δ modulator features a second-order architecture with two switched-capacitor integrators. The circuit schematic is shown in Fig. 4a. The 1-bit current-mode digital-to-analog converter (DAC) is based on switched-capacitors; this topology allows for a reduction of dead zones, thereby increasing the linearity of the circuit, and has good immunity to parasitic capacitances. Further details can be found in [23].

Each ItoF converter is based on the architecture previously proposed in [15] and comprises three main blocks: 1) an input current conveyor; 2) an integrating capacitor; 3) a continuous time comparator. The circuit schematic is shown in Fig. 4b. Together with an external counter (realized with an FPGA as explained in the next section), the ItoF converter forms a complete ADC. The current conveyor buffers the current flowing from the drain terminal of the nanowires where the voltage is fixed due to the low input impedance of the conveyor. The operational amplifier used in the current conveyor is designed with large gain to ensure higher precision for the current conveyor.

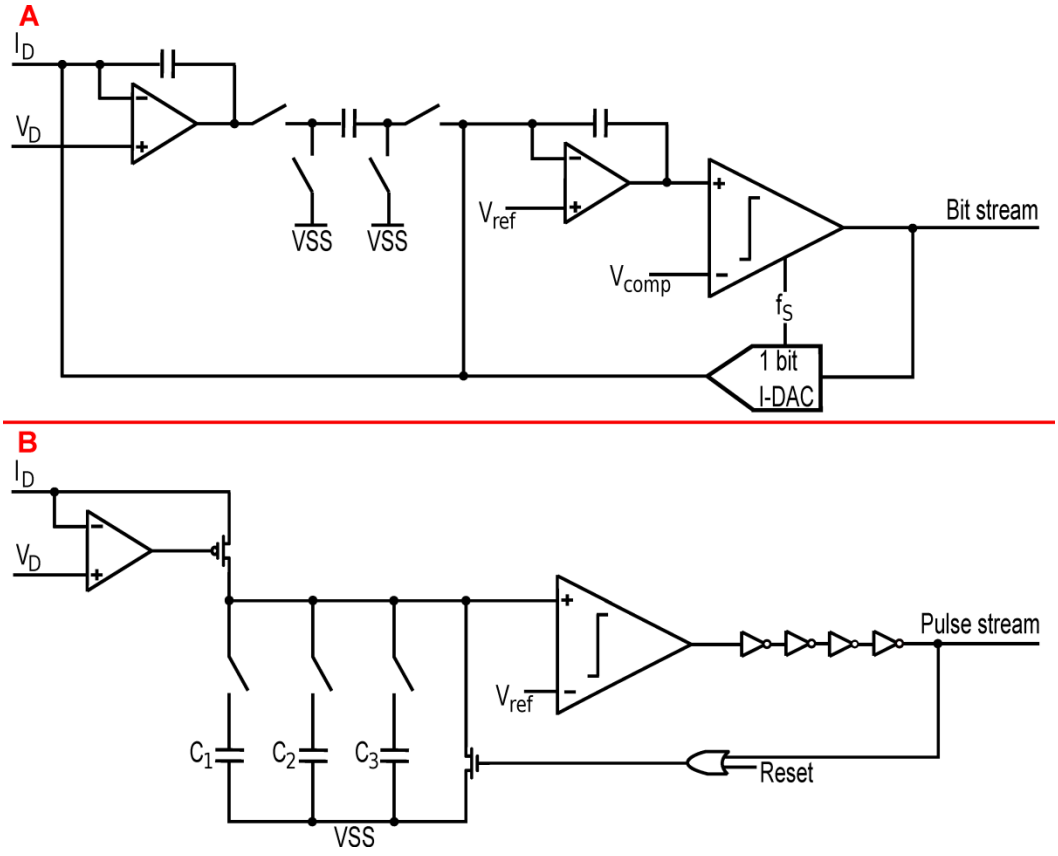


Fig. 4. (a) Circuit schematic of the second-order sigma-delta converter. (b) Circuit schematic of the current-to-frequency converter.

The current is then integrated on a capacitor and converted to a voltage. This voltage is compared to a reference voltage, V_{ref} by the comparator: each time it crosses V_{ref}, a reset pulse is created, and the integrating capacitor is discharged; after reset, the current integration starts again. As a result, a train of output pulses is generated, all having the same time duration: the frequency of this sequence of pulses encodes the input measurement signal. Neglecting all parasitic terms, i.e., parasitic capacitances, leakage currents, comparator delay, the output frequency, f_{out}, is proportional to the input current to be measured, I_{in}, as given by the equation:

$$f_{out} \approx \frac{I_{in}}{C_{\text{int}} V_{ref}} \tag{2}$$

where C_{int} is the integrating capacitor. Three different integrating capacitors were included on the chip and can be adjusted via digital switches according to different current ranges: 200 fF (for the range 200 pA to 400 nA), 1pF (for the range 400 nA to 1.4 µA) and 10 pF (for the range 1.4 µA to 3 µA). The current ranges were chosen in order to achieve a uniform resolution of about 11bits over the whole input current range: by having three different integrating capacitors, the range of the measured input current can be extended without losing in resolution, which renders the system suitable for a wider range of applications. If only the smallest capacitor, i.e., 200 fF, were used for the whole current range (up to 3 µA), errors would have appeared in the measured frequency, because the delay of the comparator would start to

significantly contribute, and fractions of the input current would not be correctly integrated anymore [15]. Such errors would finally appear as an increased noise level in the measured current.

After the generation of an output pulse an inverter chain is extending the time duration of each pulse (about 30ns in this design) in order to fully discharge the integrating capacitor. Moreover, a reset phase has been implemented so that any unwanted charge that might still be in the integrating capacitor can be removed before each measurement to avoid unprecise conversions.

The chip has been fabricated in a 0.35-µm CMOS technology. Its total area is 3.4 mm x 4 mm. The chip has been wire bonded to a custom designed PCB that allows for connection of the chip to the main PCB and has then been covered with epoxy to protect the bond wires. Fig. 3b shows a picture of the fabricated chip, wire-bonded to the PCB, and a micrograph of the chip itself on the right side. The chip includes other test circuitry, which was used in other projects. The circuitry for the SiNWs is highlighted in Fig. 3b.

4.6 FPGA and software

The outputs of the Σ-Δ modulators and ItoF converters are acquired by an FPGA (as sketched in Fig. 2). A Xilinx FPGA (XC3S1600E) was used in this work. As already mentioned, the output of the ItoF converter is a sequence of pulses, with the same duration, whose frequency encodes the input signal. A simple digital counter is thus sufficient to extract the relevant information. Since the pulses are very short in duration, a 125 MHz clock oscillator was used to generate the clock of the FPGA. The system works in two modes: a slow-mode with a conversion period of 100 ms, and a fast-mode with a conversion period of 1 ms.

The output of the Σ-Δ modulator is a train of pulses, the length of which is different. The pulses are sampled at a fixed frequency (1 MHz), and the input is encoded in the pulse width. A decimation filter is needed to extract the relevant information.

The processed data are then sent to a host PC via USB by using a programmable, low-power USB 2.0 peripheral controller. The power supply is also provided with the same USB port.

A Graphical User Interface (GUI) was developed by using the programming language C#. The developed program allows for real-time plotting and recording of the received data on the host PC. The data are saved in a file using a high-priority thread. The real-time plots are updated every 1 s.

4.7 Results and discussions

4.7.1 Electrical characterization

First, an electrical characterization of the CMOS chip alone was carried out. The measurements were done by using a test channel in the chip that can be directly connected to an external current source. In the same way, a three-point calibration was performed for the ItoF converters in order to improve the accuracy of the conversion. According to eq. (2), the ratio between the measured frequency and current is anti-proportional to the integrating capacitance. However, as already mentioned, parasitics (mostly the parasitic capacitance at the input of the comparator) can introduce errors. Using a simple calibration scheme this kind of error can be drastically reduced. A summary of some important parameters is shown in Table 1.

Table 1. Summary of the main results of the electrical characterization of the CMOS chip. More details about the sigma-delta converter can be found in [23].

	Slow mode (t_{int} = 100ms)			Fast mode (t_{int} = 1ms)			Sigma Delta
C_{int}	200fF	1pF	10pF	200fF	1pF	10pF	
Current range (nA)	0.2 - 400	400-1400	1400-3000	0.2 - 400	400-1400	1400-3000	± 2000
SNDR (dB)	67.6	67.2	68.8	45.4	41.3	45.4	81.8
ENOB	10.9	10.9	11.1	7.3	6.6	7.3	13.3
THD (dB)	-69.3	-68.1	-70.7	-50.4	-48.2	-49.1	-80.2
Total RMS noise (pA)	8.2	10.5	38	64	74	164	250
Size (μm²)	490 x 105						660 x 203
Power consumption (mW)	0.228						1.2

The ItoF converter achieves a resolution of about 11 effective number of bits (ENOBs) (in the slow-mode) and of 7 ENOBs (in the fast-mode) in the current range from 200 pA to 3 μA. The highest measured noise in the slow-mode is 38 pA_{RMS} with an integrating capacitor of 10 pF and input current of 3 μA. The noise decreases to only 8.2 pA for a smaller input current (50 nA) by using the 200 fF integrating capacitor. To verify the design considerations described before, the RMS noise was also measured for an input current of 3 μA by using the small integrating capacitor (200 fF): the value was 98 pA, which confirms the need of bigger integrating capacitors when low noise levels and wide input current ranges are required. As it can be expected, the noise performance decreases in the fast-mode. However, the measured RMS noise values are still significantly smaller than most current changes due to sensing events (see next section). Moreover, the slow-mode is the most commonly used mode, since the SiNWs are mostly biased with DC. An application of the fast mode is shown in the next section, where it can also be seen that an increased noise level may not impose limits on the experiment.

The Σ-Δ modulator provides a much higher resolution (more than 13 bits) in a very wide input current range (±2 μA) at a signal bandwidth of 1 kHz; it requires, however, a 1 MHz clock and an oversampling ratio of 256. The RMS noise level increases to 250 pA. Lower noise levels can be achieved by trading bandwidth with input current range; as an example, the RMS noise decreases to 16.8 pA in the ±100 nA range at a bandwidth of 1 kHz, whereas the RMS noise is only 220 fA in the smallest range of ±100 pA at a bandwidth of 100 Hz [23].

The noise of the SiNW FETs used for this paper was previously characterized, and the results described in [24] show that the best noise characteristics of roughly 10 pA_{RMS} (in a 100 Hz bandwidth) can be achieved with SiNWs having a width of 1μm. It has been also shown that the noise decreases as the square root of the SiNW area increases, i.e., the product of the width and length. Since the SiNWs can have widths as small as 100nm, the noise level can increase to about 40 pA_{RMS}. The results presented in the next section show that currents in the range of few nA up to few hundreds of nA were measured using the SiNWs. In this current range the noise of the CMOS readout (in slow-mode) is around 10 pA_{RMS} (see above and Table 1) or lower, meaning that the CMOS chip and circuitry is not limiting the performance of the overall sensor system.

The maximal measured power consumption of the CMOS chip was 35 mW.

4.7.2 Experimental results

The SiNW array was first used to measure different pH values. This experiment does not require a chemical functionalization of the NW surface, since the pH sensitivity comes from the OH groups of the oxide surface: at high pH values (>7) the OH groups get deprotonated, and provide negative charges on the gate, whereas at low pH values (<7) the OH groups get protonated and provide positive charges on the gate. Since the SiNWs used in this work are p-channel devices, pH-values larger than 7 will increase the threshold voltage (and the current I_d for a fixed V_{lg}), whereas pH-values lower than 7 will decrease the threshold voltage (and the current I_d for a fixed V_{lg}).

As a consequence, there are two ways to perform a pH measurement: 1) monitoring the NW threshold voltage; 2) monitoring the current flowing through the NW. For the first method, the $I_d(V_{lg})$ curve of the NW FET was acquired for each pH value: V_{lg} was swept, while I_d is measured. The voltage across the NW (V_{sd}) was kept constant (in all experiments of this work, this voltage was fixed to 1 V). The voltage at the reference electrode (a standard Ag/AgCl electrode) was applied through a DAQ card controlled by a LabView interface. pH-measurement results using this method are shown in Fig. 5, where the measured current for different pH values (3 to 10) has been plotted on a logarithmic scale as a function of the applied voltage at the reference electrode. The curves resemble typical $I_d(V_g)$ curves of a MOSFET showing that the electrical characteristics of fabricated SiNWs are similar to normal MOSFETs. There is an almost linear shift of the $I_d(V_{lg})$ curves obtained at the

different pH values (~50 mV/pH), which confirms the theory laid out above and demonstrates the sensing capability of the fabricated SiNWs.

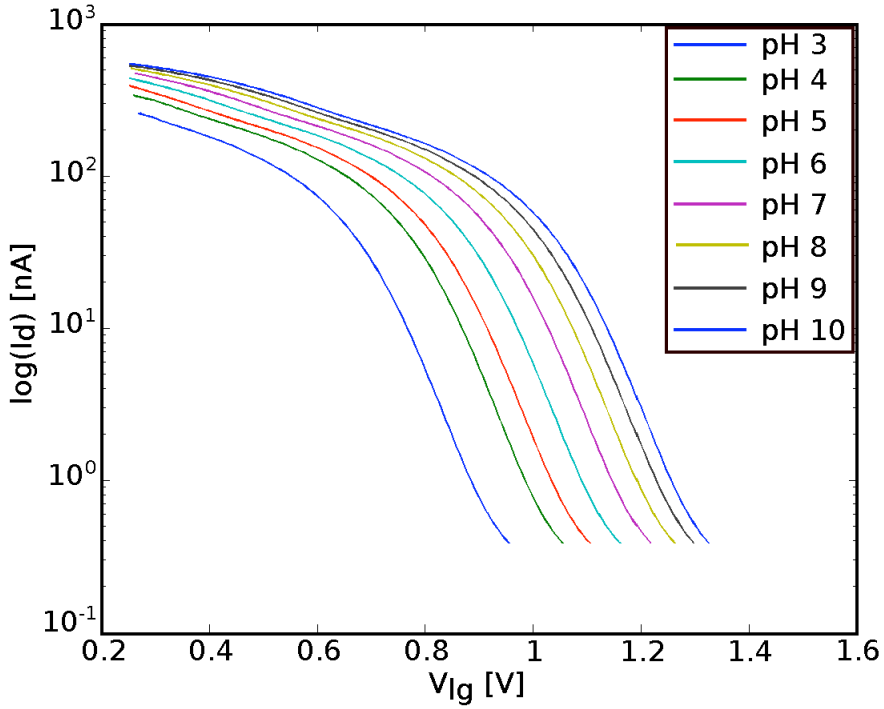


Fig. 5. Measured currents (plotted in log scale) of one of the SiNW FETs as a function of the voltage applied to the reference electrode in the liquid solution ($I_d(V_{lg})$ curve) and for different pH values. The plot shows the threshold voltage shifts induced by the pH changes.

For the second measurement method, V_{lg}, also has been kept constant, and I_d, was continuously measured over time while the pH-values of the solution have been varied. For this method, V_{lg} has been chosen according to the obtained $I_d(V_{lg})$ curve, in such way that the SiNW FET is biased in the linear region to obtain a high current change for a given threshold voltage shift. The experimental results are shown in Fig. 6a, where the current changes, ΔI_d, per pH value are plotted versus time (the current measured for a pH of 3 has been taken as the zero level). In this measurement, seven NWs have been measured simultaneously and in real time (without multiplexing) The top three curves were acquired with Σ-Δ modulators, whereas the four curves at the bottom were acquired with ItoF converters; in both cases the noise level was low. The temporal resolution of 100ms was high enough, and the response time of the sensors was only limited by transport of the liquid solution through the tubing system. As described in Section 2.1, the measured SiNWs have different widths, and therefore different current responses are obtained. A time delay difference between the top three curves and the four bottom ones can be observed. The SiNWs have been arranged into four sub-arrays in the corner regions of the chips as described in Section 2.1 and shown in Fig. 3a. The analyte supplied to all NWs flows through the same microfluidic channel. The liquid enters at the right side of the chip and flows then back to the left side. The top three curves were obtained

from NWs on the right side, whereas the four bottom curves were obtained from NWs on the left side; since the flow rate in the channel is not high (~60µl/min), the liquid solution reaches the left side of the chip few seconds after flowing over the NWs on the right side, which explains the time delay difference that is visible in Fig. 6a. Although this time consideration may not be relevant for pH measurements, it is a demonstration of the advantage of recording from several NW sensors simultaneously and in real-time. Experiments can be envisioned, which include the measurement of kinetics of different chemical reactions happening on the sensor array.

Another time-resolved experiment is shown in Fig. 6b: the measurement of different KCl concentrations. The current through the NWs is monitored for different concentrations of KCl over time. The oxide surface of the SiNWs responds to the concentration of Cl^- ions similarly as a negative gate voltage, so that larger concentrations of KCl will increase the current, in analogy to increasing pH values [25]. The result shown in Fig. 6b was obtained from five SiNWs, by changing repeatedly the concentration of KCl, starting at 1 mM, then applying 10 mM, then going back to 1 mM and so on. The current measured at a concentration of 1 mM was used as baseline and all measured current values (ΔI_d) are plotted in reference to the current value at 1 mM.

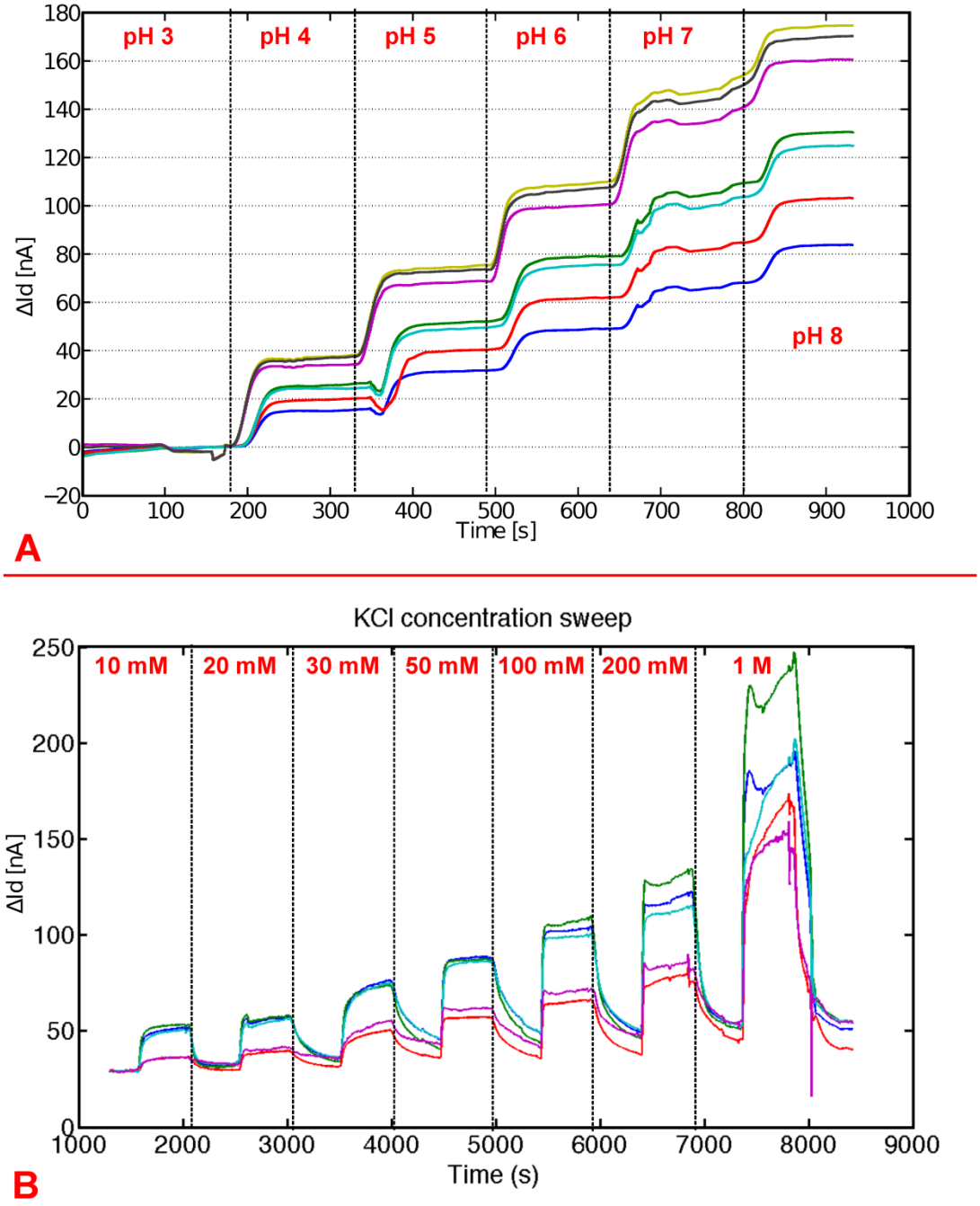


Fig. 6. (a) Current changes over time as a function of different pH values and measured simultaneously from 7 SiNWs; the current measured at pH 3 is taken as the zero level and the current differences, ΔI_d, in reference to the current values at pH 3 are plotted. (b) Current change over time as a function of different concentrations of KCl and measured simultaneously from 5 SiNWs; the measurement started with a concentration of 1 mM and after each concentration increase, a solution of 1 mM was used to purge the system; the currents measured at 1 mM were taken as baseline and only the respective current differences, ΔI_d, were plotted.

In principle, the experiment described above could be performed together with the preceding pH experiments provided that two separate microchannels and the respective SiNWs sub-arrays would have been used (see Fig. 3a). It recently has been shown that SiNWs can be functionalized to achieve specificity to selected ions [25]. Another possibility includes using a subset of SiNWs as control or reference, so that differential measurements can be performed. A combination of sensor arrays with CMOS readout chips offers the potential to perform simultaneous detection of different analytes and real-time differential measurements, which are important features for, e.g., small and reliable diagnostic devices.

Fig. 7 shows results of an experiment in which solutions of different pH were dosed to the sensors while the threshold voltages were monitored with the same procedure explained for Fig. 5. After acquiring the $I_d(V_{lg})$ curve of each SiNW FET, the threshold voltage was computed by extracting the maximum value of the second derivative of the $I_d(V_{lg})$ curve [26]. The extracted values were then plotted for each SiNW (twenty in this case) and each pH value. The corresponding 3D representation constitutes a sensing map, which allows for facile identification of defective sensors as well as for statistical analysis of the sensor array properties: (1) the spread of the SiNW FET threshold voltages over different NWs can be computed (10% in the experiment shown in Fig. 7) to evaluate, e.g., variations in the fabrication process; (2) the responses of different sensors can be compared and averaged (~50mV/pH for the experiment shown in Fig. 7). To attain meaningful statistical analysis results, it is important that all sensors are evaluated at the same time and under the same conditions. It is desirable to perform rapid characterization of the sensor array before each set of measurements at sufficient resolution. The use of the CMOS readout helps to accomplish rapid characterization, a resolution of 7 bits can be ensured even by using only 1ms as integration time (Table 1), so that several $I_d(V_{lg})$ curves could be acquired in a few hundred ms with good resolution.

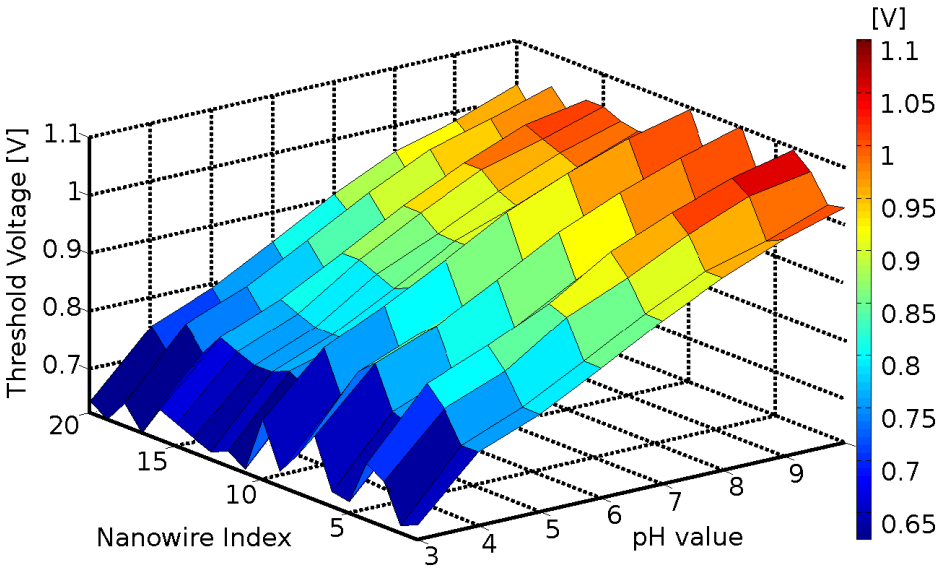


Fig. 7. Sensing map: the threshold voltages of 20 SiNWs has been extracted and plotted for each SiNW and different pH value. Spreads in the threshold voltages of different SiNWs and in the corresponding sensor response can be evaluated. The color bar shows the value of the threshold voltage.

Another important application for the fast-mode feature provided by the CMOS readout system is to characterize the quality of the interface between the SiNW and the top oxide (Fig. 1b). Hysteresis is caused by charges that get trapped at the interface between the SiNW and the top oxide [27]. Hysteresis results in different threshold voltage measurements depending on the voltage sweep direction applied to the reference electrode (V_{lg}). The characterization of hysteresis is important in order to detect hysteresis-induced threshold voltage shifts in the FET, which are indistinguishable with sensing events. By sweeping the gate voltage quickly enough, fewer charges get trapped at the SiNW-top oxide interface, so that the shift in the threshold voltage is smaller [27]. The result of such an experiment is shown in Fig. 8, where two $I_d(V_{lg})$ curves are plotted: the red one is obtained by sweeping V_{lg} in 10s, whereas the blue one is obtained by sweeping V_{lg} in 125ms (the sweeping direction of V_{lg} is also highlighted). By comparing the threshold voltage shifts, ΔV_{th}, of the two curves (for increasing V_{lg}), the density of trapping sites, N_{trap}, can be evaluated, according to the following equation [27]:

$$N_{trap} = \frac{\varepsilon_0 \kappa \Delta V_{th}}{qx} \approx 10^{11} cm^{-2} \qquad (3)$$

where x is the experimentally determined distance from the gate electrode, q is the fundamental charge, ε_0 is the permittivity of the free space and κ is the dielectric constant of the high-k layer. A large N_{trap} (>10^{12}) will result in a sensor with large hysteresis [27].

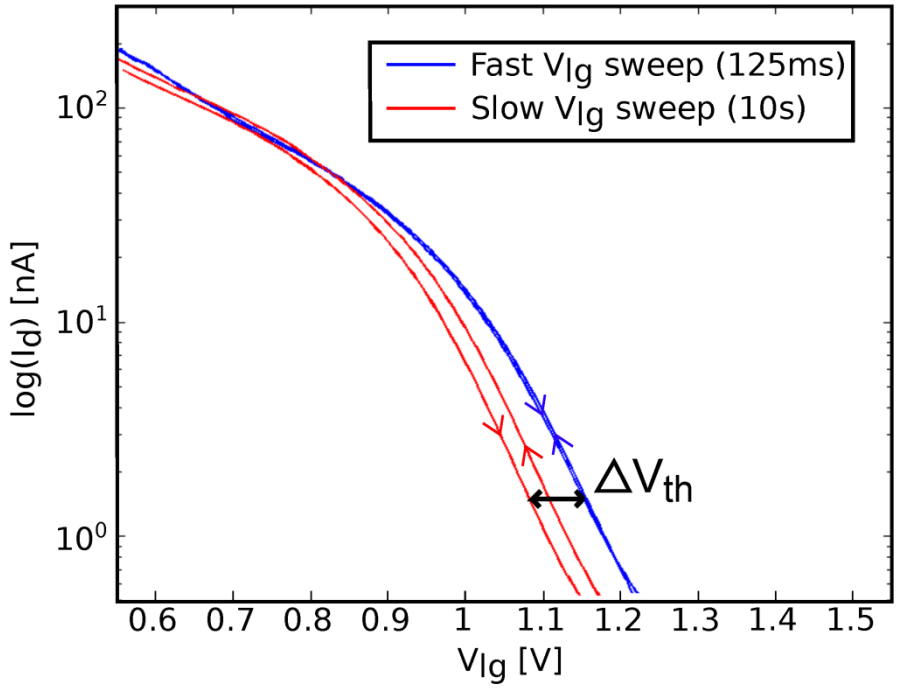


Fig. 8. $I_d(V_{lg})$ curve of one SiNW (in log scale) acquired by sweeping the gate voltage, V_{lg}, at different speeds; the arrows on the curves indicate the sweep direction of V_{lg} in order to evaluate the hysteresis of the sensor and the effect of increasing the sweeping speed. Moreover, by extracting the threshold voltage difference between fast and the sweeps, ΔV_{th}, the interface between SiNW and the top oxide can be characterized.

The use of the fast and parallel readout capabilities of the CMOS chip provides means to rapidly characterize the sensor array and helps to reduce the effects of non-ideal sensor behavior such as hysteresis. By using high-k dielectrics, such as HfO_2, and/or atomic layer deposition methods for the top oxide layer, which yields a higher quality oxide, the hysteresis of SiNW FETs can be reduced to almost zero as shown in Fig. 8. Reducing hysteresis renders the sensor response more precise and reproducible. It is important to note that hysteresis and other mechanistic effects cannot be detected in simply monitoring conductance changes of SiNWs, as commonly done in previous publications [18-20].

Furthermore, it has been shown that the dominating noise in a SiNW FET is the 1/f noise, whose gate-referred voltage power spectral density, S_{Vlg}, can be computed as [24]:

$$S_{V_{lg}} = \frac{q^2 N_{trap}}{WLC_{ox}^2} \frac{1}{f} \tag{4}$$

where q is the elementary charge, W and L denote width and length of the SiNW, and C_{ox} is the gate oxide capacitance per unit area. Equation 4 shows how a characterization of N_{trap} is also important in order to characterize the noise performance of the SiNW sensor.

To further validate our findings on the trapped charges, a study with twenty SiNWs was conducted: all $I_d(V_{lg})$ curves of the different SiNWs were simultaneously acquired with the CMOS chip; the time for sweeping the gate voltage, V_{lg}, was set to 10s and then to 125ms; the threshold voltages were then computed for each SiNW for both sweep directions and speed modes as explained before. The difference between $V_{th,up}$ and $V_{th,dn}$ has been used to quantify the hysteresis of the SiNW FET. The graph in Fig. 9 shows the average values and standard deviations (error bars) of this threshold voltage difference for twenty SiNWs in the fast and the slow mode. Moreover, this experiment was repeated at different pH values (from 3 to 10) in order to assess the validity of the method also for different chemical conditions at the SiNW surface. As it can be seen in Fig. 9, the hysteresis is consistently and significantly reduced (up to 80%) in the fast-mode sweep of V_{lg} for all SiNWs and for all pH values.

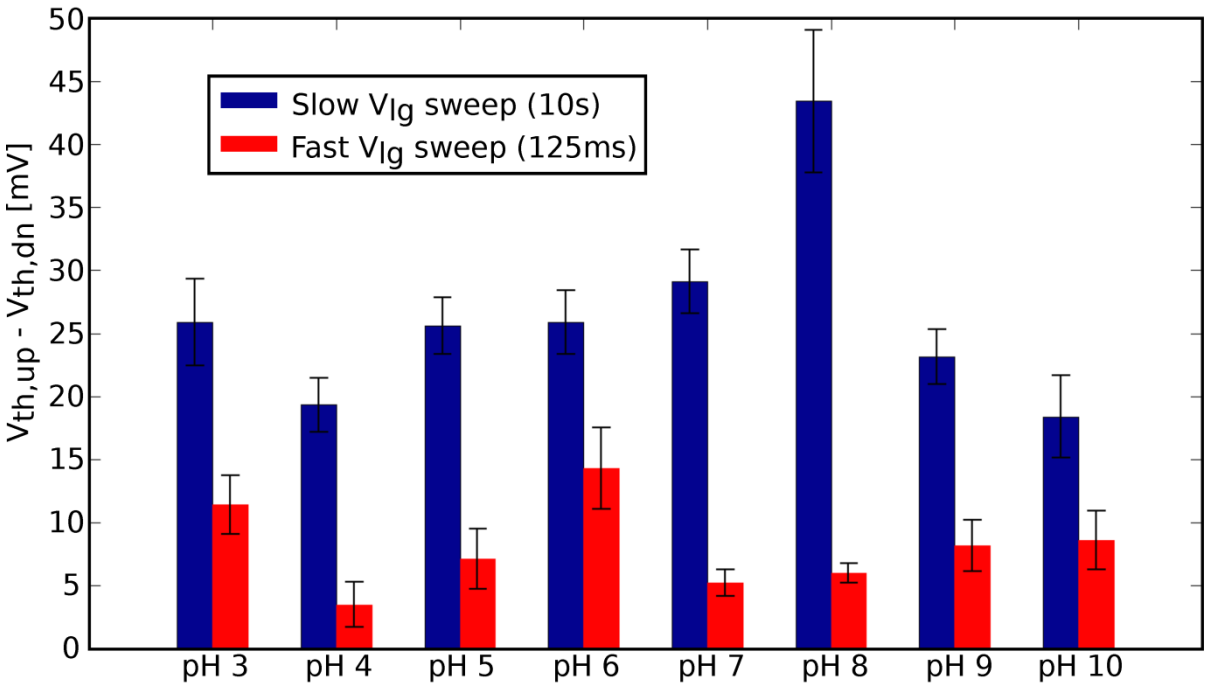


Fig. 9. Average values and standard deviations (error bars) over 20 SiNWs of the difference between threshold voltages for upward sweeping of V_{lg} ($V_{th,up}$) and threshold voltages for downward sweeping of V_{lg} ($V_{th,dn}$), for fast and slow V_{lg} sweeps at different pH values. The difference between $V_{th,up}$ and $V_{th,dn}$ can be used to characterize the hysteresis of the SiNW sensor. The plot shows, how a fast V_{lg} sweep can consistently and significantly reduce hysteresis effects.

4.8 Conclusions

We have presented a micro/nanosensor system featuring an array of SiNW FETs, combined with a CMOS-based readout chip. A dedicated readout channel can be assigned to each nanowire so that leakage current issues due to sensor multiplexing can be avoided.
The CMOS chip includes two commonly used current measurement circuit schemes: a sigma-delta (Σ-Δ) modulator and a current-to-frequency (ItoF) converter. Both architectures have been implemented and tested with SiNWs. The ItoF converter is a simpler architecture requiring less area and consuming less power. The Σ-Δ, on the other hand, provides higher resolution and accuracy at the cost of increased circuit complexity in terms of area, power consumption and output signal decoding, i.e., complex decimation filters are needed. Both architectures provided low noise levels. The ItoF converter is a versatile circuit, since it can cover a very wide input current range while featuring a very good resolution of 11 bits and noise levels down to 8.2 pA_{RMS}; these features, combined with a more compact and scalable design that consumes less power, make the ItoF converter an ideal NW-based sensor readout scheme, so that the ItoF will be chosen as the only architecture in a redesign of the presented CMOS chip.
The general advantages of using a CMOS readout chip were also demonstrated: (1) different sensor outputs were acquired simultaneously, so that highly time resolved experiments can be performed and enough sensor signals can be acquired to obtain meaningful statistics; (2) by exploiting the high temporal and signal resolution and low noise levels of the CMOS chip, even for fast voltage sweeps, non-idealities of the SiNW FET sensors, e.g., hysteresis, could be reduced (up to 80%). The importance of monitoring not only SiNW FET conductance but also threshold voltages has been demonstrated.
We believe that nanowire-based systems with integrated circuits may prove very useful for future portable sensor or diagnostic devices capable of simultaneously detecting different analytes with high sensitivity, high resolution and fast response time. Integrated electronics will help to improve overall sensor system performance and efficiently process the acquired sensor data.

Acknowledgments

This work was part of the project “NanowireSensor”, funded by the Swiss federal research initiative Nano-Tera under reference number 611_61, and was also financially supported by the EU FP7 ERC Advanced Grant “NeuroCMOS” under contract number AdG 267351. Amir Shadmani received individual support through the Marie Curie Research Training Network “EngCaBra” (grant agreement no. 264417). The authors would like to thank Alexander Stettler, at BEL, ETH Zürich, for the CMOS chips packaging.

References

[1] G.J. Zhang, J.H. Chua, R.E. Chee, A. Agarwal, S.M. Wong, K. Buddharaju, N. Balasubramanian, Highly Sensitive Measurements of PNA-DNA Hybridization Using Oxide-etched Silicon Nanowire Biosensors, J. Biosens. Bioelectron. 23 (2008) 1701-1707.

[2] Y. Cui, Q.Q. Wei, H.K. Park, C.M. Lieber, Nanowire Nanosensors for Highly Sensitive and Selective Detection of Biological and Chemical Species, Science 293 (2001) 1289-1292.

[3] I. Park, Z. Li, X. Li, A. Pisano, R. S. Williams, Towards the Silicon Nanowire-based Sensor for Intracellular Biochemical Detection, J. Biosens. Bioelectron. 22 (2007) 2065-2070.

[4] M. Kwiat, F. Patolsky, Interfacing Biomolecules, Cells and Tissues with Nanowire-based Electrical Devices, in: N. Eliaz (ed.), Applications of Electrochemistry and Nanotechnology in Biology and Medicine II, Springer Science+Business Media, 2012, pp. 67-104.

[5] P. Bergveld, Development of an Ion-sensitive Solid-state Device for Neurophysiological Measurements, IEEE Trans. Biomed. Eng. (1970) 70-71.

[6] P. Bergveld, Thirty Years of ISFETOLOGY, What Happened in the Past 30 Years and What May Happen in the Next 30 Years, Sensors and Actuators B 88 (2003) 1-20.

[7] P. Livi, K. Bedner, A. Tarasov, M. Wipf, Y. Chen, C. Schönenberger, A. Hierlemann, A Verilog-A Model for Silicon Nanowire Biosensors: From Theory to Verification, Sensors and Actuators B 179 (2013) 293-300.

[8] G. Ferrari, M. Farina, G. Guagliardo, M. Carminati, M. Sampietro, Ultra-low-noise CMOS Current Preamplifier from DC to 1MHz, IEEE Electron. Letters 45 (2009) 1278-1280.

[9] P. Weerakoon, E. Culurciello, K. G. Klemic, and F. J. Sigworth, An Integrated Patch-Clamp Potentiostat With Electrode Compensation", IEEE Trans. BioCAS 3 (2009) 117-125.

[10] G. Ferri, P. De Laurentiis, A. D'Amico, C. Di Natale, A Low-voltage Integrated CMOS Analog Lock-in Amplifier Prototype for LAPS Applications, Sensors and Actuators A 92 (2001) 263-272.

[11] C. Yang, S. R. Jadhav, R. M. Worden, A. J. Mason, Compact Low-Power Impedance-to-Digital Converter for Sensor Array Microsystems, IEEE JSSC 44 (2010) 2844-2855.

[12] L. Shepherd, P. Georgiou, C. Toumazou, A Novel Voltage-clamped CMOS ISFET Sensor Interface, in Proceedings of IEEE ISCAS (2007).

[13] M. Bennati, F. Thei, M. Rossi, M. Crescentini, G. D'Avino, A. Baschirotto, M. Tartagni, A Sub-pA Delta-Sigma Current Amplifier for Single Molecule Nanosensors, in Dig. Tech. Papers of IEEE ISSCC (2009) 348-349.

[14] M. Crescentini, M. Rossi, M. Bennati, F. Thei, A. Baschirotto, M. Tartagni, A Nanosensor Interface Based on Delta-Sigma Arrays, in Proceedings of Int. Workshop on Advances in Sensors and Interfaces (2009) 163-167.

[15] M. Schienle, C. Paulus, A. Frey, F. Hofmann, B. Holzapfl, P. Schindler-Bauer, R. Thewes, A Fully Electronic DNA Sensor With 128 Positions and In-Pixel A/D Conversion, IEEE JSSC 39 (2004) 2438-2445.

[16] A. Flammini, D. Marioli, A. Taroni, A Low-Cost Interface to High-Value Resistive Sensors Varying Over a Wide Range, IEEE Trans. On Instrumentation and Measurement 53 (2004) 1052-1056.

[17] M. M. Ahmadi, G. A. Juliene, Current-Mirror-Based Potentiostat for Three-Electrode Amperometric Electrochemical Sensors, IEEE Trans. Circuits Syst. I 56 (2009) 1339-1348.

[18] H. Liu, T. H. Teo, Y. P. Zhang, A Low-Power Wide-Range Interface Circuit for Nanowire Sensor Array Based on Resistance-to-Frequency Conversion Technique, in Proceedings of ISIC (2009) 13-16.

[19] K. T. C. Chai, K. Choe, O. D. Bernal, P. K. Gopalakrishnan, G. J. Zhang, T. G. Kang, M. Je, A 64-Channel Readout ASIC for Nanowire Biosensor Array with Electrical Calibration Scheme, in Proceedings of IEEE EMBS (2010) 3491-3494.

[20] G. J. Zhang, K. T. C. Chai, H. Z. H. Luo, J. M. Huang, I. Guang Kai Tay, A. Eu-Jin Lim, M. Je, Multiplexed Detection of Cardiac Biomarkers in Serum with Nanowire Arrays Using Readout ASIC, J. Biosens. Bioelectron. 35 (2012) 218-223.

[21] K. Bedner, V. Guzenko, A. Tarasov, M. Wipf, R. L. Stoop, D. Just, S. Rigante, W. Fu, R. Minamisawa, C. David, M. Calame, J. Gobrecht, C. Schönenberger, pH Response of Silicon Nanowire Sensors: Impact of Nanowire Width and Gate Oxide, Sensors and Materials 25 (2013) 567-576.

[22] O. Knopfmacher, A. Tarasov, W. Fu, M. Wipf, B. Niesen, M. Calame, C. Schönenberger, Nernst Limit in Dual-Gated Si-Nanowire FET Sensors, Nano Letters 10 (2010) 2268-74.

[23] J. Rothe, O. Frey, A. Stettler, Y. Chen, A. Hierlemann, A Fully Integrated CMOS Microsystem for Electrochemical Measurements on 32x32 Working Electrodes at 90 Frames per Second, ACS (2014) accepted.

[24] K. Bedner, V. Guzenko, A. Tarasov, M. Wipf, R. L. Stoop, S. Rigante, J. Brunner, W. Fu, C. David, M. Calame, J. Gobrecht, C. Schönenberger, Investigation of the Dominant 1/f Noise Source in Silicon Nanowire Sensors, Sensors and Actuators B 191 (2014) 270-275.

[25] M. Wipf, R. L. Stoop, A. Tarasov, K. Bedner, W. Fu, I. A. Wright, C. J. Martin, E. C. Constable, M. Calame, C. Schönenberger, Selective Sodium Sensing with Gold-Coated Silicon Nanowire Field-Effect Transistors in a Differential Setup, ACS Nano 7 (2013) 5978-5983.

[26] A. Ortiz-Conde, F. J. Garcia-Sanchez, J. Muci, A. T. Barrios, J .J. Liou, C. S. Ho, Revisiting MOSFET Threshold Voltage Extraction Methods, Microelectronics Reliability 53 (2013) 90-104.

[27] C. D. Young, Y. Zhao, D. Heh, R. Choi, B. Hun Lee, G. Bersuker, Pulsed Id–Vg Methodology and Its Application to Electron-Trapping Characterization and Defect Density Profiling, IEEE Trans. On Electron Devices 56 (2009) 1322-1329.

CHAPTER 5

Adding the ‘heart’ to hanging drop networks for microphysiological multi-tissue experiments

Saeed Rismani Yazdi, **Amir Shadmani**, Sebastian C. Bürgel, Patrick M. Misun, Andreas Hierlemann and Olivier Frey

ETH Zurich, Department of Biosystems Science and Engineering, Bio Engineering Laboratory, Mattenstrasse 26, CH-4058 Basel, Switzerland

Published in:

Journal of Lab on a Chip

Volume 15, Issue 21, 15 September 2015, Pages 4138–4147

5.1 Abstract

Microfluidic hanging-drop networks enable culturing and analysis of 3D microtissue spheroids derived from different cell types under controlled perfusion and investigating inter-tissue communication in multi-tissue formats. In this paper we introduce a compact on-chip pumping approach for flow control in hanging-drop networks. The pump includes one pneumatic chamber located directly above one of the hanging drops and uses the surface tension at the liquid–air-interface for flow actuation. Control of the pneumatic protocol provides a wide range of unidirectional pulsatile and continuous flow profiles. With the proposed concept several independent hanging-drop networks can be operated in parallel with only one single pneumatic actuation line at high fidelity. Closed-loop medium circulation between different organ models for multi-tissue formats and multiple simultaneous assays in parallel are possible. Finally, we implemented a real-time feedback control-loop of the pump actuation based on the beating of a human iPS-derived cardiac microtissue cultured in the same system. This configuration allows for simulating physiological effects on the heart and their impact on flow circulation between the organ models on chip.

5.2 Introduction

The design of microphysiological systems constitutes a next step towards more biomimetic in vitro models [1]. Such systems comprise conventional or advanced three-dimensional cell cultures and a fluidic routing scheme, which allows for interconnecting the different organ models for controlled perfusion and metabolite transfer [2–8]. The growing interest in these systems is due to their potential to enable better predictions of the impact of compounds on the human body and their potential to enable better understanding – in a more systemic way – of how different organs interact with each other under different experimental conditions.

One possible realization of a microphysiological system includes configurable hanging-drop networks (HDNs) that host microtissue spheroids. These systems are aimed at realizing complex multi-organ systems and, in particular, at enabling associated continuous tissue–tissue interactions and molecule exchange [9]. Spheroids as three-dimensional tissue models currently experience large interest, as they have been demonstrated to better represent in vivo like conditions with respect to cell morphology and functionality compared to standard 2D cell cultures [10]. Further, spheroids simply form by self-aggregation of cells in solution and are easy to handle by using conventional pipetting tools. Spheroids can be derived from different cell types so that a multitude of different organ models based on the same fabrication technology becomes available. Example organ models include liver [11], pancreas [12], heart [13], brain [14], as well as a large variety of spherical tumor models [15–17]. Spheroids are predominantly formed by using the hanging-drop technique, in which a drop of cell suspension is pipetted onto a substrate that is subsequently inverted. The cells then sediment to the liquid–air interface of the now hanging drop and start to attach to each other and form the spherical 3D tissue constructs as no surface is present to which they could adhere.

Hanging-drop networks provide an ideal formation and culturing environment for spheroids and include microfluidic channels and other units to interconnect the different drops so as to obtain functional drop networks. Flow through a network of hanging drops can be precisely controlled, which substantially increases the experimental options in using spheroids and paves the way to complex multi-tissue, or so-called “body-on-a-chip” formats. The microfluidics of HDN systems have been realized by patterning hydrophobic rim structures on the surface of a PDMS substrate that is afterwards turned upside down. The rim structures delimit wetted areas and prevent the liquid from spreading over the whole surface. By using circular structures it is therefore possible to define drops; narrow grooves are used for realizing the interconnecting channels, so that bottom-open, complex microfluidic networks of hanging drops can be obtained.

Until now, medium flow in such systems has been controlled by using external pumps and tubing that were connected to the devices. Tubing, however, increases medium volume, bears the risk of compound adsorption at internal surfaces and compromises the ease-of-use and the potential for parallelized experiments. A compact on-chip pump solution would help to overcome these limitations. The integration of a pump into the HDNs should, however, not significantly increase device complexity. The pump should be simple and robust in operation and controllable through a minimal number of control lines. There is a plethora of on-chip micropump designs that have been reported for different applications in the past [18–20]. The vast majority of those, however, was designed for driving liquid flow in closed microfluidic systems so that they are of limited use for open microfluidic systems, including the hanging-drop networks.

Here, we present a novel integrated pump concept specifically developed for microfluidic hanging-drop networks. The pump itself is based on a hanging drop and relies on surface tensions at the liquid–air interface to drive liquid through hanging-drop networks. The pump is actuated via a single pneumatic control line and allows for realizing a variety of different flow schemes including physiological pulsatile flow. Pumping units can be operated in several networks in parallel, while all of them rely on the same single actuation line. Larger HDNs can be perfused by using more than one pump stage in serial configurations. Finally, we implemented a feedback control, in which the pump rate has been coupled to the spontaneous contraction of a human iPS-derived cardiac microtissue, which has been included in the system. The aim was to be able to also mimic physiologically induced alterations of the pump rate upon temperature change or compound dosage as they occur in vivo. We then show how external stimuli, e.g., through adding drugs, affect the overall medium flow and related drug distribution in the “body-on-achip” multi-tissue HDN system.

5.3 Results and discussion

5.3.1 Device architecture

Fig. 1a illustrates the design of a simple microfluidic hanging-drop loop with an integrated pump. The loop consists of 10 hanging drops in a series, each with 3.5 mm base diameter (circular areas) arranged on a 4.5 mm grid (384- well-plate format). The hanging drops are connected through 200 µm-wide microchannels. Hydrophobic PDMS rims confine the liquid and allow for building up the completely open hanging-drop network at the bottom of a substrate. For more details on the concept see reference Frey et al. [9]. Three of the hanging drops form a pneumatically driven hanging-drop pump (cross-section AA′). The other seven hanging drops can be used to host the microtissue spheroids for investigations (cross-section BB′).

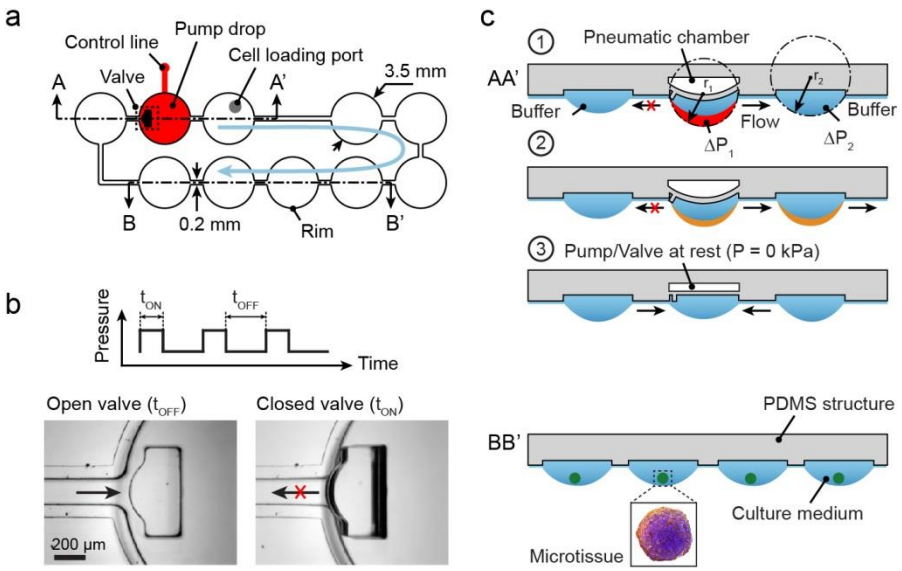


Fig. 1. Architecture and concept of the integrated micropump in a microfluidic loop of hanging drops. (a) Top view of the design of the hanging-drop network. The pump consists of three drops; seven other drops can be used to culture spheroids. (b) The protocol of the applied pressure for pump actuation is indicated as a square wave and consists of an OFF and ON state, each with a defined duration. The ON state can vary in pressure. Photographs show the open and closed states of the integrated valve. (c) Cross-section AA' (indicated in Fig. 1a) schematically illustrates the sequence of pump actuation (see text for details); BB' the drops for spheroid culturing.

5.3.2 Pump concept

The pump is operated by pressurizing a pneumatic chamber (typically 20–60 kPa) located above the center pump drop through a pressure control line (indicated in red in Fig. 1a). The increase of pressure bulges the membrane between pneumatic chamber and drop ceiling (500 µm thick) and closes, at the same time, an integrated valve at the inlet channel of the drop (Fig. 1b). The valve consists of a small PDMS

structure with a size of 0.5 mm × 1.2 mm and the same height as the rim structure. The PDMS structure is localized at the edge of the membrane at a distance of 25 µm (membrane in not-deflected position) from the channel inlet; it tilts and blocks the inlet of the channel upon membrane deflection. The blockage increases the fluid resistance and substantially reduces liquid flow to and from the left of the pump drop. Position and shape of the PDMS block have been optimized with respect to optimal inlet closure.

The membrane deflection virtually increases the volume of the drop (indicated in red in Fig. 1c1). When the drop gets bigger (towards a hemisphere), its radius gets smaller. Based on Young–Laplace's equation, $\Delta P = 2\gamma/r$, the decrease of the radius of the air–liquid interface results in a pressure increase. The resulting pressure difference between drop 1 and drop 2 ($\Delta P1 > \Delta P2$) and the simultaneous closure of the valve that prevents back-flow induce a directional flow from the center pump drop to the right neighboring drop (Fig. 1c2, transferred volume shown in orange). The flow is maintained until both radii are the same, and the pressure is equilibrated.

Upon release of the pneumatic pressure, the pump drop membrane returns to the resting state, and liquid from both neighboring drops flows into the center drop thereby restoring equilibrium conditions (Fig. 1c3). This sequence is repeated with each pump stroke, and a uniform flow through the hanging-drop loop in generated. The pump actuation protocol (shown as a square wave function in Fig. 1b) consists of two states maintained for two different durations. The time t_{ON} refers to the time when a certain pressure is applied to the pneumatic chamber. In this state the membrane is deflected, and the valve is closed. t_{OFF} stands for the time at resting state when no pressure is applied. The rather thick – and therefore stiff – PDMS membrane ensures short transition times. Different flow schemes can be programmed by varying pressure and state durations.

5.3.3 Integration of flow buffers

The two drops up- and downstream of the center pump drop are buffers (diameter of 3.5 mm) that act as low-pass filters to reduce flow pulsation. Due to the two discrete membrane states and the design of the valve, the flow is intrinsically pulsatile, which needs to be controlled, or pulsatility needs to be reduced for some applications. Fig. 2 illustrates measured position and calculated flow rate of a tracked particle in the channel after the pump drop versus time using t_{ON} = 50 ms and t_{OFF} = 50 ms as actuation protocol. Without buffer (Fig. 2a and c), a large back flow, which is substantially larger than the resulting net flow was observed for every pump stroke. In the presented example, peak-to-peak values larger than 120 µl min^{-1} were observed at a net flow of 5.6 µl min^{-1}. By integrating the two buffer drops before and after thepump drop the pulsation was drastically reduced (1.9 µl min^{-1}), while the net flow only slight changed compared to non-buffered flow (5.3 µl min^{-1}, Fig. 2b and d with zoomed view of the curve).

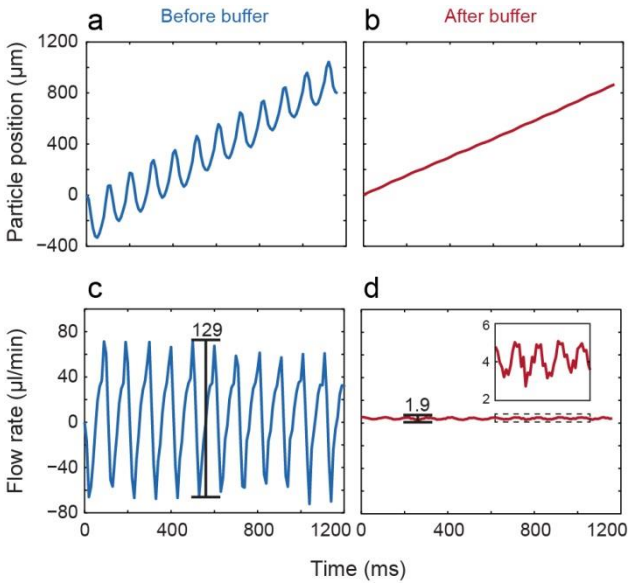


Fig. 2. Influence of buffers. Tracked particle position and corresponding flow rate before (a & c) and after the buffer drop (b & d).

5.3.4 Pump characteristics

Table 1 presents flow rates and peak-to-peak flow pulsation values for different pump actuation protocols. The two input values, t_{ON} and t_{OFF}, were varied within a range of 25 ms and 2 s. In all cases, a constant flow rate was achieved after approximately 20 pump strokes. The flow rate was adjustable in a range of 0.5–8 µl min^{-1} at 40 kPa of applied pressure. Flow rates were in general decreasing with increasing actuation time t_{ON}, when keeping t_{OFF} constant. High frequency modes produced much lower pulsation compared to actuation frequencies in the single-digit Hz range. All values presented here were recorded with buffers. It has to be mentioned that these values apply to the specific 10-drop loop depicted in Fig. 1a. For larger networks with, for example, higher flow resistances, lower flow rates are to be expected (see below).

5.3.5 Influence of the applied pressure

The pressure applied to the pneumatic chamber influences the flow rate. A higher pressure results in a stronger deflection of the membrane and, hence, a larger virtual volume that is added to the pump drop. Higher stroke volumes are therefore expected. For increasing pressures, from 20 to 60 kPa, we observed a linear increase of the flow rates for both actuation protocols that are shown in the graph. Thus, control of the applied pressure provides an additional independent parameter for flow rate adjustment.

Table 1. Measured flow rates and pulsation amplitudes for higher and lower pump frequencies (mean values, n>3). All the experiments were conducted with a total liquid volume of 80 µl, which produces a drop height of 820 µm. The applied pressure was 40 kPa.

t_{OFF} (ms)	t_{ON} (ms)						
	25	50	100	200	500	1000	2000
Flow rate (µl/min)							
25	1.6	1.5	1.0	0.6			
50	4.5	5.3	4.9	4.1			
200				7.1	7.3	6.2	4.6
500				4.3	6.2	5.3	4.4
Peak-to-peak pulsation amplitude (µl/min)							
25	1.4	2.9	6.1	7.1			
50	5.8	10.5	16.3	20.6			
200				40	47	55	61
500				41	50	62	69

5.3.6 Influence of drop volume

We further characterized the relation between flow rate and initial volume of the pump drop. We varied the total volume loaded into the 10-drop loop at the beginning of the experiment between 60 µl and 90 µl and measured resulting flow rates for one frequently used actuation protocol (t_{ON} = t_{OFF} = 200 ms: applied pressure of 60 kPa). Note that all drops in the loop have equal size at equilibrium as a consequence of the Young–Laplace law. Moreover, it has to be mentioned that each of the 10 drops consists of two volume parts; the fixed cylindrical recess volume (450 µm deep, 3.5 mm diameter) and the variable hanging-drop volume protruding from the rim structure. Approximately 45 µl volume are contained in the recess and channels. Varying the total chip volume between 60, 70, 80 and 90 µl results in drop heights of 540, 700, 820 and 920 µm, respectively. The flow rate variations in dependence of the overall drop size are small. For volumes between 70 and 90 µl the flow rate variation is negligible, for a lower volume (60 µl) the flow rate decreases by approximately 10%. These findings are important in view of liquid evaporation and small drop volume reductions that may occur for experiments lasting several days; also for these extended time scales a reasonably constant flow rate could be maintained.

5.3.7 Pulsatile flow profiles

An important advantage of the proposed pump concept is that it enables both, constant and pulsatile perfusion at similar flow rates by modifying the pump actuation protocol. This becomes especially interesting, if the pulsatile pump characteristics of the heart should be reproduced to investigate their influence on the behavior of other cell types. The buffer drops enable controlled reduction of the pulsation and allow for simulating different locations and conditions in the body, as blood flow right after the heart and in arteries is strongly pulsatile and becomes more uniform in finer vessels and capillaries.

Fig. 3 shows measured flow rates over time for 3 different pump actuation protocols with fixed t_{ON} = 200 ms and varying t_{OFF} (50, 200 and 500 ms). Depending on the applied protocol, different amplitudes and pulsation frequencies can be achieved. For all three actuation protocols the overall net flow rate is positive, although there is a flow reversal during the pulse. The figure demonstrates that we are able to generate pulsatile flow of varying characteristic by adjusting the pump actuation protocols.

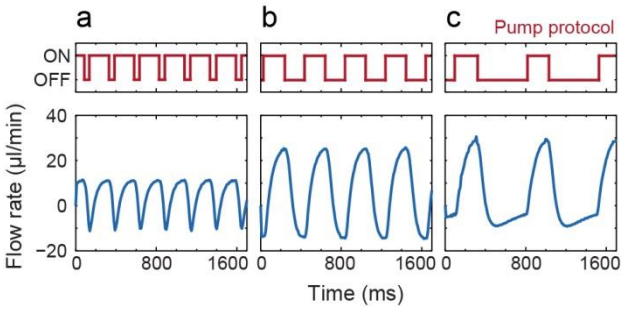


Fig. 3. Pulsatile flow. Tuning the shape of the pulsatile flow by regulating the pump actuation protocol t_{ON} = 200 ms and t_{OFF} = 50, 200 and 500 ms. Applied pressure was 40 kPa. Resulting net flow rates were (a) 4.11 µl/min, (b) 7.07 µl/min, and (c) 4.3 µl/min.

5.3.8 Parallelization

The proposed pump concept requires only a single pneumatic actuation line for flow actuation in a closed hanging-drop loop. In a next step, we designed a platform consisting of 8 independent hanging-drop loops (Fig. 4a). Their architecture is the same as that in Fig. 1a. The 8 loops have been placed side-by-side so that all drops and inlet ports are located on a 384-well plate grid (i.e., 4.5 mm pitch). This arrangement makes the device compatible with multichannel pipets (9 mm pitch between inlet ports of neighboring loops) to facilitate convenient medium and cell suspension loading and automated readout.

The four left loops are mirrored with respect to the ones on the right. Each loop consists of 3 sub-networks, shown in 3 different colors in loop number 5: a pump unit (3 drops), a 2-drop unit and a 5-drop unit. The pump unit includes the pump drop (red) and the 2 adjacent buffer drops (green). The other seven drops have been arranged in groups of 2 and 5 drops in series (pink and blue color), and can be loaded with microtissues derived from different cell types so that multitissue experiments can be carried out. The 3 sub-networks are initially separated from each other through capillary stop valves.9 After cell suspensions have been loaded into the microfluidic network via loading ports, the sub-networks have been fluidically interconnected through opening of the capillary stop valves by supplying a small amount of liquid through the respective connecting ports. A fully loaded chip with differently colored liquids for better illustration is shown in Fig. 4b.

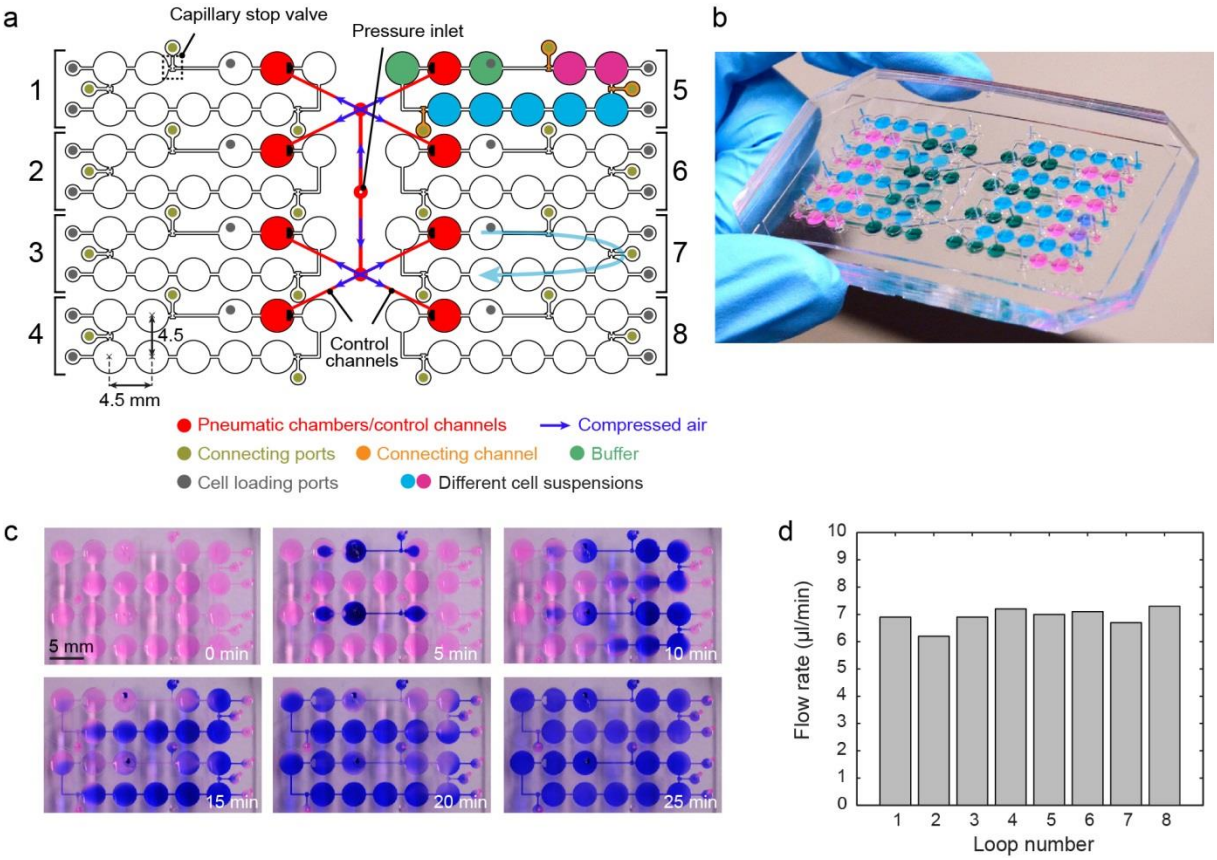


Fig. 4. Parallel network. (a) Design of the integrated pumping system in parallel hanging-drop networks. (b) Photograph of the device filled with differently colored liquids as schematically shown in (a). The chip was rotated by 180°. (c) Time-lapse images of blue ink circulating in 2 independent loops of the 8-loop platform using the integrated micropump. Each loop was initially loaded with diluted red dye in DI water, and before the pump sequence was started (t = 0 min), 4 µl of blue ink were added to the buffer drop after the pump drop through the loading port using a multi-channel pipet (t_{ON} = t_{OFF} = 50 ms). (d) Flow rates simultaneously generated in 8 independent HDN loops (t_{ON} = t_{OFF} = 200 ms; applied pressure of 40 kPa; device was filled with 80 µl DI water with suspended beads).

The integrated pump is identical to the one presented above. The eight pneumatic chambers, however, are symmetrically connected to a single pressure inlet in the center through control channels (shown in red). All pneumatic chambers and control channels are identical and have the same dimensions. This layout allows for uniform distribution of compressed air and ensures equal actuation frequencies and membrane deflection heights and, consequently, identical flow rates in all eight loops. With the proposed layout, eight independent hanging-drop loops can be operated in parallel with only one pneumatic control line. Thus, eight multi-tissue experiments can be carried out at the same time in the same environment, which then allow for parallel investigation of, for example, the effects of different drug candidates and/or concentrations under identical culturing conditions. For illustration purposes we added blue ink in two of the loops, which was then circulated with the same flow rate (Fig. 4c). Once the pump actuation started, a clockwise flow was generated in both loops, and the blue ink was transported along the drops and channels through the system. We measured the flow rate variation among the different loops on the same platform. The relative variation was below 10% (Fig. 4d), which demonstrates the

performance uniformity of the integrated pumping system in controlling the flow rate in different subnetworks.

5.3.9 Large networks

We further investigated, if the proposed pumping scheme is able to drive larger hanging-drop networks. Fig. 5a shows two arrays of 4 × 6 drops. The arrays are again separated into two sub-networks, shown in 2 different colors (pink 2 × 4, blue 4 × 4) to represent different spheroid types. Two adjacent pump lanes were used to operate the arrays. Since the network is larger compared to the 10-drop loop and consists of a higher number of drops and channels, the total hydraulic resistance is higher. Thus, three identical micropump drops were integrated in series in a row (red). In addition, we implemented two buffer drops before and after the pump drops (green). The circulation in the lower and upper networks is intended to be counter-clock wise and clock wise, respectively. In each drop row, the flow rate is a fourth of the overall flow generated by the pump due to the flow splitting.

The three integrated pump drops of both arrays are actuated in a defined sequential protocol (Fig. 5b). Compared to the single-drop pump, an additional parameter, t_{DELAY}, was required to define the time shift between the pump strokes of the three individual pump drops.

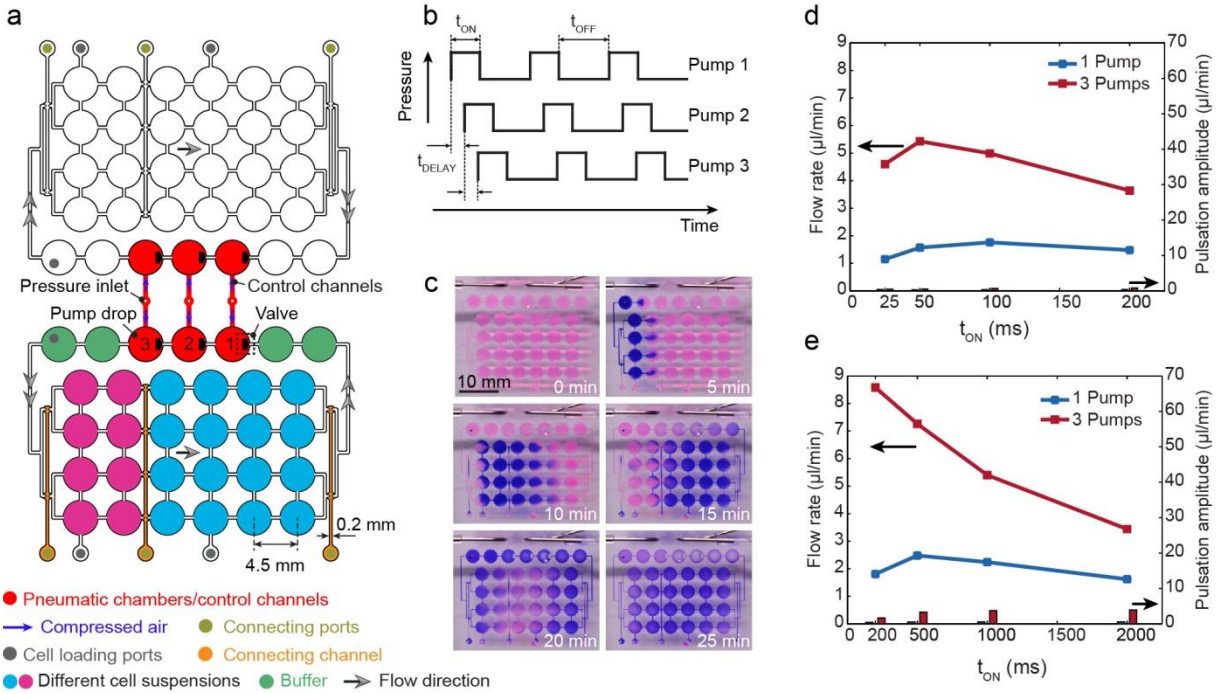


Fig. 5. Large networks. (a) Layout of the integrated pump system of a large hanging-drop array of 24 drops. (b) Actuation protocol of the three micropumps in series. (c) Time-lapse images showing the circulation of a small volume of blue ink, added to the buffer drops at the left within one of the hanging-drop subnetworks (t_{ON} = t_{OFF} = 200 ms, applied pressure of 40 kPa, initial volume of 250 µl, see also Supplementary Movie 3). Characteristic flow rates and pulsation amplitudes in the 4×6 hanging-drop subnetwork in dependence of the pump actuation protocol are presented in (d) for t_{OFF} = t_{DELAY} = 50 ms and in (e) for t_{OFF} = t_{DELAY} = 200 ms.

The 3-drop pump was able to drive liquid through the whole array as illustrated in Fig. 5c by using ink. For characterization, different actuation modes have been investigated. First, t_{OFF} and t_{DELAY} were set to $t_{OFF} = t_{DELAY} = 50$ ms and $t_{OFF} = t_{DELAY} = 200$ ms and t_{ON} was varied. Then, pump tests were performed by activating only a single or all three pumps for comparison (Fig. 5d and e). By using only one pump, flow rates of up to 3 µl min−1 could be achieved (blue lines). By actuating all 3 pump drops, we were able to cover a range of flow rates from 2 µl min−1 to up to 9 µl min−1. By integrating 2 buffer drops before and after the pump drops, flow pulsation could be strongly reduced for all actuation protocols.

5.3.10 Real-time closed-loop pump regulation using a beating cardiac microtissue

The previous results illustrated the performance of the integrated hanging-drop pump with respect to flow and pulsation control. In the human body, blood flow is generated by the heart and regulated mainly over the beating frequency in response to nutrient and oxygen demands in the periphery of the body. In order to simulate such physiological behavior, we developed a feedback control loop, which has been used to synchronize pulsatile pump actuation with the beating of a human iPS-derived cardiac microtissue (hCdMT), which was cultured in the very same hanging-drop loop and which was capable to reproduce a physiological response upon temperature changes and drug exposure. Pump actuation was triggered for every microtissue contraction, and any variation in the microtissue beating rate was directly translated into pump actuation frequency. The flow characteristics and profile then changed accordingly in real-time.

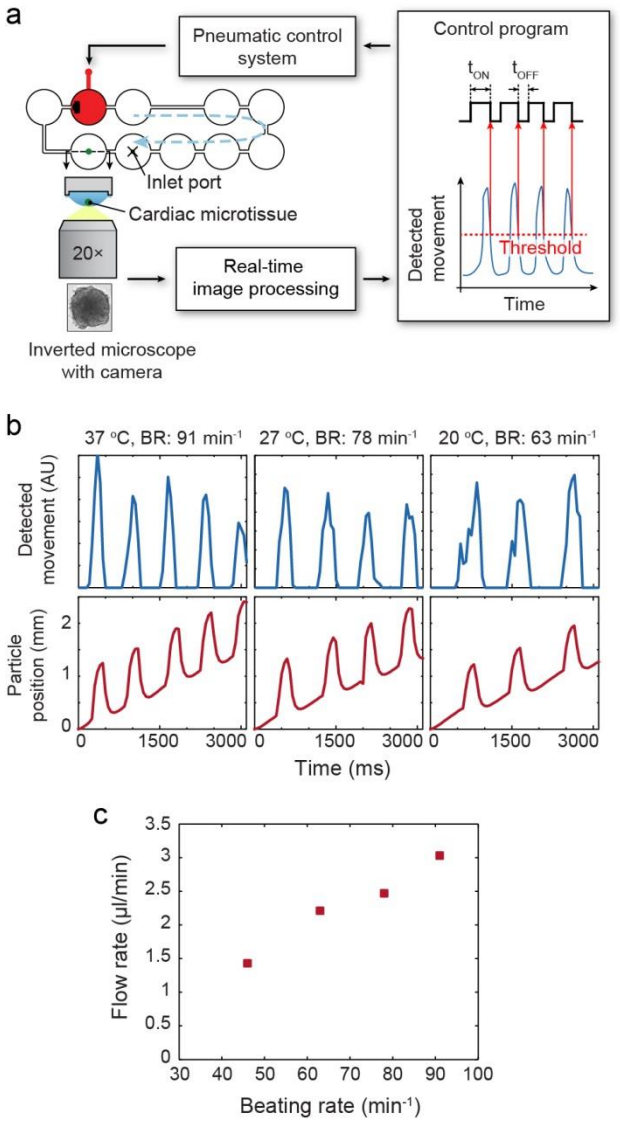


Fig. 6. Regulated flow driven by a cardiac microtissue. (a) Schematic view of the setup with feedback loop. b) Synchronized beating of the hCdMT and flow (particle position) generated by the integrated micropump at 3 different culture medium temperatures corresponding to 3 different beating rates. (c) Relation between beating rate and generated flow rate (applied pressure of 20 kPa, initial volume of 80 µl).

One hCdMT was placed in the indicated hanging drop of the single hanging-drop loop with one pump unit (Fig. 6a). A high-speed camera was used to monitor the hCdMT, and online image processing was performed to detect the contractions. The output was continuously used to control the pump actuation protocol; t_{OFF} was kept constant at 200 ms, whereas t_{ON} was adapted with regards to the contraction characteristics of the cardiac microtissue. Fig. 6b presents time courses of the image processing output showing contraction peaks and particle traces in the liquid flow in the channel after the pump unit. The pulsatile flow-peaks nicely correlate with beating of the cardiac microtissue. The hCdMT did not move due to liquid drag of the pulsatile flow, which would otherwise have interfered with the detection of microtissue contraction movements.

We then continuously decreased the temperature from 37 °C to 20 °C in the HDN setup, which induced a reduction of cardiac beating frequency [13]. As expected, the slower beating did result in correlated slower pump actuation and, hence, flow pulsation. The flow rate in the hanging-drop loop was reduced accordingly (Fig. 6c).

In a second experiment in the same setup, we infused isoproterenol into the drop preceding the drop hosting the cardiac microtissue (inlet port indicated in Fig. 6a). Isoproterenol is a known beta-adrenergic receptor agonist, which increases the heart rate, as already has been demonstrated with hCdMTs [13]. The goal here was to closely simulate spreading and clearance of the drug in the system in dependence of the effect of the drug on the heart beating or liquid pumping rate by using the implemented feedback. The infusion of 2 µl of bare medium did not affect the beating rate (Fig. 7a). Upon infusing a 2 µl aliquot of isoproterenol (5 µM) we observed a clear increase in beating rate (Fig. 7b) as soon as the drug reached the cardiac microtissue. The larger beating rate produced an increased liquid flow, which rapidly transported more of the compound towards the drop containing the cardiac microtissue. The sustained larger pump rate in the loop then promotes clearance of the drug from the drop with the hCdMT and transport to downstream drops and dilution in the overall medium volume of the liquid-phase loop. As a result we observed an expected reduction of the beating rate to approximately the initial value within about 20 minutes. The integration of the feedback accelerated this clearance and distribution effect, because the pump rate was increased due to the stimulation of the hCdMT. Switching off the feedback and setting the pump actuation to a constant rate, independent of the stimulation, resulted in a longer duration of high beating rates and slower recovery of the hCdMT to the state before drug exposure (Fig. 7c).

The functional coupling of the pump rate in the microfluidic system to the contraction of the cardiac microtissue thus offers the possibility of better simulating and mimicking physiological effects. Substances, either introduced to the microphysiological system from outside or produced in the system by other organ models (e.g. metabolites) that have an effect on cardiac tissue will also directly influence medium circulation, as it would be the case in vivo. The corresponding systemic changes and impact of the respective substances can be assessed.

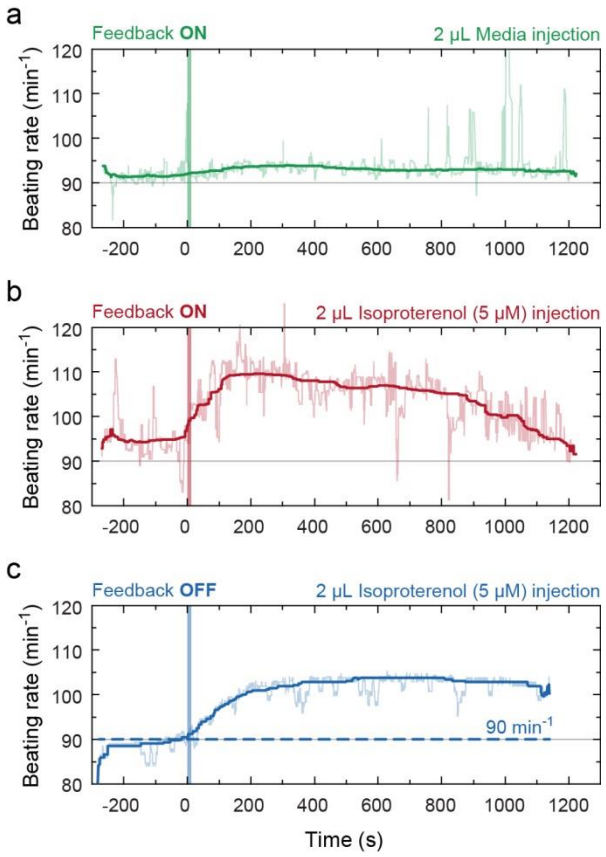


Fig. 7. Biochemical stimulation of the hCdMT in the feedback-controlled hanging-drop loop. Infusions at t=0 are indicated by a vertical line. (a) No effect on the beating rate was observed when medium alone is infused in a control experiment. (b) Increase and decrease of beating rate and the feedback-coupled perfusion rate upon adding 2 µl of Isoproterenol (5 µM in medium, 10 µl/min) in the preceding drop. The 2-µl aliquot was diluted to 1 µM when infused (10 µl drop volume). (c) Increase and slow decrease of the beating rate upon adding 2 µl of Isoproterenol without feedback control. The pump rate was constant and set to 90 strokes per minute (dashed line). (All curves have been smoothed using median filtering; applied pressure was 50 kPa, initial volume was 100 µl).

5.4 Methods

5.4.1 Microfluidic device fabrication

All microfluidic devices were fabricated by using multilayer soft lithography. The devices consist of two polylJdimethylsiloxane) layers (PDMS, Sylgard 184, Dow Corning Corp., USA), which were aligned and bonded to each other (see ESI† for fabrication process). The masters for molding both layers, the pneumatic pump layer and hanging-drop network (HDN) layer, consisted of SU-8 100 negative photoresist (MicroChem Crop., USA), which was spin coated on 4 inch silicon wafers to achieve photoresist layers of 200 and 250 µm thicknesses. After softbake of each layer, the wafers were exposed to ultraviolet light through foil masks to initiate crosslinking followed by post-exposure baking. The wafers were subsequently developed for 30 min in SU-8 developer (mrDev 600, micro resist technology GmbH, Germany). Before replica molding, wafers were coated with trichlorosilane (Sigma-Aldrich, Switzerland) to reduce adhesion.

Microfluidic devices were casted from PDMS. PDMS base and curing agent were mixed at a ratio of 10 : 1 and degassed before pouring the mixture onto the SU-8 mold. The upper pneumatic pump layer had a thickness of approx. 6 mm in order to provide the required mechanical stability for the device. The lower HDN layer was adjusted to 1 mm so that a pump membrane thickness of 500 µm was obtained. PDMS was cured at 80 °C for 2 hours. The micropatterned replicas were cut and peeled off the SU-8 mold. An inlet hole for the control line was punched into the pump layer. After oxygen plasma activation, the pump layer was precisely aligned and irreversibly bonded onto the HDN layer. Then, liquid inlet ports were punched through the bonded layers to be able to supply medium and/or cell suspensions. Prior to device usage, the surface of the HDN layer was covered with a PDMS layer with openings at the drop areas and then activated with oxygen plasma in order to increase wettability of the PDMS. In this way, only the surfaces that came in contact with liquid were rendered hydrophilic, whereas rim structures were kept hydrophobic and prevented liquid overflow. The PDMS devices were either affixed to a custom-made chip holder or placed onto PDMS spacers inside a polystyrene box.

5.4.2 Experimental setup

Characterization tests were performed on an inverted microscope (Olympus IX81 and Nikon Ti Eclipse) using a 5× (and 4×, respectively) objective in bright-field mode. To minimize evaporation of the liquid from the hanging drops during experiments, microfluidic devices were placed into a closed humidified chamber. A single tube connected the pneumatic control line to pneumatic chamberIJs). The control line was connected to a 3/2-way miniature solenoid valve (Festo AG, Germany) controlled via a data acquisition (DAQ) device (National Instruments) operated by LabVIEW programming software in initial experiments, or control commands were sent via USB to a custom-designed printed circuit board (PCB) in subsequent experiments. A pressure controller (DPI 520, GE Druck, USA) was used to regulate the applied

positive pressure. Liquid was infused through tubing by connecting one inlet port to a syringe controlled by a precision syringe pump (neMESYS, Cetoni GmbH, Germany).

5.4.3 Particle tracing

For characterization experiments, the hanging-drop loops were filled with a suspension containing micro-beads (5 µm diameter) in de-ionized water. Particles were traced along the length of the channels downstream of the pump drops by using a digital camera (ORCA-Flash 4.0, Hamamatsu, Japan) with an image acquisition rate of 100 frames per second. Imaris software was used to automatically extract the changes in particle positions and speeds. All data represent mean values of 10–20 traced particles within a defined window of interest (150 µm wide, 25 µm distance from channel walls).

5.4.4 Closed-loop actuation

The closed-loop valve controller setup included the microscope, a digital camera (Leica DFC340FX, Switzerland), a controller application on the PC, and custom-made valve driver hardware. The camera was used to acquire the microscopic images of the beating cardiac tissue, and these images were transferred to the PC. A motion detection application was implemented in C# to detect the cardiac tissue beating from the acquired images. In short, image frames were captured from a selected area on the screen (microscope live image) every 100 ms, and each frame was subtracted from a predefined image (spheroid at rest). The gray-value differences of all pixels of the obtained differential image were summed up, and a pump actuation command was sent to the valve driver when the value exceeded a user-defined threshold. All closed-loop actuation experiments were performed on a Leica DMI6000 inverted microscope comprising an environmental box for temperature control and a stage-top incubator for CO2 and humidity control (Life Imaging Services, Switzerland). For the proof-of-principle results in Fig. 6b and c, the feedback experiments needed to be split into two experimental sequences, because cardiac spheroid imaging and particle tracing could not be done simultaneously. The reason is that the respective monitoring areas are at different locations on the chip and could not be imaged in the same field of view of the camera. We therefore first recorded the cardiac beating for 30 seconds, while the pump was operating in feedback mode. We then replayed the movie on the screen right after with the same feedback settings to image the particles and analyze the flow. No change in beating behavior was observed during that time. All other feedback experiments were executed in real time.

5.4.5 Spheroid cultures

Human iPS-derived cardiac microtissue spheroids and cardiac maintenance medium were obtained from InSphero AG, Zurich, Switzerland. For experiments, a pre-formed spheroid was inserted into one of the hanging drops by manual pipetting. The characteristic beating frequency of those spheroids was in the range of 60 to 90 bpm at 37 °C, which is very similar to the range of beating frequencies of human hearts.

5.5 Conclusion

Design, fabrication, and characterization of a novel micropump for completely open microfluidic systems have been presented. The concept of using the liquid–air interface forces to actuate pressure-driven flow enables seamless integration of the pump into hanging drop networks without increasing fabrication complexity. Pump operation can be controlled via a single pneumatic line, and parallel actuation of several drop arrangements on the same chip is possible. Simultaneous unidirectional fluid flow in up to 8 independent hanging-drop networks as well as fluid flow in larger hanging drop arrays has been presented. The developed technique offers great potential for parallel and multi-tissue experiments in hanging drop networks.

By using the pump in feedback with a beating cardiac microtissue in the HDN, the function and characteristic behavior of a heart can be simulated, which constitutes a further step towards microphysiological, or so-called “body-on-achip” systems. This approach opens a route to investigate the impact of compounds on the heart itself and, additionally, the impact on other organ models in the system as a consequence of alterations in liquid circulation. Temperature reduction in the system, e.g., reduced the cardiac tissue beating frequency, which in turn reduced the pump rate and reproduced the respective physiological effect. We also biochemically stimulated the heart tissue in the HDN system and could not only see the direct effect on the tissue – an increased beating rate – but also its functional effect on the whole system – in this case a faster clearance of the stimulating drug from the heart environment as a results of the increased pump rate.

Very different concepts have been applied before to integrate heart functions on-chip [21–23]. The approach proposed here provides seamless integration of a 3D heart organ model, capable to closely mimic in vivo like conditions and tissue responses upon, e.g., dosage of compounds, as well as the major function of the heart – pumping of liquid to the different organ models. This holds particularly true, as the two features are interdependent in an organism: any response of cardiac tissue to a substance will inherently alter the pumping characteristics and the supply of liquid and nutrients to the other organs.

Acknowledgements

Financial support was provided by a European Union FP7 grant ‘Body on a chip’ ICT-FET-296257, an individual Ambizione Grant 142440 of the Swiss National Science Foundation for Olivier Frey and the ERC Advanced Grant “NeuroCMOS” under contract number AdG 267351. Amir Shadmani received individual support through the Marie Curie Research Training Network “EngCaBra” (grant agreement no. 264417). We would like to thank InSphero AG, Switzerland, for providing human iPS-derived cardiac microtissues (hCdMT) and culture medium. We also thank D-BSSE Single Cell Facility and the Cleanroom Facility at ETH Zurich for assistance in microscopy and microfabrication. The authors are grateful to Christian Zuppinger, University of Bern, for suggesting the use of isoproterenol for cardiac tissue stimulation.

References

[1] N. Alépée, A. Bahinski, M. Daneshian, B. De Wever, E. Fritsche, A. Goldberg, J. Hansmann, T. Hartung, J. Haycock, H. T. Hogberg, L. Hoelting, J. M. Kelm, S. Kadereit, E. McVey, R. Landsiedel, M. Leist, M. Lübberstedt, F. Noor, C. Pellevoisin, D. Petersohn, U. Pfannenbecker, K. Reisinger, T. Ramirez, B. Rothen-Rutishauser, M. Schäfer-Korting, K. Zeilinger and M.-G. Zurich, State-of-the-Art of 3D Cultures (Organs-on-a-Chip) in Safety Testing and Pathophysiology, ALTEX, 2014, 4(31), 441–477.

[2] M. B. Esch, A. S. T. Smith, J.-M. Prot, C. Oleaga, J. J. Hickman and M. L. Shuler, Adv. Drug Delivery Rev., 2014, 69–70, 158–169.

[3] S. N. Bhatia and D. E. Ingber, Nat. Biotechnol., 2014, 32, 760–772.

[4] J. H. Sung, C. Kam and M. L. Shuler, Lab Chip, 2010, 10, 446–455.

[5] P. M. van Midwoud, M. T. Merema, E. Verpoorte and G. M. M. Groothuis, Lab Chip, 2010, 10, 2778–2786.

[6] C. Zhang, Z. Zhao, N. A. Abdul Rahim, D. van Noort and H. Yu, Lab Chip, 2009, 9, 3185–3192.

[7] J. P. Wikswo, E. L. Curtis, Z. E. Eagleton, B. C. Evans, A. Kole, L. H. Hofmeister and W. J. Matloff, Lab Chip, 2013, 13, 3496–3511.

[8] D. Huh, Y. Torisawa, G. A. Hamilton, H. J. Kim and D. E. Ingber, Lab Chip, 2012, 12, 2156–2164.

[9] O. Frey, P. M. Misun, D. A. Fluri, J. G. Hengstler and A. Hierlemann, Nat. Commun., 2014, 5, 4250.

[10] F. Pampaloni, E. G. Reynaud and E. H. K. Stelzer, Nat. Rev. Mol. Cell Biol., 2007, 8, 839–845.

[11] S. Messner, I. Agarkova, W. Moritz and J. M. Kelm, Arch. Toxicol., 2013, 87, 209–213.

[12] R. A. Zuellig, G. Cavallari, P. Gerber, O. Tschopp, G. A. Spinas, W. Moritz and R. Lehmann, J. Tissue Eng. Regener. Med., 2014, 4, 524–531.

[13] P. Beauchamp, W. Moritz, J. M. Kelm, N. D. Ullrich, I. Agarkova, B. Anson, T. M. Suter and C. Zuppinger, Tissue Eng., Part C, 2015, 21(8), 852–861.

[14] E. Urich, C. Patsch, S. Aigner, M. Graf, R. Iacone and P.-O. Freskgård, Sci. Rep., 2013, 3, 1500.

[15] C. R. Thoma, M. Zimmermann, I. Agarkova, J. M. Kelm and W. Krek, Adv. Drug Delivery Rev., 2014, 69–70, 29–41.

[16] A. Amann, M. Zwierzina, G. Gamerith, M. Bitsche and J. M. Huber, et. al., Development of an Innovative 3D Cell Culture System to Study Tumour - Stroma Interactions in Non-Small Cell Lung Cancer Cells, PLoS One, 2014, 9(3), e92511.

[17] M. Rimann, S. Laternser, A. Gvozdenovic, R. Muff, B. Fuchs, J. M. Kelm and U. Graf-Hausner, J. Biotechnol., 2014, 189, 129–135.

[18] B. D. Iverson and S. V. Garimella, Microfluid. Nanofluid., 2008, 5, 145–174.

[19] A. K. Au, H. Lai, B. R. Utela and A. Folch, Micromachines, 2011, 2, 179–220.

[20] C. K. Byun, K. Abi-Samra, Y.-K. Cho and S. Takayama, Electrophoresis, 2013, 245–257.

[21] A. Agarwal, J. A. Goss, A. Cho, M. L. McCain and K. K. Parker, Lab Chip, 2013, 13, 3599–3608.

[22] A. Mathur, P. Loskill, K. Shao, N. Huebsch, S. Hong, S. G. Marcus, N. Marks, M. Mandegar, B. R. Conklin, L. P. Lee and K. E. Healy, Sci. Rep., 2015, 5, 1–7.

[23] Y. Tanaka, K. Sato, T. Shimizu, M. Yamato, T. Okano and T. Kitamori, Lab Chip, 2007, 7, 207–212.

CHAPTER 6

Conclusions and Outlook

6.1 Conclusion: Multifunctional high-density MEA

The design, fabrication, characterization and application of a novel multifunctional high-density CMOS MEA has been presented. The chip featured a large microelectrode array, each electrode of which could be connected to different sensing and stimulation units. The different modalities on the chip included action-potential recording units, local-field-potential-recording units, impedance-measurement units, neurotransmitter-detection units and dual-mode voltage-only or current/voltage-controlled stimulation units. In this thesis, the design and implementation of the respective stimulation units, the stimulation artifact suppression circuitry, and the neurotransmitter detection units was covered in detail.

The stimulation units were designed to operate in two modes: voltage-control only and current/voltage control. For voltage-control only mode, a near rail-to-rail OTA was implemented, which worked in unity-gain feedback and obviated the need of large feedback resistors. The current/voltage controlled mode enables to simultaneously control the current and voltage level of the stimulation pulse and was based on a CCII+ with feedback circuits for sensing the output voltage. By exploiting the features of this stimulation circuit, the relation between current and voltage values of biphasic stimulation pulses for exciting neurons was investigated through experiments.

The artifacts caused by stimulation signals pose a serious challenge for operating the recording channel circuitry that was connected to electrodes neighboring the stimulation site. Therefore a new technique was developed, which drastically reduced stimulation artifacts in the recording channels. This technique, called “soft reset” technique, reduced the effective resistance of the MOS resistors during the stimulation period. The MOS resistors (pseudo-resistors), in turn, were used for establishing low-frequency high-pass filters in the recording channels. Measurements proved the effectiveness of applying this technique as the duration of stimulation artifacts was decreased from 13 ms to 200 µs.

While the stimulation and electrophysiological recording units were used to electrically interact with neurons, neuronal activity generally involves both, electrical and chemical processes. Chemical processes are involved in transmitting signals between neurons through the release of chemical substances, i.e., neurotransmitters. The possibility of detecting neurotransmitters in a living neuronal network provides valuable information about neural communication. Therefore, neurotransmitter

detection units were devised and implemented in the overall system chip. The neurotransmitter detection scheme was based on the Fast Scan Cyclic Voltammetry (FSCV) technique. To minimize disturbances to other sensing units on the chip, the FSCV scan signals were applied directly to the working electrodes, while the reference electrode potential was kept constant. Each neurotransmitter detection unit included a voltage copier to buffer the scanning voltage on the working electrode and to mirror the electrode current to the TIA block and, afterwards, to ADC. The designed neurotransmitter detection units were capable of detecting currents at nA to µA levels and were used to detect different concentrations of dopamine in PBS during an experiment.

The overall system chip has been used for measurements of cultures of dissociated rat cortical neurons and slices of the mouse cerebellum. All experimental procedures can also be applied to a variety of other electroactive preparations such cardiac tissues or retinae.

6.2 Conclusion: Experimental setup design

Having a reliable setup is essential to achieving optimal performance of the designed microsystem chips. During the course of this PhD work, setups for multifunctional MEA chips, nanowire sensor chips and hanging drop networks have been designed.

The setup for the multifunctional MEA chips included a custom-designed PCB for generating bias voltages and clock signals and a DAQ Card to acquire the digital data and to then transfer them to PC for visualization and data storage. One of the challenges with multifunctional MEA chip included the power drop on the digital supply rails, while the chip received configuration commands. This power drop caused random rotations in the sequence of transmitted electrophysiological data. To overcome this issue, the clock signals for data receiving and data transmission were defined to be out-of-phase by using an FPGA on the custom-designed PCB.

A setup for nanowire sensor chips was designed utilizing an FPGA and a USB controller chip. For online visualization and data storage, an interface application was developed using the C# programming language. The main requirements for this setup were robustness and portability.

In collaboration with the microfludics team, a setup for closed-loop pump actuation in hanging-drop networks was developed. The relaxed timing constraints in this project allowed for implementing all computational tasks, such as motion-detection and pump command generation, in software.

For these setups, signal and power integrity issues were carefully considered so that stable, robust and low-noise setups were obtained.

6.3 Outlook

The developed multifunctional MEA chip features the largest active area (10.9 mm^2) to date, with the largest number of sensing/stimulation modules on the same platform. These features together with high spatial resolution (13.5 µm electrode pitch) and large number (2048) of parallel electrophysiological recording channels, render the system chip a unique tool for studying electroactive cells at network and/or subcellular levels.

The current/voltage control option in the stimulation units can be used for in-depth investigations of the responses of neurons to different current and voltage levels of stimulation pulses of arbitrary shapes and pulse widths. Moreover, a new feature for stimulation buffers was included, which enables to simultaneously generate two inverted pulse waveforms around mid-supply (1.65 V). Using this feature, simultaneous stimulation pulses of opposite polarity can be applied to neighboring electrodes in order to achieve spatially more confined stimulation. The new artifact suppression feature provides the possibility to study neuronal responses in close proximity to the stimulation site, which was previously impossible due to the signal saturation of the recording channels. By exploiting this feature, the initiation of neuronal excitation can be better investigated.

The available platinum electrodes were utilized in pilot experiments of detecting neurotransmitters by using the on-chip neurotransmitter detection units. On the one hand, the designed neurotransmitter detection units provided low-noise and high-resolution current readouts, on the other hand, it became evident that metal or Pt electrodes are not the optimum choice for electrochemical detection due to the low cathodic potential range and the bio-fouling of the metal electrodes.. Carbon-based microelectrode materials are commonly used as working electrodes for electrochemical detections. In order to get optimal performance of on-chip neurotransmitter detection units, more research is required to use carbon-based or other alternative electrode materials on the MEA for electrochemical detection.

The designed setup for the multifunctional MEA needs to be further developed to achieve a stand-alone setup without the need for a DAQ Card. The new setup should be equipped with an FPGA to perform signal processing tasks such as spike sorting and to enable fast feedback experiments. Due to the increased data rate of this new MEA chip, issues of data transmission and data storage will require careful consideration.

The developed setup for the nanowire sensor chip was used in applications with collaborating research groups for silicon nanowires developed at the Danish Technical University and for nanowire-based gas sensors at ETH Zurich, where the portability of the setup allowed for usage in an underground garage to measure NO_2 levels.

For the hanging-drop pump platform, one of the major issues was the evaporation of medium, which rendered the setup only suitable for short-term measurements. Work is in progress in the group to provide automated control of the medium volume, the results of which then will be integrated to the pump platform. Moreover, feasibility studies to integrate magnetic beads and cell flow into the hanging-drop platform are under research. The possibility of having flowing beads or cells in the medium would allow for modeling immune responses of blood cells and organs.

APPENDIX A

Hanging-drop pump setup design

A. Hardware design

To drive the pneumatic pumps, the close and open commands should be sent to valves in a controlled timed manner. In this project 3/2-way miniature solenoid valves (Festo, Germany) were used with 24 VDC operating voltage. Low speed requirements of the pump operation, allowed us to push all the computation to the PC and to only send the switching bits to the valve. Initially, a DAQCard with LabView interface was developed, however, in order to have more control over the valves and to also integrate the closed-loop operation, a cheaper solution of using FT245R chip was chosen. This chip from Future Technology Devices International Ltd. is a USB-to-Parallel converter FIFO which works similar as old-days parallel ports of the PCs. This chip simplifies the USB-to-parallel designs by integrating the EEPROM, clock circuits and USB resistors onto the chip with the entire USB protocol handled on the chip. For interfacing to this chip there are two options: 1.Using specific device drivers providing 1Megabyte/second data transfer rate and 2. Using Virtual COM Port (VCP) drivers providing 300 kilobyte/second data transfer rate. In the VCP mode, the chip can be treated as a COM port device from the interfacing program. The chip can be powered from the USB bus, and the output voltage level of the chip can be adjusted to 5V / 3.3V / 2.8V / 1.8V for driving CMOS circuits with maximum output current of 24 mA on each output pin [1]. However for driving the valves, a 24 VDC signal is needed, and the valve consumes 1W power in the closed state. Therefore, current boosting and voltage level shifting is required. Fig. 1 shows the designed discrete circuit, which translates the 5 V of the FT245R output pins to 24 V with no more than 2mA loading on the output pins of the chip. The custom circuit is powered with a 24 VDC external power source. Schottky diode (D1) avoids the high-voltage formation due to switching off of the valve solenoid.

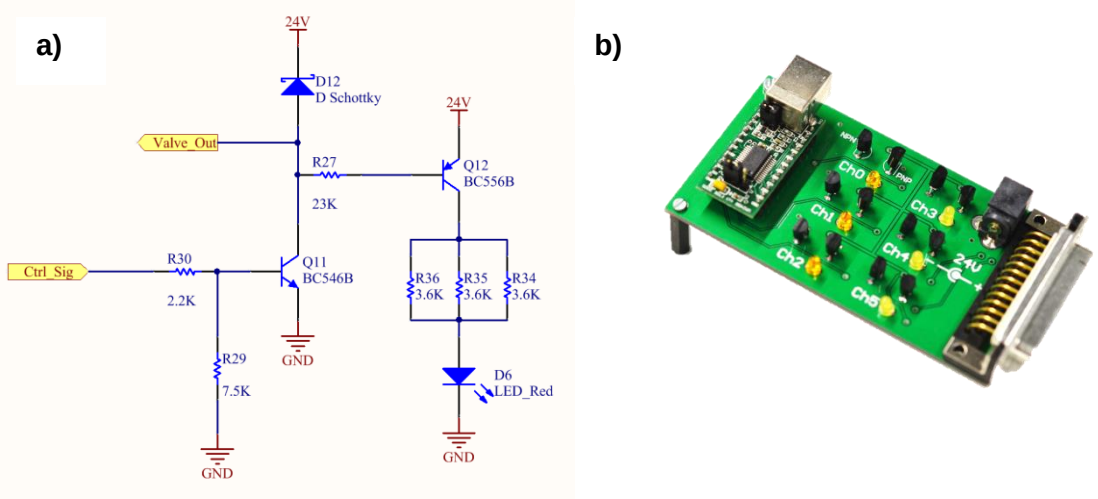


Fig. 1. a) Circuit schematic of the custom designed PCB for driving the valves (one channel shown), b) the final fabricated PCB with 6 simultaneous channels.

B. Software design

A C# interface program was developed for sending the operation commands to USB-to-Parallel chip. The on/off times of the valves are first calculated and then sent to the USB-to-Parallel using timer events. Fig. 2 shows the screenshot of the developed C# interface program for simultaneously controlling three valves.



Fig. 2. Screenshot of the developed C# interface program for controlling the valves

C. Closed loop operation

For “adding the heart to the microfluidic hanging drops”, the pump pulses need to be synchronized with the beatings of the heart spheroid placed in the hanging drop. Using an inverted microscope the spheroid image was magnified and then captured and transferred to PC in real-time, using a high-speed digital camera. On the C# interface program a motion detection code was added to detect the movements of the beating cells in the real-time spheroid image and to trigger the pump accordingly. Fig. 3 shows the block diagram of this closed loop operation. The C# program samples the image pixel information every 100 ms. Afterwards the image analysis is performed by first thresholding the image and then using subtraction technique to extract the motion information. The complete C# code is made available in the next subsection.

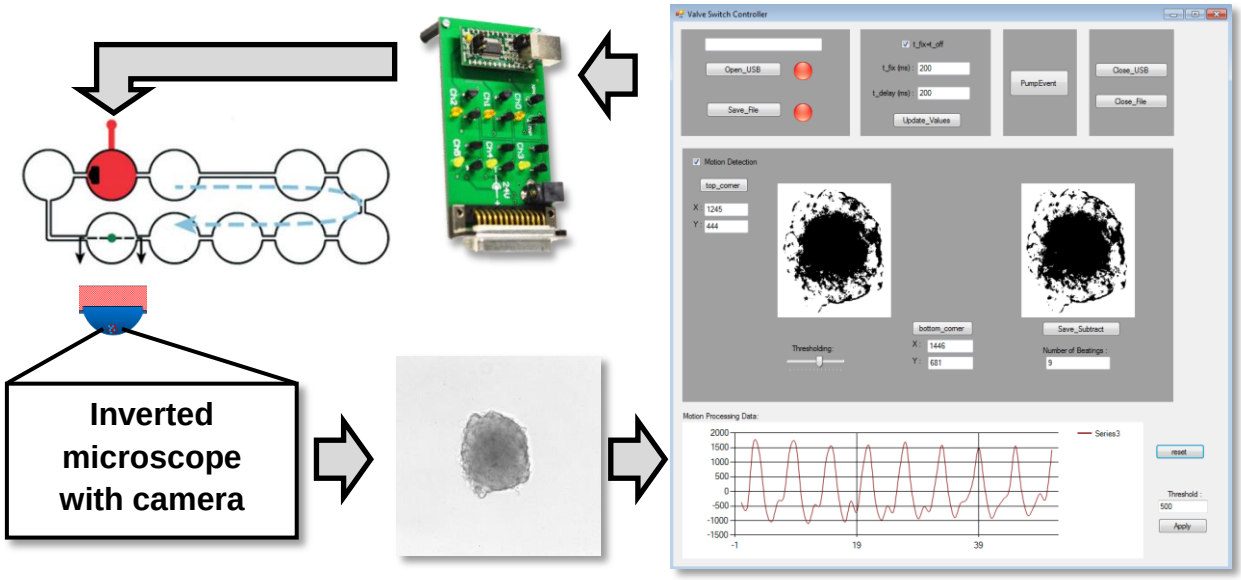


Fig. 3. Schematic view of closed-loop setup

References

[1] Future Technology Devices International Ltd. “USB Bridging Solutions”, FT245R datasheet version 2.13, 2010.

APPENDIX B

Publications

Journal Papers

[1] J. Dragas*, V. Viswam*, **A. Shadmani***, Y. Chen*, R. Bounik, A. Stettler, M. Radivojevic, S. Geissler, M. Obien, J. Müller, A. Hierlemann, "A multi-functional microelectrode array featuring 59760 electrodes, 2048 electrophysiology channels, stimulation, impedance measurement and neurotransmitter detection channels", submitted to IEEE J. Solid-State Circuits.
* Authors contributed equally

[2] **A. Shadmani**, V. Viswam, Y. Chen, R. Bounik, J. Dragas, M. Radivojevic, S. Geissler, S. Sitnikov, J. Müller, A. Hierlemann, "Integrated current/voltage stimulation buffer and artifact suppression circuits for multifunctional high-density microelectrode arrays", in preparation.

[3] P. Livi, **A. Shadmani**, m. Wipf, R. L. Stoop, J. Rothe, Y. Chen. M. Calame, C. Schoenenberger, A. Hierlemann, "Sensor system including silicon nanowire ion sensitive FET arrays and CMOS readout", *Sens. Actuators B Chem*., vol. 104, pp. 568-577, Dec. 2014.

[4] P. Livi, M. Kwiat, **A. Shadmani**, A. Pevzner, G. Navarra, J. Rothe, A. Stettler, Y. Chen, F. Patolsky, A. Hierlemann, "Monolithic integration of a silicon nanowire field-effect transistors array on a complementary metal-oxide semiconductor chip for biochemical sensor applications", Anal. Chem., vol. 87, no. 19, pp. 9982-9990, Oct. 2015.

[5] S. Rismani Yazdi, **A. Shadmani**, S.C. Bürgel, P.M. Misun, A. Hierlemann and O. Frey, "Adding the 'heart' to hanging drop networks for microphysiological multi-tissue experiments", Lab Chip, 15, 4138-4147, 2015.

[6] M. Ballini, J. Müller, P. Livi, Y. Chen, U. Frey, A. Stettler, **A. Shadmani**, V. Viswam, I. L. Jones, D. Jäckel, M. Radivojevic, M. K. Lewandowska, W. Gong, M. Fiscella, D. J. Bakkum, F. Heer and A. Hierlemann, "A 1024-channel CMOS microelectrode array with 26400 electrodes for recording and stimulation of electrogenic cells in vitro", IEEE J. Solid-State Circuits, vol. 49, no. 11, pp. 2705-2719, Nov. 2014.

[7] J. Müller, M. Ballini, P. Livi, Y. Chen, M. Radivojevic, **A. Shadmani**, V. Viswam, I.L. Jones, M. Fiscella, R. Diggelmann, A. Stettler, U. Frey, D.J. Bakkum and A. Hierlemann, "High-resolution CMOS MEA platform to study neurons at subcellular, cellular, and network levels", Lab Chip, 15, 2767-2780. 2015.

Conference Contributions

[1] V. Viswam*, J. Dragas*, **A. Shadmani***, Y. Chen*, A. Stettler, J. Müller, A. Hierlemann, "Multi-functional microelectrode array system featuring 59'760 electrodes, 2048 electrophysiology channels, impedance, and neurotransmitter measurement units", in Proceedings of the IEEE International Solid-State Circuits Conference (ISSCC), San Francisco, CA, USA, pp. 394-396, Jan. 2016.
* Authors contributed equally

[2] V. Viswam, Y. Chen, **A. Shadmani**, J. Dragas, R. Bounik, M. Radivojevic, J. Müller, A. Hierlemann, "2048 Action Potential Recording Channels with 2.4 µVrms Noise and Stimulation Artifact Suppression", in Proceedings of the Biomedical Circuits and Systems Conference (BioCAS), Shanghai, China, Oct. 2016.

[3] **A. Shadmani**, V. Viswam, J. Dragas, Y. Chen, R. Bounik, J. Müller, A. Hierlemann, "A Novel Multi-Functional High-Density MEA Platform for Studying Electrogenic Cells", in Proceedings of the 10th International Meeting on Substrate-Integrated Microelectrode Arrays, Reutlingen, Germany, 2016.

[4] **A. Shadmani**, J. Müller, P. Livi, M. Ballini, Y. Chen and A. Hierlemann, "Stimulation Artifact Suppression Techniques for High-Density Microelectrode Arrays", in Proceedings of 9th International Meeting on Substrate-Integrated Microelectrode Arrays, Reutlingen, Germany, 2014.

[5] M. Ballini, J. Müller, P. Livi, Y.Chen, U. Frey, **A. Shadmani**, I. L. Jones, W. Gong, M. Fiscella, M. Radivojevic, D. J. Bakkum, A. Stettler, F. Heer and A. Hierlemann, "A 1024-Channel CMOS Microelectrode-Array System with 26'400 Electrodes for Recording and Stimulation of Electro-active Cells In-vitro", Digest of Technical Papers of Symposium on VLSI Circuits, 2013, Kyoto, Japan, pp. C54-C55, 2013.

[6] J. Müller, M. Ballini, P. Livi, Y. Chen, **A. Shadmani**, U. Frey, I. L. Jones, M. Fiscella, M. Radivojevic, D. J. Bakkum, A. Stettler, F. Heer and A. Hierlemann, "Conferring Flexibility and Reconfigurability to a 26'400 Microelectrode CMOS Array for High Throughput Neural Recordings", Proceedings of the 17th IEEE International Conference on Solid-State Sensors, Actuators & Microsystems, Transducers, 2013, Barcelona, Spain, pp. 744-747, 2013.

[7] V. Viswam, Y. Chen, R. Bounik, **A. Shadmani**, J. Dragas, J. Müller, A. Hierlemann, "Highly Matched TΩ Pseudo-resistors for Multi-channel Neural Signal Recording Applications", in Proceedings of the 10th International Meeting on Substrate-Integrated Microelectrode Arrays, Reutlingen, Germany, 2016.

[8] R. Bounik, V. Viswam, **A. Shadmani**, J. Dragas, J. Müller, Y. Chen, A. Hierlemann, "Impedance Measurements on a Multi-functional Neural MEA Platform", in Proceedings of the 10th International Meeting on Substrate-Integrated Microelectrode Arrays, Reutlingen, Germany, 2016.

[9] J. Dragas, V. Viswam, **A. Shadmani**, Y. Chen, A. Stettler, M. Radivojevic, J. Müller, A. Hierlemann, "Multi-Functional CMOS-HDMEA Integrating 59'760 Electrodes, 2'048 Electrophysiology Readouts, Stimulation Units, Impedance and Neurotransmitter Measurement Units", in Proceedings of the 10th International

Meeting on Substrate-Integrated Microelectrode Arrays, Reutlingen, Germany, 2016.

[10] V. Viswam, R. Bounik, **A. Shadmani**, J. Dragas, A.J. Boos, A. Birchler, J. Müller, Y. Chen, A. Hierlemann, "High-Density CMOS Microelectrode Array System for Impedance Spectroscopy and Imaging of Biological Cells", Oct. 2016, Orlando, FL, USA, 2016.

[11] V. Viswam, R. Bounik, **A. Shadmani**, J. Dragas, J. Müller, Y. Chen, A. Hierlemann, "32-Channel Integrated Electrical Impedance Sensors on a Multi-Functional Neural Microelectrode Array Platform", in Proceedings of the 30th Eurosensors Conf., Sep. 2016, Budapest, Hungary.

[12] S. Rismani Yazdi, **A. Shadmani**, A. Hierlemann, O. Frey, "Microfluidic Hanging-drop Platform for Parallel Closed-loop Multi-tissue Experiments", in Proceedings of the 28th IEEE International Conference on Micro Electro Mechanical Systems (MEMS), 2015, Estoril, Portugal, pp. 535-538. 2015.

[13] S. Rismani Yazdi, **A. Shadmani**, A. Hierlemann, O. Frey, "Integration of a Peristaltic Micropump into Microfluidic Hanging Drop Networks", in Proceedings of the Micro and Nano Engineering Conference Sep. 2014, Lausanne, Switzerland, 2014.

[14] P.M. Misun, **A. Shadmani**, A. Hierlemann, O. Frey, "Integrating multi-electrode arrays in microfluidic hanging-drop networks", in Proceedings of the 30th IEEE International Conference on Micro Electro Mechanical Systems (MEMS), Las Vegas, NV, USA, pp. 91-94. Jan. 2017.

[15] S. Rismani Yazdi, S. C. Bürgel, **A. Shadmani**, A. Hierlemann, and O. Frey, "Integrated Pump for Hanging-Drop Networks Controlled by Human iPS-Derived Cardiac Microtissue", in Proceedings of the 19th International Conference on Miniaturized Systems for Chemistry and Life Sciences (µTAS), October 25-29, 2015, Gyeongju, Korea, pp. 111-113.

APPENDIX C

Personal conclusions

The main focus of my PhD project was to design and characterize new circuits for a novel multifunctional high-density CMOS MEA (NeuroCMOS) to be used for in-vitro studies of electrogenic cells. While having solid knowledge of CMOS circuits was essential to this project, creativity, innovation and a clear understanding about the field of application was also necessary for developing new tools for neuroscience research.

One of the functional blocks that I designed for NeuroCMOS chip was the stimulation buffer. After a detailed study of previous work in neural electrical stimulators, I realized that the respective circuits were based on either voltage-controlled or current-controlled circuits and that there was a great debate in comparing the pros and cons of these techniques with respect to stimulation efficiency, safety and complexity. Therefore, I decided to design a new circuit, which can combine the advantages of both techniques by simultaneously controlling both current and voltage of the output pulse. At a first glance, this goal seemed infeasible, as current and voltage are interrelated through the output load. However, since pulse waveforms are commonly used for stimulation purposes, it is possible to make a circuit that is controlled for one parameter (current or voltage) of the output pulse and is limited for the other parameter, leading to a controlled output current and voltage pulse. I started the design by implementing a positive current conveyor of type II (CCII+) for generating controlled current pulses and added a feedback path to control the output voltage levels.

Since the amplitude of stimulation pulses is orders of magnitude higher than the amplitude of neural signals, they cause artifacts in the recording channels connected to the electrodes neighboring the stimulation site. To suppress the stimulation artifacts, a novel technique was implemented, which decreased the dead time of the recording channels by increasing their high-pass cut-off frequency during the stimulation period. While similar resetting ideas have been pursued before, our circuit achieved far better results compared to previous implementations. This achievement was due to thorough analysis of the recording channels at the MOS level. In the recording channels, MOS transistors are used to generate high value resistances, i.e., pseudo-resistors, to establish low-frequency high-pass filters. These pseudo-resistors are connected to the input node of the amplifiers, which is a high-impedance node. Therefore, any charge accumulated on this node would result in offset or increased settling time. In our implemented technique, we tried to minimize the charge injection, caused by switching of the MOS transistors in pseudo-resistors, by

soft-resetting these transistors, which drastically improved the performance of the technique. Unlike previous implementations, we even considered the small leakage charge from the reverse-biased diode, formed at the bulk-source or bulk-drain junction, which would be accumulated at the high-impedance node and would result in a large offset. By using MOS transistors with isolated wells and connecting the bulk to the mid-supply voltage, the offset of the recording channels was decreased to a level that no additional offset cancellation technique was required.

During my work I tried to extensively learn about different circuit techniques and ideas and considered the possible benefits that we may gain by applying those techniques to our application. One example was the design of neurotransmitter detection circuit. For the electrochemical detection of neurotransmitters, fast-scan cyclic voltammetry technique was employed. To minimize disturbances to other measurement modalities on the chip, the scan voltage for voltammetry was planned to be directly applied to the working electrodes. Therefore, the circuits required a voltage copier to buffer the scan voltage to the electrodes, and the output current of the buffer needed to be sensed and converted to the respective voltage values. Two structures of trans-impedance amplifiers (TIA) were devised, which are suitable for different measurement scenarios. In one structure, I reused the local common mode feedback structure, similar to the OTA design used in stimulation circuits. Since the dominant pole of this OTA structure is placed at the output node, it could provide proper stability even with large electrode capacitance values, which made it suitable for driving pseudo-large electrodes consisting of many connected electrodes. Current mirrors with cascode transistors were used to sense the output current of the OTA, which was further converted to a voltage using a resistive TIA. In the second neurotransmitter detection circuit structure, a different approach was devised using a transconductance cell. In a linear transconductance cell, the output current is directly proportional to the input voltage values. Therefore, by integrating a linear transconductance cell in a unit-gain feedback opamp, there would be no need for a current-to-voltage convertor stage, and the whole feedback structure could be used as the voltage copier element. For attaining stability of this structure, a similar idea of using three-stage Miller-compensated opamps was exploited and applied to this application.

Interdisciplinary research is based upon combining a broad range of skills and expertise. Through discussions within our team, I realized that there is a great need for skills in designing experimental setups. After the design of CMOS or microfluidic chips, these devices require a robust communication with a PC to configure the device or acquire and record the data for further analysis. Since I had previous experience and skills in discrete electronics such as PCB design, signal/power integrity and EMC issues, as well as software design, I got involved in several projects to design experimental setups. One of the requirements of the users was to provide the quickest possible solutions in developing a robust experimental setup. Moreover, the software application needed to be easy-to-use to avoid consuming extra time of a potential user for learning or debugging the software.

One of my designed setups was for nanowire sensor project. The nanowire sensor chip has been developed in our lab to detect small current changes (pA range) in the nanowires. The chip incorporated a current-to-frequency and sigma delta blocks. The initial attempted setup, using data acquisition (DAQ) Card, had failed due to the high data rate of the chip. Moreover, due to the portability limitations of this setup, it could not be used at the collaborator's site, where the nanowires were fabricated. After proper analysis, I decided to include an FPGA to perform preliminary signal processing in order to decrease the output data rate. On the other hand for transferring the data to PC, there are several standard protocols such as RS232, USB or Ethernet link, providing different communication speed and complexity. The decreased data rate, after signal processing, could be transmitted to a PC by using a high-speed USB 2.0 bus. The USB port can also supply 5V with maximum of 500 mA current. Since our setup would consume less power, we could use the USB port as the supply source, eliminating the need for external power supplies. The software interface was developed using C# programming language to provide a user-friendly Windows-based application. Due to the robustness and portability of the setup together with its user-friendly interface, it has been further used in similar nanowire projects conducted at the other groups at DTU in Copenhagen and at ETH Zurich.

During the course of collaborations with microfluidic team, a setup was developed for closed-loop pump actuation in hanging-drop networks. The motion of the cardiac tissue in the hanging drop was captured by a camera installed on a microscope. A motion detection algorithm was developed in C# to detect the motions of the cardiac tissue and drive the pump accordingly. The loose timing constraints in this project allowed all computational tasks to be implemented on software, which resulted in a convenient and fast setup solution.

The main challenging setup design was the setup of NeuroCMOS chip. Due to the high clock speed of the NeuroCMOS chip, careful consideration of signal integrity issues was required to protect the sensitive analog signals from noisy digital signals. Therefore a four-layer PCB with partitioned sections for analog and digital signals was developed. However in spite of all considerations and proper filtering, the chip initially failed to perform as designed. While the normal recording operation seemed accurate, however, random shifts would occur in the time slots of the output data stream at the occasions of sending configuration commands to the chip. After thorough investigation, it was concluded that the problem was due to digital supply fluctuations at the times of sending configuration commands to the chip. Since this problem was occurring at random configuration commands, tackling this issue was rather challenging. However this problem could be solved by generating two out-of-phase clock signals for transmit and receive functions so that the voltage drop, caused by digital receiver circuits, would not affect the operation of transmitter circuits. For sending configuration commands to the chip, we developed a C# interface, which provided us with great flexibility in constructing different commands or sequences of commands.

In conclusion, performing an electronics research project requires solid knowledge of the field, clear understanding of the application and ability to devise the most suitable solution based on the requirements. By combining the techniques problem-solving capabilities that I acquired during my PhD studies, I feel confident to conduct other projects in scientific or industrial research areas.

APPENDIX D

Acknowledgments

I would like to thank Prof. Andreas Hierlemann for giving me the opportunity to pursue a Ph.D. degree in his laboratory and for the freedom he gave me to organize my work in order to tackle the technical problems and also for his great help in proofreading and correcting my manuscripts.

I also would like to thank Dr. Yihui Chen for his great technical guidance and his support throughout my Ph.D. I thank Prof. Schmid and Prof. Guiducci for being my Ph.D. examiners and also for many lessons that I have learned from Prof. Schmid.

I am thankful to Vijay Viswam, Paolo Livi and Jan Müller for the very fruitful technical discussions and very good times at work. I also thank Olivier Frey for his great support and guidance.

I am very thankful to all my colleagues and friends at BEL group who helped and supported me during my Ph.D. and provided a very nice and friendly environment to work.

I am also very grateful for the financial support. The work for this thesis was supported by the European Community through the FP7 ERC Advanced Grant “NeuroCMOS” under contract number AdG 267351 and the EU Marie Curie Initial Training Network “EngCaBra” under grant agreement FP7MC-ITN 264417.